From one to two, and up the ladder

When does a rung of the tower $\mathrm{Cl}(3n, 3n)$ carry several particles?
The second rung $\mathrm{Cl}(6, 6)$ and the rungs above it

Investigation note to *The electromagnetic sphere*

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Abstract

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1 One particle or two: the second rung

The conventions of [1] say that a constituent is a tensor factor $\mathrm{Cl}(3,3)$ of the tower, that the word never refers to a particle, and that three constituents do not describe three particles (Section 1.4 of [1]). The duplication $\mathrm{Cl}(3,3) \to \mathrm{Cl}(6,6)$ of [2] (Sections 7–9 there) is therefore not yet the emergence of two particles from one. This section asks in what precise sense, and under which added structures, the second rung $\mathrm{Cl}(6,6)$ carries two particles rather than one particle with a larger internal space. Four tests separate the two readings: whether the copies have separate mass shells, whether there is one complex structure under which both copies are complex Dirac spinors, whether some operator counts particles, and whether exchange of the copies sorts states by statistics. The answer has three parts. The graded tower describes one particle: its two mass terms anticommute, they add in quadrature, and no state has a sharp rest energy in each copy (fixed). A two-particle reading exists, and it is fixed as an algebra: the commutant of the first copy is a second, commuting $\mathrm{Cl}(3,3)$, whose generators are the graded ones dressed by the chirality of the first copy (fixed). But it needs three further structures, and none of them is selected. They are a common complex structure, which exists only on a 32-dimensional subspace, an additive Hamiltonian that is not a vector of $\mathrm{Cl}(6,6)$, and an ungraded exchange that is not the Clifford exchange (constructed; the selection of a complex structure is obstructed under the diagonal action). The Fock number of the tower counts quanta, not particles. No gauge-admissible field that commutes with the binding root changes it (fixed). It is not invariant under the interior Lorentz algebra, under the boosts of $\mathfrak{l}_G$, or under the charge J (fixed), so as a particle number it is an internal reading (interpretive). Identifiers e1-n refer to the `PASS` lines of `verify/e1_two_from_one.py` (Table 3).

Notation. $S = \mathbb{R}^8$ with the realisation of equation (1) of [1], $\omega = e_1 \cdots e_6$, $J = e_2e_3$, $T = e_1e_2e_3$, $\Omega = e_1e_4e_5e_6$, and $\gamma^\mu = (e_1, e_4, e_5, e_6)$. Appendix A of [1] records $\omega = J\Omega = \Gamma$, the chirality. On $S_2 = S \otimes S = \mathbb{R}^{64}$ the graded generators are $E_{(1,a)} = e_a \otimes \mathbf{1}$ and $E_{(2,a)} = \omega \otimes e_a$ (equation (2) of [1]), and the *ungraded* copy-2 generators are $u_a = \mathbf{1} \otimes e_a$. We write $\Gamma^{(1)} = \omega \otimes \mathbf{1}$ and $\Gamma^{(2)} = \mathbf{1} \otimes \omega$ for the volume elements of the two constituents, $J^{(1)} = J \otimes \mathbf{1}$, $J^{(2)} = \mathbf{1} \otimes J$, $T^{(1)} = T \otimes \mathbf{1}$, $T^{(2)} = \mathbf{1} \otimes T$, and $\omega_S = E_{(1,2)}E_{(1,3)}E_{(2,2)}E_{(2,3)}$ for the volume element of the four surplus directions. On vectors, K_n is the para-complex structure $Ke_i = e_{i+3}$, $Ke_{i+3} = e_i$ on each constituent; it is $-\mathcal{K}$ of Appendix A of [1]. G_n is the binding root, and $\mathfrak{l}_G$ and $\mathfrak{l}_{\mathrm{int}}$ are

the field-strength and interior Lorentz algebras of one constituent. The Fock modes are $a^{\pm}_{k,i} = \frac{1}{2}(E_{(k,i)} \pm E_{(k,i+3)})$, with $N = \sum_{k,i} a^{+}_{k,i} a^{-}_{k,i}$ and $N^{(k)}$ the part of constituent k (Proposition 7.2 of [2]). The rest masses are $m_1, m_2 \geq 0$ and $M = (m_1^2 + m_2^2)^{1/2}$.

1.1 Two copies of $\mathrm{Cl}(3,3)$ inside $\mathrm{Cl}(6,6)$

Proposition 1.1 (The commuting copy)**.** *(i) The twelve $E_{(k,a)}$ satisfy the Clifford relations of $\mathbb{R}^{6,6}$. The six u_a satisfy those of $\mathbb{R}^{3,3}$ and commute with every $E_{(1,b)}$. Moreover $u_a = \Gamma^{(1)} E_{(2,a)} = E_{(1,1)} \cdots E_{(1,6)} E_{(2,a)}$, a blade of grade 7.*

(ii) The commutant of $\mathrm{Cl}(3,3) \otimes \mathbf{1}$ in $M_{64}(\mathbb{R})$ is spanned by 64 blades of $\mathrm{Cl}(6,6)$, each equal to $\pm \mathbf{1} \otimes e_J$; it is $\mathbf{1} \otimes M_8(\mathbb{R})$, the algebra generated by the u_a. A second copy of $\mathrm{Cl}(3,3)$ commuting with the first is therefore unique up to conjugation by $\mathbf{1} \otimes GL(8,\mathbb{R})$.

(iii) The dressing is the chirality and the Fock parity of the first copy: $\Gamma^{(1)} = (J\Omega) \otimes \mathbf{1} = -(-1)^{N^{(1)}}$.

(iv) Both the ungraded product $\mathrm{Cl}(3,3) \otimes \mathrm{Cl}(3,3)$ and the graded product $\mathrm{Cl}(3,3) \hat{\otimes} \mathrm{Cl}(3,3) = \mathrm{Cl}(6,6)$ are $M_{64}(\mathbb{R})$, with the same 4096 blades as basis. The twelve elements $e_a \otimes \mathbf{1}$, u_a span no quadratic space: $(e_1 \otimes \mathbf{1} + u_1)^2 = 2\,\mathbf{1} + 2\, e_1 \otimes e_1$.

(v) At rung n the ungraded generator of constituent k, $\mathbf{1}^{\otimes(k-1)} \otimes e_a \otimes \mathbf{1}^{\otimes(n-k)}$, equals $\Gamma^{(1)} \cdots \Gamma^{(k-1)} E_{(k,a)}$, a blade of grade $6(k-1)+1$. Ungraded generators of distinct constituents commute and graded ones anticommute (checked at $n = 3, 4$).

Proof. (i) $(\omega \otimes \mathbf{1})(\omega \otimes e_a) = \omega^2 \otimes e_a = \mathbf{1} \otimes e_a$, and $\Gamma^{(1)} = E_{(1,1)} \cdots E_{(1,6)}$. (ii) Conjugation by an invertible blade multiplies every blade by ± 1 (equation (3) of [1]), so the commutant of a set of blades is spanned by the blades it contains. A blade commutes with all $E_{(1,b)}$ exactly when it is $\Gamma^{(1)\varepsilon}$ times a blade of copy 2 with grade parity ε, that is $\pm \mathbf{1} \otimes e_J$ (Proposition 7.1(iii) of [2]). The uniqueness is the Skolem–Noether theorem in $\mathbf{1} \otimes M_8(\mathbb{R}) \cong M_8(\mathbb{R})$. (iii) $\omega = J\Omega$ by moving e_1 past $e_2 e_3$, and $(-1)^{N_1+N_2+N_3} = -\omega$ is Proposition 7.2 of [2]. (iv) Distinct blades B, B' have $\mathrm{Tr}(B^{\top} B') = 0$ because $B^{\top} B'$ is $\pm$ a blade other than $\mathbf{1}$. The square is a one-line expansion. (v) $\Gamma^{(j)} = \mathbf{1}^{\otimes(j-1)} \otimes \omega \otimes \mathbf{1}^{\otimes(n-j)}$, and the ω's cancel in pairs. □

Parts (i)–(iii) restate Proposition 7.1 of [2] and Appendix A of [1] (fixed there), with the ungraded copy named. The Klein–Jordan–Wigner reading is inherited [4, 5], and so is the exponential law [3, 6]. The identification of the Klein factor with the chirality of the first copy, $u_a = \Gamma^{(1)} E_{(2,a)}$, is fixed (e1-1a to e1-1d). Two remarks organise the rest of the section. First, bivectors do not see the grading: $E_{(2,a)} E_{(2,b)} = \mathbf{1} \otimes e_a e_b$. Hence each constituent's $\mathfrak{l}^{(k)}_{\mathrm{int}}$ and $\mathfrak{l}^{(k)}_{G}$, its charge $J^{(k)} = E_{(k,2)} E_{(k,3)}$ and its number $N^{(k)}$ are the same operators in both readings, and the two Lorentz algebras commute in both. Second, the readings differ only in their odd elements, the gamma matrices and the mass terms. The algebra $M_{64}(\mathbb{R})$ is the same; what changes is which of its elements are taken as the vectors.

1.2 Rest energies: quadrature against addition

Put $H_g = m_1 E_{(1,1)} + m_2 E_{(2,1)}$, the rest Hamiltonian built from the tower's vectors, and $H_u = m_1\, e_1 \otimes \mathbf{1} + m_2\, \mathbf{1} \otimes e_1$, built from the commuting copy. On one constituent the rest flow of the real Dirac form is $\dot{\Psi} = -JH(0)\Psi$ with $JH(0) = mT$, of angular frequency m (Section 4.4 of [1]; Proposition 4.9 of [2]).

Proposition 1.2 (Graded and ungraded rest energies)**.** *(i) $H_g^2 = M^2\, \mathbf{1}$ and $\mathrm{Tr}\, H_g = 0$, so $\mathrm{spec}\, H_g = \{\pm M\}$, each of multiplicity 32.*

(ii) *With* $B = E_{(2,1)}E_{(1,1)}$, $B^2 = -\mathbf{1}$, *and* $U_\phi = \cos\phi + \sin\phi\, B$, $\tan 2\phi = m_2/m_1$: $U_\phi(ME_{(1,1)})U_\phi^{-1} = H_g$. *The graded rest Hamiltonian is one Dirac mass along a unit vector of* $\mathbb{R}^{6,6}$, *turned by a compact element of* $\mathrm{Spin}(6,6)$.

(iii) $\operatorname{spec} H_u = \{\pm m_1 \pm m_2\}$, *each of multiplicity* 16. $H_u = m_1 E_{(1,1)} + m_2\,\Gamma^{(1)}E_{(2,1)}$ *has components of grade* 1 *and* 7. *For* $m_1 \neq m_2$, *both nonzero, it has four distinct eigenvalues, so it is not conjugate to any vector of* $\mathbb{R}^{6,6}$, *which satisfies* $v^2 = \eta_2(v,v)\mathbf{1}$ *and has at most two.*

(iv) *Joint mass shells. No nonzero* ψ *satisfies* $E_{(1,1)}\psi = s_1\psi$ *and* $E_{(2,1)}\psi = s_2\psi$ *with* $s_1, s_2 = \pm 1$. *Each joint sign* (s_1, s_2) *of* $e_1 \otimes \mathbf{1}$ *and* $\mathbf{1} \otimes e_1$ *occurs on* 16 *dimensions. More generally, the graded Dirac operators* $D_k = \sum_\mu p_\mu^{(k)} E_{(k,\mu)}$ *anticommute, with* $(D_1 + D_2)^2 = (p^{(1)} \cdot p^{(1)} + p^{(2)} \cdot p^{(2)})\mathbf{1}$, *while their ungraded counterparts commute.*

(v) *Rest flows.* $J^{(1)}$ *and* $J^{(2)}$ *commute with* H_g *and* H_u. *The ungraded generator* $Z_u = m_1 T^{(1)} + m_2 T^{(2)}$ *is a sum of two commuting complex structures, with eigenvalues* $i(\pm m_1 \pm m_2)$: $e^{-\lambda Z_u}$ *winds a two-torus with angular frequencies* m_1 *and* m_2. *The graded generator* $Z_g = J^{(1)}H_g$ *has* $Z_g^2 = -M^2\mathbf{1}$, *so* $e^{-\lambda Z_g} = \cos(M\lambda) - \sin(M\lambda)Z_g/M$ *has the single frequency* M.

(vi) *At rung* n *the graded* $E_{(k,1)}$ *anticommute and the ungraded ones commute, with traceless products over every nonempty subset. Hence* $\operatorname{spec}(\sum_k m_k E_{(k,1)}) = \{\pm(\sum_k m_k^2)^{1/2}\}$, *while the ungraded sum has spectrum* $\{\sum_k \pm m_k\}$, *each value of multiplicity* 4^n.

Proof. (i) $E_{(1,1)}$ and $E_{(2,1)}$ are anticommuting traceless involutions. (ii) $U_\phi X U_\phi^{-1} = \cos^2\phi\, X + \sin\phi\cos\phi\,[B, X] - \sin^2\phi\, BXB$, with $[B, E_{(1,1)}] = 2E_{(2,1)}$ and $BE_{(1,1)}B = E_{(1,1)}$. (iii) Commuting involutions with traceless products have all four joint signs with equal multiplicity. (iv) If $E_{(1,1)}\psi = s_1\psi$ and $E_{(2,1)}\psi = s_2\psi$, then $s_1 s_2 \psi = E_{(1,1)}E_{(2,1)}\psi = -E_{(2,1)}E_{(1,1)}\psi = -s_1 s_2 \psi$. Generators of distinct constituents anticommute, so the cross terms of $(D_1 + D_2)^2$ cancel. (v) $T^{(k)} = J^{(k)} \times$ (the mass term of copy k), both $T^{(k)}$ square to $-\mathbf{1}$, and they commute. $J^{(1)}$ commutes with $E_{(1,1)}$ and $E_{(2,1)}$ and squares to $-\mathbf{1}$, so $(J^{(1)}H_g)^2 = -H_g^2$. (vi) As (i) and (iii), with 2^n joint signs. □

Parts (i) and (iii) at $m_1 = m_2$ restate Proposition 8.11(iv) of [2]. The general masses, the rotation (ii), the joint mass shells (iv), the flows (v) and the ladder (vi) are fixed (e1-2a to e1-2g). The free-particle facts behind (iv) are inherited. A system of two particles has two mass-shell conditions, one per particle, imposed simultaneously, as in the two-body Dirac (Breit) equation [8]. A single particle in a doubled momentum space has one, $p^{(1)} \cdot p^{(1)} + p^{(2)} \cdot p^{(2)} = M^2$. By (ii) and (iv) the graded tower is of the second kind. Its two copies have no joint mass shell, its rest energy is one value M, and the second copy enters the one particle as further momentum and mass directions. The ungraded operators are of the first kind: the energies add, and the pair threshold is $m_1 + m_2$.

1.3 Complex structures: the two-particle space and the selected unit

The paper reads $J = e_2 e_3$ as the imaginary unit of one constituent (Theorem 4.1 of [1]). A complex two-particle space needs one imaginary unit acting as J on both factors.

Proposition 1.3 (The complex two-particle space)**.** *(i)* $J^{(1)}$ *and* $J^{(2)} = E_{(2,2)}E_{(2,3)}$ *commute and square to* $-\mathbf{1}$, *and* $J^{(1)}J^{(2)} = \omega_S$ *is a symmetric traceless involution. The subspace* $W = \{\psi : J^{(1)}\psi = J^{(2)}\psi\} = \ker(\omega_S + \mathbf{1})$ *has dimension* 32, *and its orthogonal complement is* $\overline{W} = \ker(J^{(1)} + J^{(2)}) = \ker(\omega_S - \mathbf{1}) = \operatorname{im}(J^{(1)} - J^{(2)})$. *With* $J := J^{(1)}|_W = J^{(2)}|_W$ *as imaginary unit,* $W \cong S \otimes_{\mathbb{C}} S$, *and* $\overline{W} \cong S \otimes_{\mathbb{C}} \overline{S}$.

(ii) *A blade of* $\mathrm{Cl}(6,6)$ *with an even number of surplus indices preserves* W *and* $\overline{W}$. *A blade with an odd number exchanges them, for instance* $\mathbf{1} \otimes e_2$ *(charge conjugation* $C = e_2$ *of copy 2 alone, Theorem 5.1 of [1]) and* $E_{(1,1)}E_{(2,2)}$.

(iii) $J^{(1)}$ *and* $J^{(2)}$ *commute with* $\mathfrak{l}^{(1)}_{\mathrm{int}}$, $\mathfrak{l}^{(2)}_{\mathrm{int}}$, *the eight graded and the eight ungraded interior gammas,* H_g, H_u *and the interior copy rotation* $M_{12} = \frac{1}{2}\sum_\mu \eta_{\mu\mu} E_{(1,\mu)}E_{(2,\mu)}$. *All of these are* $\mathbb{C}$*-linear on* W *and on* $\overline{W}$.

(iv) On W *the surplus part of the copy rotation vanishes, since* $E_{(1,3)}E_{(2,3)} = E_{(1,2)}E_{(2,2)}\omega_S$, *so the copy rotation* $C_b = \frac{1}{2}\sum_a \eta_{aa} E_{(1,a)}E_{(2,a)}$ *of Proposition 7.3(ii) of [2] acts on* W *as* M_{12}, $\mathbb{C}$*-linearly. The four cross surplus bivectors* $E_{(1,s)}E_{(2,t)}$ *(*$s,t \in \{2,3\}$*) preserve* W *but anticommute with* $J^{(1)}$ *and* $J^{(2)}$, *so they act on* W *antilinearly.*

Proof. (i) Bivectors of distinct constituents commute; $(J^{(1)}J^{(2)})^2 = (J^{(1)})^2(J^{(2)})^2 = \mathbf{1}$ and $\mathrm{Tr}(J \otimes J) = (\mathrm{Tr}\, J)^2 = 0$. $J^{(1)}\psi = J^{(2)}\psi$ is equivalent to $J^{(1)}J^{(2)}\psi = -\psi$. $J^{(1)} - J^{(2)}$ is antisymmetric, so its kernel is the orthogonal complement of its image, and $\omega_S(J^{(1)} - J^{(2)}) = J^{(1)} - J^{(2)}$. The canonical map $S \otimes_{\mathbb{R}} S \to S \otimes_{\mathbb{C}} S$ has kernel $\mathrm{im}(J^{(1)} - J^{(2)})$, spanned by $Jx \otimes y - x \otimes Jy$, so it restricts to an isomorphism on W; the case of $\overline{W}$ is the same with $-J$ on the second factor. (ii) By equation (3) of [1] a blade commutes with the surplus four-blade ω_S iff it shares an even number of indices with it, and otherwise anticommutes. (iii) The blade rule again: none of these contains a surplus index, except $J^{(k)}$ itself and M_{12}, whose terms share no index with $J^{(1)}$ or $J^{(2)}$. (iv) $E_{(1,2)}E_{(2,2)}\omega_S = -E_{(2,2)}E_{(1,3)}E_{(2,2)}E_{(2,3)} = E_{(1,3)}E_{(2,3)}$, by the anticommutation of distinct generators. On W, $\omega_S = -\mathbf{1}$, so the two surplus terms of C_b cancel. A cross surplus bivector shares one index with each $J^{(k)}$. □

The tensor-product linear algebra of (i) is inherited. Parts (ii)–(iv) are fixed (e1-3a, e1-3b, e1-3f). With the external i of Proposition 8.11(i) of [2], $J^{(1)} + J^{(2)}$ is $2J$ on W and 0 on $\overline{W}$. So W carries total charge ± 2, two particles of equal charge, and $\overline{W}$ the neutral pairs, among them the particle–antiparticle state $\psi \otimes C\phi$ of that proposition (restated; the charge reading interpretive). The question is now whether anything in the algebra selects W and the complex structure on it.

Proposition 1.4 (Which imaginary unit the second rung selects)**.** *(i) Per-particle selection. The commutant of the eight ungraded gammas* $\gamma^\mu \otimes \mathbf{1}$, $\mathbf{1} \otimes \gamma^\mu$ *is* $\mathbb{H} \otimes \mathbb{H} \cong M_4(\mathbb{R})$, *spanned by* 16 *blades. Adding the two temporal volume elements* $T^{(1)}, T^{(2)}$ *(the datum of Remark 4.2 of [1] for each copy) cuts it to* $\mathrm{span}\{\mathbf{1}, J^{(1)}, J^{(2)}, \omega_S\} \cong \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C}$. *Its solutions of* $x^2 = -\mathbf{1}$ *are exactly* $\pm J^{(1)}$ *and* $\pm J^{(2)}$, *four isolated points, none of which acts as* J *on both factors of* S_2.

(ii) The spectator action, $k_I = 1$. *The selected unit of Theorem 4.1 of [1] at* $n = 2$, *the product of the eight internal generators, is* $J_{\mathrm{sel}} = J \otimes \omega = J^{(1)}\Gamma^{(2)}$ *(Proposition 7.11(iv) of [2]). It commutes with* $J^{(1)}$ *and* $J^{(2)}$ *and so preserves* W. *It equals* $J^{(1)}$ *on* $\{\Gamma^{(2)} = 1\}$ *and* $-J^{(1)}$ *on* $\{\Gamma^{(2)} = -1\}$, *and* $J^{(2)}$ *on* $\{\Gamma^{(2)}\omega_S = -1\}$ *and* $-J^{(2)}$ *on* $\{\Gamma^{(2)}\omega_S = 1\}$, 32 *dimensions each; on* W *it is* J *on* 16 *and* $-J$ *on* 16 *dimensions. It anticommutes with every generator of copy 2, graded or ungraded. So no gamma matrix of copy 2 is* J_{sel}*-linear, and neither* H_g *nor* H_u *is for* $m_2 \neq 0$; $m_1 E_{(1,1)}$ *is.*

(iii) The diagonal action, $k_I = 2$. *The commutant of the eight graded interior gammas* $E_{(k,\mu)}$ *is spanned by* 16 *blades (*$M_2(\mathbb{H})$*). Adding the six surplus bivectors cuts it to* $\mathrm{span}\{\mathbf{1}, \omega_S\} \cong \mathbb{R} \oplus \mathbb{R}$, *with* $\omega_S^2 = +\mathbf{1}$. *No complex structure commutes with the gammas of both copies and the surplus bivectors. On* W *alone the restrictions of the same operators have commutant* $\mathbb{R}$.

(iv) Up the ladder, under the diagonal action of rung n *the same commutant is* $\mathrm{span}\{\mathbf{1}, \omega_S\}$ *with* $\omega_S = J^{(1)} \cdots J^{(n)}$ *and* $\omega_S^2 = (-1)^n$ *(checked at* $n = 2, 3$ *for the commutant and* $n = 2, 3, 4$ *for the square). An imaginary unit is selected exactly at odd* n. *The common subspace* $J^{(1)} = \cdots = J^{(n)}$, *the complex* n*-fold product* $S \otimes_{\mathbb{C}} \cdots \otimes_{\mathbb{C}} S$, *has dimension* $2 \cdot 4^n$.

Proof. (i) By the blade argument of Proposition 1.1(ii), since $\mathbf{1} \otimes \gamma^\mu = \Gamma^{(1)} E_{(2,\mu)}$ and $T^{(2)} = \Gamma^{(1)} E_{(2,1)} E_{(2,2)} E_{(2,3)}$ are blades. Equivalently, the commutant of the gammas on S is the quaternion algebra of Remark 4.2 of [1], and that of $\{\gamma^\mu, T\}$ is $\mathbb{R} \oplus \mathbb{R} J$. For $x = a + bJ^{(1)} + cJ^{(2)} + d\,\omega_S$, $x^2 = (a^2 - b^2 - c^2 + d^2) + 2(ab - cd)J^{(1)} + 2(ac - bd)J^{(2)} + 2(ad + bc)\omega_S$, and $x^2 = -\mathbf{1}$ has exactly the four solutions listed. (ii) J_{sel} is a blade of grade 8 containing every copy-2 index, so it anticommutes with each $E_{(2,a)}$; it commutes with $\Gamma^{(1)}$, so it also anticommutes with each $u_a = \Gamma^{(1)} E_{(2,a)}$. $\Gamma^{(2)}$ commutes with $J^{(1)}$ and $J^{(2)}$, and $\operatorname{Tr} \Gamma^{(2)} = 0$, also on W. Finally $J_{\mathrm{sel}} \mp J^{(2)} = J^{(1)}(\Gamma^{(2)} \pm \omega_S)$ with $J^{(1)}$ invertible, and $\Gamma^{(2)}\omega_S$ is a traceless involution. (iii) A blade commuting with all eight interior generators meets them in all or none of them. Meeting all of them requires odd grade, and meeting none requires even grade. Commuting with every surplus bivector forces it to meet the four surplus directions in all or none. Both conditions together leave $\mathbf{1}$ and ω_S. The algebra generated contains the central idempotent $\frac{1}{2}(\mathbf{1} - \omega_S)$, so its commutant on W is the compression of $\operatorname{span}\{\mathbf{1}, \omega_S\}$, which is $\mathbb{R}$. (iv) The same parity count with $4n$ interior and $2n$ surplus generators: a blade containing all $4n$ interior generators has grade $4n$ or $6n$, both even, so it fails. The square follows from equation (4) of [1], and the dimension is $8^n/2^{n-1}$, because the $n-1$ involutions $J^{(k)}J^{(k+1)}$ commute and have traceless products. □

Part (i) is fixed, with the quaternion commutant of one copy restated from Remark 4.2 of [1]. Part (ii) restates Theorem 4.1 of [1] and Proposition 7.11(iv) of [2] (fixed there), and the comparison with W is fixed. Part (iii) is obstructed under the stated hypotheses (e1-3c to e1-3e, e1-3g). Part (iv) restates the selection at odd n of Proposition 10.10(ii) of [1] (fixed there) and adds that nothing is selected at even n (fixed; e1-3h).

Corollary 1.5 (No selected two-particle complex structure)**.** *At the second rung none of the selection principles of [1] gives a complex structure under which both copies are complex Dirac spinors with $\mathbb{C}$-linear gamma matrices. Per copy they give $\pm J^{(1)}$ and $\pm J^{(2)}$ separately. The spectator action gives J_{sel}, under which copy 2 carries no linear Dirac operator, so that copy 2 is internal. The diagonal action gives none. What the diagonal action does select is the splitting $S_2 = W \oplus \overline{W}$, the eigenspaces of the one nontrivial element ω_S of its commutant. The complex structure $J|_W$ that makes W the two-particle space commutes with every gamma matrix, with both Hamiltonians and with the copy rotation, but not with the surplus $\mathfrak{so}(4)$. It is an added datum.*

The corollary is fixed as a statement about these selection principles (Proposition 1.4). That the choice of W, rather than $\overline{W}$, and of $J|_W$ is what turns the second rung into a two-particle space is interpretive. This also bears on the phrase "two particles from one". The paper's own second quantisation of one constituent (Theorem 5.7 of [1], Proposition 3.9 of [2]) already has two-excitation states of one Dirac mode, a level of dimension 6 with charges $q = \pm 2$ once each and $q = 0$ four times. In ordinary physics that is where several particles live (inherited [9]). It does not need the second rung.

1.4 The Fock reading: quanta, not particles

The tower has its own number operator. S_n is the Fock space of $3n$ fermionic modes, and N counts them (Proposition 7.2 of [2]). We ask which elements of $\mathfrak{so}(3n, 3n)$, the bivectors, conserve N.

Proposition 1.6 (Bivectors and the number of quanta)**.** *(i) $N = \frac{3n}{2} - \frac{1}{2}\sum_{k,i} E_{(k,i)}E_{(k,i+3)}$ is diagonal, with spectrum $0, \ldots, 3n$; at $n = 2$ the multiplicities are $1, 6, 15, 20, 15, 6, 1$. $N = N^{(1)} + N^{(2)}$, the same operator in the graded and ungraded readings. $(-1)^N = \Gamma^{(1)}\Gamma^{(2)}$ is the volume element of $\mathrm{Cl}(6, 6)$.*

(ii) On vectors $[N, v] = K_n v$*. A bivector* X *commutes with* N *iff* $\operatorname{ad} X$ *commutes with* K_n*. These form* $\mathfrak{gl}(L_+) \cong \mathfrak{gl}(3n, \mathbb{R})$*, of dimension* $9n^2$ *(*36 *at* $n = 2$*), the stabiliser of the transverse maximal isotropic subspaces* $L_\pm = \ker(K_n \mp \mathbf{1})$*. The remaining* $3n(3n-1)$ *directions (*30 *at* $n = 2$*) are* $\Lambda^2 L_+ \oplus \Lambda^2 L_-$*, on which* $[N, X] = \pm 2X$*: they create or annihilate a pair of quanta.*

(iii) $(-1)^N$ *commutes with every bivector and anticommutes with every generator. No element of* $\mathfrak{so}(3n, 3n)$ *changes* N *by an odd number.*

Proof. (i) $N_i = \frac{1}{2}(\mathbf{1} - e_i e_{i+3})$, and the pole operators are diagonal (Appendix A of [1]). (ii) $[N, a^\pm_{k,i}] = \pm a^\pm_{k,i}$, and $a^\pm_{k,i}$ is proportional to $e_i \pm e_{i+3}$ in constituent k, the ± 1 eigenvectors of K. $N - \frac{3n}{2}$ is a bivector, so $[N, X]$ is one, and the spin representation of $\mathfrak{so}(V_n)$ is faithful. Hence $[N, X] = 0$ iff $[\operatorname{ad} N, \operatorname{ad} X] = [K_n, \operatorname{ad} X] = 0$. Under $\operatorname{ad} K_n$, $\mathfrak{so}(V_n) \cong \Lambda^2 V_n = \Lambda^2 L_+ \oplus (L_+ \otimes L_-) \oplus \Lambda^2 L_-$ has eigenvalues $2, 0, -2$, and $L_\pm$ are isotropic because $\eta(e_i \pm e_{i+3}, e_j \pm e_{j+3}) = \delta_{ij} - \delta_{ij}$. (iii) Grade parity. □

Part (i) restates Proposition 7.2 of [2]. The rest is the standard structure of the spin representation on the exterior algebra of a maximal isotropic subspace [3], inherited, with its realisation in the tower fixed (e1-4a, e1-4b, e1-4g). The question is which of these directions are physical.

Theorem 1.7 (Compact fields conserve the quanta)**.** *Let* $X \in \operatorname{Cent}(G_n) \cap \mathfrak{so}(3n, 3n)$*, the algebra* $\mathfrak{so}(3n, \mathbb{C})$ *of binding-compatible bivectors (Theorem 17.19 of [1]; Proposition 7.8 of [2]). Then* $K_n X K_n = -X^\top$*, the Cartan involution. Consequently:*

(i) X *commutes with* N *iff* X *is compact (antisymmetric), that is iff it is a paired rotation* $\left(\begin{smallmatrix} R & 0 \\ 0 & R \end{smallmatrix}\right)$*. These form* $\mathfrak{so}(3n)$*, of dimension* $\frac{3}{2}n(3n-1)$*.*

(ii) The non-compact part, the paired boosts $\left(\begin{smallmatrix} 0 & S \\ -S & 0 \end{smallmatrix}\right)$*, anticommutes with* K_n*. A nonzero paired boost has nonzero components in both* $\Lambda^2 L_+$ *and* $\Lambda^2 L_-$*, so it changes* N *by* $+2$ *and by* -2 *and conserves no part of it.*

(iii) By Corollary 2.4 of [1] no gauge boson is a boost. Hence no gauge-admissible, binding-compatible field creates or annihilates a pair of quanta, at any rung.

Proof. On vectors $\eta G_n = -K_n$ and $G_n \eta = -\eta G_n$. For X commuting with G_n, $K_n X K_n = \eta G_n X \eta G_n = -\eta G_n X G_n \eta = -\eta X G_n^2 \eta = \eta X \eta$, and $\eta X \eta = -X^\top$ for $X \in \mathfrak{so}(\eta)$. (i) follows from Proposition 1.6(ii). The block forms are those of Proposition 7.8(i) of [2]. (ii) X maps $L_\pm$ to $L_\mp$, so its components lie in $\Lambda^2 L_+ \oplus \Lambda^2 L_-$. G_n commutes with X, maps L_+ onto L_- (because $G_n K_n = -K_n G_n$) and normalises $\mathfrak{so}(\eta)$, so conjugation by G_n carries the $\Lambda^2 L_+$ component of X to its $\Lambda^2 L_-$ component. They vanish together. (iii) Gauge admissibility is compactness, the criterion of Section 2.2 of [1]. □

The theorem is fixed. It was checked exactly on vectors at $n = 1, \ldots, 4$, where $\dim \operatorname{Cent}(K_n) \cap \mathfrak{so}(3n, 3n) = 9, 36, 81, 144$, $\dim \operatorname{Cent}(G_n) \cap \mathfrak{so}(3n, 3n) = 6, 30, 72, 132$ and the compact part has half of it. It was checked entry by entry on S_n at $n = 1, \ldots, 4$ for every paired rotation $e_i e_j - e_{i+3} e_{j+3}$ and every paired boost $e_i e_{j+3} - e_j e_{i+3}$, within one constituent and across two (e1-4c, e1-4d). At $n = 4$, S_4 has dimension 4096, and the generators were handled as signed permutations.

Proposition 1.8 (The quanta and the Lorentz algebras)**.** *The number* N *behaves as follows under the copy algebra and under the two Lorentz algebras of one constituent.*

(i) The copy rotation C_b *commutes with the total number* N*, but it does not commute with the number* $N^{(1)}$ *of the first constituent. The quarter turn*

$$U = e^{\pi C_b/2} = \tfrac{1}{8} \textstyle\prod_a (\mathbf{1} + \eta_{aa} E_{(1,a)} E_{(2,a)})$$

satisfies $UN^{(1)}U^{-1} = N^{(2)}$. The spin lift of the copy boost $\varepsilon \otimes G$ changes N by ± 2 only, with both signs.

(ii) In $\mathfrak{l}_{\text{int}}$ the elements commuting with N form $\mathbb{R}\, e_1e_4$, a boost (the pole operator $\mathbf{1} - 2N_1$). The rotations e_4e_5, e_4e_6, e_5e_6, the boosts e_1e_5, e_1e_6 and the charge $J = e_2e_3$ do not commute with N.

(iii) In $\mathfrak{l}_G$ the elements commuting with N form its compact $\mathfrak{so}(3)$, the paired rotations. No nonzero boost of $\mathfrak{l}_G$ commutes with N.

Proof. (i) C_b is a paired rotation (Theorem 1.7(i)). Its six terms commute and square to $-\frac{1}{4}$, so $e^{\theta C_b} = \prod_a (\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\,\eta_{aa}E_{(1,a)}E_{(2,a)})$, which at $\theta = \frac{\pi}{2}$ is U; it carries $E_{(1,a)}$ to $\pm E_{(2,a)}$ (Proposition 7.3(ii) of [2]). The copy boost is a paired boost (Proposition 7.8(iii) of [2]); apply Theorem 1.7(ii). (ii) $\mathfrak{l}_{\text{int}} \cap \mathfrak{gl}(3,\mathbb{R})$ is computed: of the six planes in $\langle e_1, e_4, e_5, e_6\rangle$ only $\langle e_1, e_4\rangle$ is K-invariant, and J involves the modes $a_2^+a_3^+$. (iii) is Theorem 1.7 at $n = 1$, with $\mathfrak{l}_G = \mathrm{Cent}(G) \cap \mathfrak{so}(3,3)$ (Theorem 12.1 of [1]). □

All three parts are fixed (e1-4e, e1-4f), and (i) gives a precise sense to the copy rotation as a hop of quanta between the constituents at fixed total number. Part (ii) corrects an easy expectation. The Fock number is not invariant under the spatial rotations of the interior four-plane, only under one of its boosts, and it does not commute with the charge J. The quanta counted by N are therefore not Lorentz-invariant particles. They are invariant under the compact part of $\mathfrak{l}_G$, the rotations of the field-strength reading, and under nothing that contains a boost or the electric charge. Reading N as a particle number is an internal reading tied to the split structure K (interpretive). What Theorem 1.7 does establish is the following. If quanta are counted by N, the only processes that create them in pairs from the binding-compatible algebra are the paired boosts. These are excluded as gauge fields, so pair creation of quanta would need a non-compact, non-gauge interaction (fixed, with the identification of quanta interpretive). This agrees with the statement of Section 8.5 of [2] that duplication in the tower's sense is not pair creation.

1.5 Exchange and statistics

Two swaps act on S_2: the ungraded swap $P(x \otimes y) = y \otimes x$, and the Koszul swap $\tau_K = D\,P$ of Proposition 7.3(iii) of [2], where $D = (-1)^{\mathsf{p}\otimes\mathsf{p}}$ and $\omega = (-1)^{\mathsf{p}}$ on the basis of equation (1) of [1].

Proposition 1.9 (Two exchanges)**.** *(i) $P(e_a \otimes \mathbf{1})P = u_a = \Gamma^{(1)}E_{(2,a)}$. So P exchanges the ungraded copies and maps the vectors $E_{(1,a)}$ of $\mathrm{Cl}(6,6)$ to elements of grade 7. It commutes with the diagonal Lorentz generators $\Delta_{\mu\nu}$, with N, with ω_S (so it preserves W) and with H_u at $m_1 = m_2$, but not with H_g.*

(ii) $\tau_K E_{(1,a)}\tau_K = E_{(2,a)}$. τ_K commutes with $\Delta_{\mu\nu}$, N, ω_S and H_g at $m_1 = m_2$, but not with H_u.

(iii) $D = \frac{1}{2}(\mathbf{1} + \Gamma^{(1)} + \Gamma^{(2)} - \Gamma^{(1)}\Gamma^{(2)})$. So $\tau_K = P$ except on the sector $\Gamma^{(1)} = \Gamma^{(2)} = -1$, where $\tau_K = -P$.

Proof. (i) $P(A \otimes B)P = B \otimes A$. Diagonal bivectors, $N = N^{(1)} + N^{(2)}$, $\omega_S = J \otimes J$ and H_u at equal masses are symmetric under it, while $P(\omega \otimes e_1)P = e_1 \otimes \omega$. (ii) The first identity restates Proposition 7.3(iii) of [2]. Hence $\tau_K\Gamma^{(1)}\tau_K = \Gamma^{(2)}$ and $\tau_K u_1 \tau_K = e_1 \otimes \omega \neq u_1$, while bivectors $E_{(1,a)}E_{(1,b)}$ go to $E_{(2,a)}E_{(2,b)}$. (iii) D is diagonal with entry -1 exactly where both ω-eigenvalues are -1. □

Each swap is a symmetry of its own reading and not of the other: P of the additive Hamiltonian, τ_K of the graded one (fixed; e1-5a, e1-5b). For the Lorentz content let $C = -\sum_{\mu<\nu}\eta^{\mu\mu}\eta^{\nu\nu}\Delta_{\mu\nu}^2$ be the diagonal Casimir, normalised as in Section 10.2 of [1]. It is $\frac{3}{2}$ on

sector	on $W = S \otimes_{\mathbb{C}} S$ (32)	on $\overline{W} = S \otimes_{\mathbb{C}} \overline{S}$ (32)	on S_2 (64)
$P = -1$	$0^4\ 3^8$	$0^2\ 3^8\ 4^6$	$0^6\ 3^{16}\ 4^6$
$P = +1$	$3^8\ 4^{12}$	$0^2\ 3^8\ 4^6$	$0^2\ 3^{16}\ 4^{18}$
$\tau_K = -1$	$0^2\ 3^8\ 4^6$	$0^2\ 3^8\ 4^6$	$0^4\ 3^{16}\ 4^{12}$
$\tau_K = +1$	$0^2\ 3^8\ 4^6$	$0^2\ 3^8\ 4^6$	$0^4\ 3^{16}\ 4^{12}$

Table 1: Casimir content c^d ($C = c$ on d real dimensions) of the exchange sectors under the diagonal interior Lorentz algebra. Checks e1-5d to e1-5f.

one constituent, it satisfies $C(C-3)(C-4) = 0$ on S_2, and it takes the values 0, 3 and 4 on $(0,0)$, $(\frac{1}{2},\frac{1}{2})$ and $(1,0) \oplus (0,1)$. On one constituent each boost is $J\Gamma$ times its dual rotation, $e_1e_4 = J\Gamma\, e_5e_6$ and cyclically, so $\Gamma = \pm 1$ are the two Weyl halves (e1-5c).

Proposition 1.10 (Statistics sectors)**.** *The dimensions of the eigenspaces of C in the sectors of P and τ_K are those of Table 1. In particular:*

(i) On W, P is $\mathbb{C}$-linear. Its -1 eigenspace, $\Lambda^2_{\mathbb{C}}S$, carries $2(0,0)\oplus(\frac{1}{2},\frac{1}{2})$, and its $+1$ eigenspace, $\mathrm{Sym}^2_{\mathbb{C}}S$, carries $(1,0)\oplus(0,1)\oplus(\frac{1}{2},\frac{1}{2})$. In the chirality sectors, $\Gamma^{(1)} = \Gamma^{(2)} = \pm 1$ gives spin one in $P = +1$ and spin zero in $P = -1$, while the mixed sector (16 dimensions, $C = 3$) splits $8 + 8$.

(ii) On W, τ_K is $\mathbb{C}$-linear, and its two eigenspaces have the same Casimir content, $C = 0, 3, 4$ on $2, 8, 6$ dimensions. The spin-one part of $\tau_K = +1$ lies in $\Gamma^{(1)} = \Gamma^{(2)} = +1$, and that of $\tau_K = -1$ in $\Gamma^{(1)} = \Gamma^{(2)} = -1$: the two sectors are mirror images.

(iii) On $\overline{W}$, P is antilinear, $PJ^{(1)} = -J^{(1)}P$, and $J^{(1)}$ exchanges its two eigenspaces. They carry the same content.

Proof. All entries are traces of products of commuting integer matrices: the projectors $\frac{1}{2}(\mathbf{1} \pm X)$ for $X \in \{\omega_S, P, \tau_K, \Gamma^{(1)}, \Gamma^{(2)}\}$ and the spectral projectors of C (e1-5d to e1-5f). (i) $PJ^{(1)}P = J^{(2)}$, which equals $J^{(1)}$ on W. (ii) Use Proposition 1.9(iii). (iii) $J^{(2)} = -J^{(1)}$ on $\overline{W}$, and $J^{(1)}$ commutes with C. □

The decompositions of Λ^2 and Sym^2 of a Dirac spinor are inherited [7]; their appearance on W, and only on W, is fixed. The Koszul column restates Proposition 7.10(iii) of [2] on S_2 (fixed there) and refines it on W. The two τ_K sectors are isomorphic as real representations, since the realification of $(1,0)$ is that of $(0,1)$, but they are mirror images once J is the imaginary unit (fixed). So the ungraded swap sorts the complex two-particle space by statistics in the textbook way. Identical fermions, antisymmetric under P, have only scalar and vector content at a single momentum mode (inherited [10]). The Clifford exchange τ_K does not sort by Lorentz content at all. Its eigenvalue is a chirality-dependent sign, not a statistics label. Fermi statistics for two copies is therefore a constructed requirement: antisymmetry under P on W. The algebra supplies τ_K, the exchange that preserves the vectors of $\mathrm{Cl}(6,6)$, and τ_K is blind to it.

1.6 Answer: what makes the second rung carry two particles

Table 2 collects the structures. Read together they answer the question in four sentences. (1) As given by the tower, the second rung carries one particle. Its vectors $E_{(k,a)}$ give one Dirac clock along a unit vector of $\mathbb{R}^{6,6}$, of mass $M = (m_1^2 + m_2^2)^{1/2}$, with no joint mass shell of the copies. Under the selected unit J_{sel} copy 2 carries no linear Dirac operator and is internal, so the rung is one particle with a larger internal space (fixed; Propositions 1.2 and 1.4). (2) The algebra also contains, uniquely up to conjugation, a second commuting copy, the generators dressed by the chirality of the first. With it the copies have separate mass shells and additive energies (fixed;

structure	what it does at $n = 2$	class
graded vectors $E_{(2,a)}$	one Dirac clock of mass M; no joint mass shell; τ_K as exchange	fixed (they define $\mathrm{Cl}(6,6)$)
commuting copy $u_a = \Gamma^{(1)} E_{(2,a)}$	two mass shells, energies $\pm m_1 \pm m_2$	fixed as an algebra (unique up to $\mathbf{1} \otimes GL(8,\mathbb{R})$); use as vectors constructed
common complex structure $J\|_W$	$W = S \otimes_{\mathbb{C}} S$, 32 dimensions, total charge ± 2	$W \oplus \overline{W}$ fixed by ω_S; $J\|_W$ constructed; selection obstructed ($k_I = 2$), copy 2 internal ($k_I = 1$)
additive Hamiltonian H_u	threshold $m_1 + m_2$; two-torus rest flow	constructed; not a vector (grades 1 and 7)
ungraded exchange P	$\Lambda^2_{\mathbb{C}} S = 2(0,0) \oplus (\frac{1}{2}, \frac{1}{2})$ on W	constructed; τ_K fixed but statistics-blind
Fock number N	counts quanta; conserved by paired rotations, changed by ± 2 by paired boosts	fixed; as a particle number interpretive (not Lorentz- or J-invariant)

Table 2: Structures that bear on a two-particle reading of the second rung.

Propositions 1.1 and 1.2). (3) Three further structures make that a two-particle system in the sense of ordinary quantum mechanics: the subspace W with the complex structure $J|_W$, the additive Hamiltonian H_u, and the ungraded exchange P with antisymmetry. None of them is selected. W is singled out only together with $\overline{W}$, and the diagonal action selects no imaginary unit at even rungs (constructed; obstructed as a selection; Corollary 1.5 and Proposition 1.10). (4) The tower's own counting operator N counts quanta of the split structure, not Lorentz-invariant particles. No gauge-admissible, binding-compatible field changes it (fixed; Theorem 1.7 and Proposition 1.8).

Which process takes one to two. Three candidates have been tested. The step from rung 1 to rung 2 is a change of arena, not a dynamical process. In the graded reading it enlarges the particle and does not add one. In the Fock reading the Lie algebra $\mathfrak{so}(3n, 3n)$ changes N only by 0 or ± 2, and among binding-compatible fields only the paired boosts do it, which are not gauge fields. A step from one quantum to two would need an odd element, which is a field operator and not a symmetry (fixed; Proposition 1.6 and Theorem 1.7). In the two-particle reading the step is the tensor embedding $\psi \mapsto \psi \otimes \phi_0$, which intertwines $A \mapsto A \otimes \mathbf{1}$ but needs a state ϕ_0 of the second particle that nothing selects (constructed). In ordinary physics the process is creation from a field, and the paper's second quantisation of one constituent already contains it (Theorem 5.7 of [1]). Whether some mechanism of the arena selects W and $J|_W$ dynamically is open.

Up the ladder. What has been proved at every rung is this. The commuting copies exist, dressed by $\Gamma^{(1)} \cdots \Gamma^{(k-1)}$. The graded masses add in quadrature and the ungraded ones additively. The rule of Theorem 1.7 holds. The common complex subspace $J^{(1)} = \cdots = J^{(n)}$ has dimension $2 \cdot 4^n$. The diagonal action selects an imaginary unit, $\omega_S = J^{(1)} \cdots J^{(n)}$, exactly at odd n. On the common subspace it acts as $(-1)^{(n-1)/2} J$, so at odd rungs the selected unit and the common complex structure agree up to sign there, while at even rungs there is no selected unit to compare (fixed; e1-1d, e1-2f, e1-3h, e1-4c, e1-4d). The physics of the odd rungs, whose diagonal sector is the vector-like matter of Section 10 of [1], is not examined here.

statement	class	checks
Proposition 1.1: commuting copy, commutant $\mathbf{1} \otimes M_8(\mathbb{R})$, grade-7 generators, chirality dressing, rung n	restated (RE Prop. 7.1, ES17 App. A); dressing fixed; exponential law inherited	e1-1a to e1-1d
Proposition 1.2: quadrature against addition, rotated clock, joint mass shells, rest flows, rung n	fixed; restated at $m_1 = m_2$ (RE Prop. 8.11(iv)); two-body kinematics inherited	e1-2a to e1-2g
Proposition 1.3: $W = S \otimes_{\mathbb{C}} S$, ω_S, what preserves W, $C_b\|_W = M_{12}\|_W$	fixed; tensor algebra inherited	e1-3a, e1-3b, e1-3f
Proposition 1.4: per-particle units $\pm J^{(k)}$; $J_{\mathrm{sel}} = J^{(1)}\Gamma^{(2)}$; no unit at $k_I = 2$; parity of n	fixed; restated (ES17 Thm 4.1, Prop. 10.10; RE Prop. 7.11); obstructed	e1-3c to e1-3e, e1-3g, e1-3h
Corollary 1.5: no selected two-particle complex structure	fixed (as stated); reading interpretive	—
Proposition 1.6: $\mathrm{ad}\, N = K_n$, $\mathfrak{gl}(3n, \mathbb{R})$, pairs, parity	restated (RE Prop. 7.2); inherited; realisation fixed	e1-4a, e1-4b, e1-4g
Theorem 1.7: compact $\Leftrightarrow$ conserves N; boosts change it by ± 2; no gauge field creates pairs	fixed	e1-4c, e1-4d
Proposition 1.8: copy rotation hops quanta; N against $\mathfrak{l}_{\mathrm{int}}$, J, $\mathfrak{l}_G$	fixed; particle reading interpretive	e1-4e, e1-4f
Proposition 1.9: P against τ_K	fixed; τ_K restated (RE Prop. 7.3)	e1-5a, e1-5b
Proposition 1.10, Table 1: statistics sectors	fixed; Dirac⊗Dirac inherited; τ_K on S_2 restated (RE Prop. 7.10)	e1-5c to e1-5f
Section 1.6: which process; up the ladder; dynamical selection of W	constructed; fixed; open	see text

Table 3: Claims of Section 1 and their classes. The checks are the `PASS` lines printed by the script `verify/e1_two_from_one.py`.

2 How one becomes two: processes

Section 1 asked which structures make a rung carry several particles. This section asks which processes take one to two. The word "particle" has four precise senses in the arena, and each has its own process. A particle can be a spin module of a Lorentz algebra, and then the process is a vertex, an equivariant map. It can be a mass shell, and then the process is a decay above a threshold. It can be a field, as in the field-strength reading of [2], and then the process is a duplication of fields. And it can be a quantum of the Fock structure of the tower, and then the process is a change of the number N. The answers, in order, are these. (1) Under the common interior Lorentz action no equivariant map takes a spin-$\frac{1}{2}$ module into the space of two constituents, in either direction. Maps from a fermion to a fermion and a boson exist, and Clifford multiplication provides them (the selection rule inherited, the counts in the realisation fixed; Section 2.1). (2) The graded composite of two masses lies below the threshold of the free pair at every momentum, and the deficit is proportional to the product of the masses (fixed; its reading as a binding energy interpretive; Section 2.2). (3) One null field never becomes a massive pair covariantly, and two non-collinear null fields always form one (restated from [2]; Section 2.3). (4) Boosts create pairs of Fock quanta, with the amplitudes of a fermionic Bogoliubov transformation, and so do the compact generators that do not commute with the binding root, among them the spatial rotations of the interior four-plane and the charge J. The compact generators that commute with the binding root never do (fixed; Section 2.4). Under the common binding action a Lorentz-invariant number of quanta exists exactly at even rungs, and its quanta are shared equally by two constituents (fixed; Section 2.5). (5) The copy rotation moves one quantum from one constituent to another and never doubles it; at unequal rates it does so by Rabi mixing (fixed; Section 2.6). Section 2.7 collects the processes with their classes. Identifiers e2-n refer to the `PASS` lines of `verify/e2_processes.py` (Table 5).

Notation. As in Section 1: $S = \mathbb{R}^8$ with the realisation of equation (1) of [1], the tower generators $E_{(k,a)} = \omega^{\otimes(k-1)} \otimes e_a \otimes \mathbf{1}^{\otimes(n-k)}$ on $S_n = S^{\otimes n}$ (equation (2) of [1]), $V = \mathbb{R}^{3,3}$, $V_n = \mathbb{R}^n \otimes V$, $\eta_n = \mathbf{1}_n \otimes \eta$, $G_n = \mathbf{1}_n \otimes G$ and $K_n = \mathbf{1}_n \otimes K$ with $Ke_i = e_{i+3}$, $Ke_{i+3} = e_i$. The spin lift of $A \in \mathfrak{so}(\eta_n)$ is $\sigma(A) = \frac{1}{4}\sum_{a,b} A_{ab}\eta_{bb}E_aE_b$, so that $[\sigma(A), E_c] = \sum_a A_{ac}E_a$. The interior Lorentz algebra $\mathfrak{l}_{\text{int}}$ has the rotations $S_1 = \frac{1}{2}e_5e_6$, $S_2 = \frac{1}{2}e_6e_4$, $S_3 = \frac{1}{2}e_4e_5$ and the boosts $T_a = \frac{1}{2}e_1e_{a+3}$; it acts on the interior four-plane $E = \langle e_1, e_4, e_5, e_6\rangle$ with $\gamma^\mu = (e_1, e_4, e_5, e_6)$, and trivially on the surplus pair $\langle e_2, e_3\rangle$ (Section 4 of [1]). The field-strength algebra $\mathfrak{l}_G$, the commutant of G in $\mathfrak{so}(3,3)$ (Theorem 12.1 of [1]), has the boosts $K_i = \frac{1}{2}(e_ke_{j+3} - e_je_{k+3})$ and the rotations $L_k = -[K_i, K_j]$, for (i,j,k) cyclic. $J = e_2e_3$. The Fock modes are $a^{\pm}_{k,i} = \frac{1}{2}(E_{(k,i)} \pm E_{(k,i+3)})$, $N^{(k)} = \sum_i a^+_{k,i}a^-_{k,i}$ and $N = \sum_k N^{(k)}$ (Proposition 7.2 of [2]); $a^-_{k,i}$ is the transpose of $a^+_{k,i}$. On S the basis vectors of the realisation are the Fock states, and $|n_1n_2n_3\rangle$ denotes the one with $N_i = n_i$; the Fock vacuum $|000\rangle$ is the first basis vector, and $|0\rangle$ is the first basis vector of S_n (e2-1). $\varepsilon = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ acts on the copy index. The copy rotation on vectors is $\varepsilon \otimes \mathbf{1}_6$ and the copy boost is $\varepsilon \otimes G$ (Section 7 of [2]). We write

$$\hat{C} := \sigma(\varepsilon \otimes \mathbf{1}_6) = -\tfrac{1}{2}\textstyle\sum_a \eta_{aa}E_{(1,a)}E_{(2,a)},$$

the negative of C_b of Proposition 7.3(ii) of [2], which lifts the copy rotation with the opposite orientation. Dimensions of spaces of equivariant maps are real dimensions.

2.1 Vertices

A vertex that takes one particle into two is an equivariant linear map from the module of the one into the module of the two. The ungraded and graded copies of $\mathrm{Cl}(3,3)$ in $\mathrm{Cl}(6,6)$ have the same bivectors, $E_{(2,a)}E_{(2,b)} = \mathbf{1} \otimes e_ae_b$ (e2-1; Proposition 1.1), so every Lorentz action built from bivectors is the same operator on $S_2 = S \otimes S$ in both readings, and the statements below hold for both.

Proposition 2.1 (Vertices)**.** *Let $\mathfrak{l}_{\text{int}}$ act on S_n diagonally, on every constituent, unless stated otherwise.*

(i) $\dim \mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S_2) = \dim \mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S_2, S) = 0$. *The rotation $e^{2\pi S_3}$ is $-\mathbf{1}$ on S and $(-1)^n\mathbf{1}$ on S_n, so $\mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S_n) = 0$ for every even n. At $n = 3$, $\dim \mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S_3) = 160$.*

(ii) The spaces of $\mathfrak{l}_{\text{int}}$-equivariant maps have the dimensions

$$\dim \mathrm{End}_{\mathfrak{l}_{\text{int}}}(S) = 8, \quad \dim \mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S \otimes E) = 8, \quad \dim \mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S \otimes V) = 24,$$

$$\dim \mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S \otimes \Lambda^2 E) = 8, \qquad \dim \mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S \otimes \Lambda^2 V) = 32.$$

(iii) The Clifford multiplications $c_1(\psi) = \sum_\mu \eta_{\mu\mu}\gamma_\mu\psi \otimes e_\mu$ and $c_2(\psi) = \sum_{\mu<\nu} \eta_{\mu\mu}\eta_{\nu\nu}\gamma_\mu\gamma_\nu\psi \otimes e_\mu \wedge e_\nu$ are $\mathfrak{l}_{\text{int}}$-equivariant, and $\mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S \otimes E) = c_1 \circ \mathrm{End}_{\mathfrak{l}_{\text{int}}}(S)$, $\mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S \otimes \Lambda^2 E) = c_2 \circ \mathrm{End}_{\mathfrak{l}_{\text{int}}}(S)$. Without the factors $\eta_{\mu\mu}$, c_1 is not equivariant.

(iv) Under the spectator action, $\mathfrak{l}_{\text{int}}$ on constituent 1 alone, $\dim \mathrm{Hom}(S, S_2) = 64$. Under the mixed action, $\mathfrak{l}_{\text{int}}$ on constituent 1 and $\mathfrak{l}_G$ on constituent 2, $\dim \mathrm{Hom}(S, S_2) = 16$ and $\dim \mathrm{Hom}(S, S \otimes V^{(2)}) = 8$, where $V^{(2)}$ is the vector space of constituent 2. On S the $\mathfrak{l}_G$ Casimir $\sum_i K_i^2 - \sum_k L_k^2$ equals $3\,\mathbf{1}$, and $e^{2\pi L_3} = \mathbf{1}$.

Proof. (i) $e^{2\pi S_3} = e^{\pi e_4e_5} = \cos\pi + \sin\pi\, e_4e_5 = -\mathbf{1}$ since $(e_4e_5)^2 = -\mathbf{1}$. The diagonal lift is a product of n commuting copies, one per constituent. An equivariant T intertwines the group elements, so $T(-\psi) = (-1)^nT\psi$, and $T = 0$ for even n; the same holds for $\mathrm{Hom}(S_n, S)$. For $n = 3$, $S \otimes \mathbb{C} = 2(\frac{1}{2}, 0) \oplus 2(0, \frac{1}{2})$ by (ii), and the joint weights of the commuting pair (S_3, T_3) on $S_3 \otimes \mathbb{C}$ contain $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$ forty times each, so $\dim_{\mathbb{C}} \mathrm{Hom} = 2 \cdot 40 + 2 \cdot 40$. (ii) Over $\mathbb{C}$, $\mathfrak{l}_{\text{int}} \cong \mathfrak{sl}(2, \mathbb{C})$ and $S \otimes \mathbb{C} = 2(\frac{1}{2}, 0) \oplus 2(0, \frac{1}{2})$, $E \otimes \mathbb{C} = (\frac{1}{2}, \frac{1}{2})$, $\Lambda^2E \otimes \mathbb{C} = (1, 0) \oplus (0, 1)$, while

$V = E \oplus 2\cdot\mathbf{1}$ and $\Lambda^2 V = \Lambda^2 E \oplus 2E \oplus \mathbf{1}$. The Clebsch–Gordan rule gives $2^2 + 2^2 = 8$ for each of $\mathrm{End}(S)$, $S \to S \otimes E$ and $S \to S \otimes \Lambda^2 E$, hence $8 + 2\cdot 8$ and $8 + 2\cdot 8 + 8$. (iii) $\sum_\mu \eta_{\mu\mu}\gamma_\mu \otimes e_\mu$ is the invariant element of $\mathrm{Cl}(E) \otimes E$ defined by η; the equivariance, its failure without η, and the ranks 8 of the composites are computed exactly. (iv) Constituent 2 is a trivial module under the spectator action, so $S_2 \cong 8S$ and the count is $8\cdot 8$. On S the $\mathfrak{l}_G$ Casimir is $3\,\mathbf{1}$, the value on a four-vector, so $S \otimes \mathbb{C}$ is two copies of $(\frac{1}{2}, \frac{1}{2})$, and $\dim \mathrm{Hom}(S, S \otimes 2E) = 2\cdot 8$. V is the field-strength module of $\mathfrak{l}_G$ (Theorem 12.3 of [1]), $V \otimes \mathbb{C} = (1,0) \oplus (0,1)$, so $\mathrm{Hom}(S, S \otimes V^{(2)}) \cong \mathrm{Hom}(S, S \otimes \Lambda^2 E)$. □

Part (i) is the angular-momentum selection rule: a spin-$\frac{1}{2}$ state cannot become a state of integer spin, and two spin-$\frac{1}{2}$ constituents have integer spin [16, 7]. It restates Proposition 10.9 of [1] and Proposition 8.11(v) of [2] as a statement about vertices (inherited; verified in the realisation, e2-2, e2-6). Parts (ii) and (iii) are inherited Clebsch–Gordan counts, fixed in the realisation (e2-3, e2-4). The vertex c_1 is the structure of the coupling $\bar\psi\gamma^\mu\psi A_\mu$, and c_2 that of the Pauli coupling $\bar\psi\gamma^\mu\gamma^\nu\psi F_{\mu\nu}$. In both the boson is an added tensor factor E or $\Lambda^2 E$, not a second constituent. Part (iv) is fixed (e2-5). Under the mixed action the second constituent itself is the boson. A fermion of constituent 1 has an equivariant vertex into $S \otimes V^{(2)}$, where $V^{(2)}$ is read through $\mathfrak{l}_G$ as a field strength, the frame field of constituent 2 in the field-strength reading. The mixed action is one of the five Lorentz actions at $n = 2$ (Theorem 10.8 of [1]; Section 7.5 of [2]), and nothing selects it, so this vertex is constructed. Under the diagonal action a fermion has vertices into the odd rungs only, and the first is $n = 3$.

2.2 Thresholds

The free Dirac Hamiltonian of one constituent is $H(\mathbf{p}) = me_1 + \sum_a p_a e_1 e_{a+3}$, with $H(\mathbf{p})^2 = E^2\mathbf{1}$, $E = (m^2 + |\mathbf{p}|^2)^{1/2}$ (equation (8) of [1]). For two constituents with masses $m_1, m_2 \geq 0$ and momenta $\mathbf{p}, \mathbf{q}$ put $E_1 = (m_1^2 + |\mathbf{p}|^2)^{1/2}$, $E_2 = (m_2^2 + |\mathbf{q}|^2)^{1/2}$,

$$a = E_1^2 + E_2^2, \qquad b = E_1^2 E_2^2 - m_1^2 m_2^2 = m_1^2|\mathbf{q}|^2 + m_2^2|\mathbf{p}|^2 + |\mathbf{p}|^2|\mathbf{q}|^2.$$

The ungraded two-particle operator is $H_u = H(\mathbf{p}) \otimes \mathbf{1} + \mathbf{1} \otimes H(\mathbf{q})$. The graded composite is $H_g = H^{(1)}(\mathbf{p}) + H^{(2)}(\mathbf{q})$, with $H^{(k)}(\mathbf{p}) = m_k E_{(k,1)} + \sum_a p_a E_{(k,1)} E_{(k,a+3)}$ built from the tower's vectors.

Proposition 2.2 (Thresholds)**.** *(i)* $\operatorname{spec} H_u = \{\pm E_1 \pm E_2\}$*, the four sign choices on* 16 *dimensions each, and* $E_1 + E_2 \geq m_1 + m_2$*, with equality iff* $\mathbf{p} = \mathbf{q} = 0$*.*

(ii) $H_g^2 = a\,\mathbf{1} + X$ *with* X *traceless and* $X^2 = 4b\,\mathbf{1}$*, and* $E_{(1,1)}E_{(2,1)}$ *anticommutes with* H_g*. Hence* $\operatorname{spec} H_g = \{\pm E_+, \pm E_-\}$ *with*

$$E_\pm^2 = E_1^2 + E_2^2 \pm 2\big(E_1^2 E_2^2 - m_1^2 m_2^2\big)^{1/2},$$

each on 16 *dimensions when* $b > 0$*, and* E_+ *is the largest eigenvalue. At rest* $\operatorname{spec} H_g = \{\pm M\}$*,* $M = (m_1^2 + m_2^2)^{1/2}$*, each on* 32 *dimensions.*

(iii) $\max(E_1, E_2) \leq E_+ \leq E_1 + E_2$*, and* $(E_1 + E_2)^2 - E_+^2 = 2\big((b + m_1^2 m_2^2)^{1/2} - b^{1/2}\big)$*, which is positive iff* $m_1 m_2 \neq 0$*. If* $m_1 m_2 = 0$*, then* $E_\pm = |E_1 \pm E_2|$*: the graded composite has the additive spectrum.*

Proof. (i) $H(\mathbf{p}) \otimes \mathbf{1}$ and $\mathbf{1} \otimes H(\mathbf{q})$ commute, are diagonalisable with eigenvalues $\pm E_1$ and $\pm E_2$ on four dimensions each, and the joint eigenspaces have dimension $4\cdot 4$. $E_k \geq m_k$. (ii) Write $H^{(1)} = m_1 A + P$ and $H^{(2)} = m_2 B + Q$ with $A = E_{(1,1)}$, $B = E_{(2,1)}$, and P, Q the momentum terms. Each $H^{(k)}$ squares to E_k^2, as on one constituent. A anticommutes with P and B, and commutes with Q; B anticommutes with Q and commutes with P; P and Q commute. Hence

$X = \{H^{(1)}, H^{(2)}\} = 2(m_1 AQ + m_2 PB + PQ)$. The squares of the three terms are $m_1^2|\mathbf{q}|^2$, $m_2^2|\mathbf{p}|^2$ and $|\mathbf{p}|^2|\mathbf{q}|^2$, and the three mutual anticommutators vanish by the same rules, so $X^2 = 4b$. Each term is a product of distinct blades, so X is traceless, and H_g^2 has the eigenvalues $a \pm 2b^{1/2}$ on 32 dimensions each. $W = E_{(1,1)}E_{(2,1)}$ anticommutes with A, B and every term of P and Q by the blade rule (equation (3) of [1]), and $W^2 = -\mathbf{1}$, so $WH_gW^{-1} = -H_g$; W commutes with X, so each eigenspace of H_g^2 carries $\pm E_\pm$ on equal halves. (iii) $E_+^2 - E_1^2 = E_2^2 + 2b^{1/2} \geq 0$, and $(E_1 + E_2)^2 = a + 2E_1E_2 = a + 2(b + m_1^2 m_2^2)^{1/2}$. □

Part (i) is the threshold of two free particles, inherited: a particle of mass M at rest decays into two of masses m_1, m_2 only if $M \geq m_1 + m_2$, since the final energy at zero total momentum is $E_1(\mathbf{p}) + E_2(-\mathbf{p}) \geq m_1 + m_2$ (e2-7). Part (ii) extends Proposition 8.11(iv) of [2] and Proposition 1.2(i) from rest to all momenta. The closed form and (iii) are fixed and elementary (e2-8: $(H_g^2 - a)^2 = 4b$ exactly at four integer data, the inequalities in 200 random samples to 10^{-9}). Two consequences. First, the graded composite of mass $M = (m_1^2 + m_2^2)^{1/2}$ lies below the threshold $m_1 + m_2$ of the free pair with the same masses, and it stays below it at every momentum. The deficit is controlled by the product $m_1 m_2$ and disappears when one constituent is massless. Second, the graded composite therefore cannot dissociate into the free pair of the ungraded reading, as a bound state cannot dissociate below its threshold [14] (inherited). Reading $m_1 + m_2 - M$ as a binding energy, $(2 - \sqrt{2})\, m$ at equal masses, a fraction $1 - 1/\sqrt{2} \approx 0.293$ of the free pair, is interpretive. Both operators act on the same space S_2. Nothing in the tower turns one into the other, so "dissociation" here compares two readings and is not a dynamical process.

2.3 Fields: one null field and two

At the level of fields three results of [2] answer the question directly. We restate them as statements about one and two.

Corollary 2.3 (One field into two)**.** *(i) One null field does not become a massive pair. Every map commuting with the diagonal $SO(3,\mathbb{C})$ sends $(z, 0)$, z null, to $(\alpha z, \beta z)$, whose Gram matrix Q vanishes, so $M_{\text{pair}} = \frac{1}{2}(\sigma_1 + \sigma_2) = 0$ (Proposition 8.5(iii) and Theorem 8.6(ii) of [2]).*

(ii) Two null fields that are not collinear do: $M_{\text{pair}} = |z_1 \cdot z_2| > 0$, and the copy rotation can move the whole of it into rest energy of the two constituents, as in $\gamma\gamma \to e^+e^-$ (Theorem 8.6(iv),(v) and Example 8.7 of [2]; [15]).

(iii) The closure moves nothing between constituents: for every Hamiltonian the per-constituent invariants of the momentum fields are constant, so no closure dynamics fills an empty constituent or makes a null one massive (Corollary 8.3 of [2]).

All three are restated (fixed there; re-verified for the Gram matrices, e2-9). Read as one and two: at the field level "one becomes two" covariantly only as "two become two". A single null field is shared between the constituents as literal replicas, and a massive pair needs a second, non-collinear field or a non-covariant third body that takes the momentum (Example 8.8 of [2]).

2.4 Quanta: pairs emerge under boosts

The tower carries a particle number of its own, the Fock number N. We now ask which Lie-algebra elements change it, and with which amplitudes. The first lemma refines Proposition 1.6 and Theorem 1.7.

Lemma 2.4 (The number and the split structure)**.** *(i) $N = \frac{3n}{2}\mathbf{1} + \sigma(K_n)$, and $\operatorname{ad} N = K_n$ on V_n. $(-1)^N = -\omega$ at $n = 1$ and is the volume element of $\mathrm{Cl}(6,6)$ at $n = 2$; it commutes with every bivector.*

(ii) $\mathfrak{so}(3n,3n) = \mathrm{Cent}(K_n) \oplus \mathrm{Anti}(K_n)$*, the parts commuting and anticommuting with* K_n*, of dimensions* $9n^2$ *and* $9n^2-3n$*. The lift of an element of* $\mathrm{Cent}(K_n)$ *commutes with* N*. The lift of an element of* $\mathrm{Anti}(K_n)$ *maps each eigenspace* $\{N=k\}$ *into* $\{N=k-2\}\oplus\{N=k+2\}$*: it changes* N *by exactly* ± 2*.*

(iii) A compact element of $\mathfrak{so}(3n,3n)$ *commutes with* N *iff it commutes with* G_n*, iff it is a paired rotation* $\left(\begin{smallmatrix} R & 0 \\ 0 & R \end{smallmatrix}\right)$*,* $R \in \mathfrak{so}(3n)$*, in the basis of the temporal directions followed by their partners.*

(iv) Let $X \in \mathfrak{so}(3n,3n)$ *be compact or Euclidean-symmetric (a pure boost). If* $X \notin \mathrm{Cent}(K_n)$*, then* $\sigma(X)|0\rangle$ *has a nonzero component with* $N=2$*:* $e^{t\sigma(X)}|0\rangle$ *contains a pair at first order in* t*.*

Proof. (i) and (ii) restate Proposition 1.6: $N_i = \frac{1}{2}(\mathbf{1} - e_i e_{i+3})$ and $\sigma(K) = -\frac{1}{2}\sum_i e_i e_{i+3}$, K is an η-antisymmetric involution, and $\mathrm{ad}\,K$ has eigenvalues $0, \pm 2$ on $\Lambda^2 V_n$. (iii) A compact element is $X = \mathrm{diag}(A, D)$ with $A, D \in \mathfrak{so}(3n)$. In the same basis $K_n = \left(\begin{smallmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$ and $G_n = \left(\begin{smallmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$, and both $K_n X K_n^{-1}$ and $G_n X G_n^{-1}$ equal $\mathrm{diag}(D, A)$. So each commutation is $A = D$. By (ii) commuting with K_n is commuting with N, because the spin representation is faithful on bivectors. (iv) A compact X is a combination of blades $e_a e_b$ with $\eta_{aa} = \eta_{bb}$, which are antisymmetric matrices, and a Euclidean-symmetric one of blades with $\eta_{aa} = -\eta_{bb}$, which are symmetric. So $\sigma(X)^\top = \mp\sigma(X)$. Write $\sigma(X) = R + Z + L$ with $[N, R] = 2R$, $[N, Z] = 0$, $[N, L] = -2L$, by (ii). Transposition exchanges raising and lowering parts ($a^- = (a^+)^\top$), so $L = \mp R^\top$, and $X \notin \mathrm{Cent}(K_n)$ gives $R \neq 0$. $R = \sum_{i<j} c_{ij} a_i^+ a_j^+$ in the $3n$ modes, and the states $a_i^+ a_j^+|0\rangle$, $i < j$, are linearly independent, so $R|0\rangle \neq 0$; it is the $N = 2$ component of $\sigma(X)|0\rangle$. □

(i) and (ii) are inherited, the spin representation on the exterior algebra of a maximal isotropic subspace [3], and fixed in the realisation at $n = 1, 2, 3$ (e2-10, e2-11). (iii) and (iv) are fixed (e2-12, e2-26). Part (iii) sharpens Theorem 1.7, which works inside $\mathrm{Cent}(G_n)$: among all compact elements, the number-conserving ones are exactly the compact binding-compatible ones. By (iv) a compact element that does not commute with G_n creates a pair from the vacuum. Proposition 2.5 gives the examples, and e2-13 records which generators of $\mathfrak{l}_{\mathrm{int}}$, $\mathfrak{l}_G$ and the copy algebras lie where: $\mathfrak{l}_{\mathrm{int}} \cap \mathrm{Cent}(K) = \mathbb{R}\,T_1$, the pole boost; every other generator of $\mathfrak{l}_{\mathrm{int}}$, and J, has a pair part; $\mathfrak{l}_G \cap \mathrm{Cent}(K)$ is spanned by the L_k, and the K_i lie in $\mathrm{Anti}(K)$; the copy rotation $\varepsilon \otimes \mathbf{1}_6$ lies in $\mathrm{Cent}(K_2)$ and the copy boost $\varepsilon \otimes G$ in $\mathrm{Anti}(K_2)$.

Proposition 2.5 (Pair amplitudes on one constituent)**.** *(i) Interior boost.* $\mathrm{ad}\,T_2$ *has eigenvalues* ± 1 *and* 0*, so* $U_\zeta = e^{\zeta T_2}$ *is the boost of rapidity* ζ *in the plane* (e_1, e_5)*.* $U_\zeta|000\rangle = \cosh\frac{\zeta}{2}\,|000\rangle + \sinh\frac{\zeta}{2}\,|110\rangle$ *and* $\|U_\zeta|000\rangle\|^2 = \cosh\zeta$*.* U_ζ *is symmetric and not orthogonal.*

(ii) Field-strength boost. $\mathrm{ad}\,K_1$ *has eigenvalues* ± 1 *(twice) and* 0 *(twice), and* $e^{\zeta K_1}|000\rangle = \cosh\zeta\,|000\rangle - \sinh\zeta\,|011\rangle$*. The rotations* L_k *commute with* N*.*

(iii) Compact generators outside $\mathrm{Cent}(G)$*.* $e^{\theta S_1}|000\rangle = \cos\frac{\theta}{2}\,|000\rangle + \sin\frac{\theta}{2}\,|011\rangle$ *and* $e^{\theta J}|000\rangle = \cos\theta\,|000\rangle + \sin\theta\,|011\rangle$*. A spatial rotation through* π *about the axis of* S_1 *carries the vacuum to a state of two quanta.*

Proof. Each generator is a sum of commuting blades B with $B^2 = \pm\mathbf{1}$, so its exponential is a product of factors $\cos t + \sin t\, B$ or $\cosh t + \sinh t\, B$. In the realisation, with $\varepsilon = \sigma_x\sigma_z$ the matrix used on the copy index, $e_1e_5 = \varepsilon\otimes\varepsilon\otimes\mathbf{1}$, $e_5e_6 = \mathbf{1}\otimes\sigma_x\otimes\varepsilon$ and $e_2e_3 = \mathbf{1}\otimes\varepsilon\otimes\sigma_x$. Each maps $|000\rangle$ to the basis vector with two occupation numbers flipped, $|110\rangle$, $|011\rangle$ and $|011\rangle$, and $(e_1e_5)^2 = \mathbf{1}$, $(e_5e_6)^2 = (e_2e_3)^2 = -\mathbf{1}$. The terms $\frac{1}{2}e_3e_5$ and $-\frac{1}{2}e_2e_6$ of K_1 commute, square to $\frac{1}{4}\mathbf{1}$ and both flip modes 2 and 3, so $e^{\zeta K_1}|000\rangle = (\cosh^2\frac{\zeta}{2} + \sinh^2\frac{\zeta}{2}\,s)|000\rangle + \cosh\frac{\zeta}{2}\sinh\frac{\zeta}{2}\,(e_3e_5 - e_2e_6)|000\rangle$ with $s = -e_3e_5e_2e_6$ on $|000\rangle$; the signs, $s = 1$ and $(e_3e_5 - e_2e_6)|000\rangle = -2|011\rangle$, are computed (e2-14 to e2-16). □

All three parts are fixed (e2-14 to e2-16, at $\zeta = 0.3, 1, 2.5$ and $\theta = 0.4, 1.3, \pi$, to 10^{-11}). The amplitudes have the form of a fermionic Bogoliubov transformation [11, 12], inherited. A boost of rapidity ζ turns the vacuum into $\cos\vartheta\,|0\rangle + \sin\vartheta\,|\text{pair}\rangle$ after normalisation, with $\tan\vartheta = \tanh\frac{\zeta}{2}$ for T_2 and $\tan\vartheta = \tanh\zeta$ for K_1. The normalised persistence $\cos^2\vartheta$ is $\cosh^2\frac{\zeta}{2}/\cosh\zeta$ and $\cosh^2\zeta/\cosh 2\zeta$, and it decreases to $\frac{1}{2}$ as $\zeta \to \infty$. The pair probability is bounded by $\frac{1}{2}$ per pair of modes, the Pauli bound of a fermionic Bogoliubov transformation, whereas the bosonic one is unbounded. The normalisation is needed because a boost does not preserve the Euclidean inner product of the realisation, which is the Fock inner product ($a_i^- = (a_i^+)^\top$): in the split realisation boosts are symmetric and rotations antisymmetric, and the Lorentz group, being non-compact, has no finite-dimensional unitary representation other than the trivial one [7] (inherited). In ordinary field theory the free-particle number is Poincaré invariant, and a vacuum turns into pairs only for an observer whose modes are not related to the inertial ones by a Poincaré transformation, as in the Unruh effect [13]. Here the inertial boosts of the interior four-plane already mix the modes, because N is built from the split structure K and not from the Lorentz algebra. The parallel with the Unruh effect is interpretive.

Part (iii) corrects an easy expectation. Compactness does not protect the number of quanta. A spatial rotation of the interior four-plane and the charge J, both compact, create pairs unitarily, and at $\theta = \pi$, resp. $\theta = \frac{\pi}{2}$, the vacuum is turned wholly into a pair. What protects N is commutation with the binding root (Lemma 2.4(iii)).

Proposition 2.6 (Pairs across constituents)**.** *At $n = 2$, let $U_\zeta = e^{\zeta\sigma(\varepsilon\otimes G)}$, the copy boost of rapidity ζ.*

(i) $\sigma(\varepsilon\otimes G) = -\sum_{i=1}^{3}(a_{1,i}^+ a_{2,i}^+ + a_{2,i}^- a_{1,i}^-)$. *Hence* $U_\zeta|0\rangle = \prod_i(\cosh\zeta - \sinh\zeta\, a_{1,i}^+ a_{2,i}^+)|0\rangle$.

(ii) $\langle 0|U_\zeta|0\rangle = \cosh^3\zeta$*; the weight of* $U_\zeta|0\rangle$ *in* $\{N = 2k\}$ *is* $\binom{3}{k}\cosh^{2(3-k)}\zeta\,\sinh^{2k}\zeta$*, of total* $\cosh^3 2\zeta$*, and there is none in odd* N.

(iii) $N^{(1)} = N^{(2)}$ *on* $U_\zeta|0\rangle$*: every pair that emerges has one quantum in each constituent.*

Proof. (i) $\varepsilon\otimes G$ anticommutes with K_2, so its lift lies in $\Lambda^2 L_+ \oplus \Lambda^2 L_-$ (Proposition 1.6(ii)), and it pairs the modes $(1,i)$ and $(2,i)$ because $\varepsilon\otimes G$ maps the plane $(e_i^{(1)}, e_{i+3}^{(1)})$ to $(e_i^{(2)}, e_{i+3}^{(2)})$. The coefficients are computed exactly, with $\sigma(\varepsilon\otimes G)$ symmetric. The three terms commute. On the two-dimensional span of $|0\rangle$ and $a_{1,i}^+ a_{2,i}^+|0\rangle$ the i-th term acts as $-\sigma_x$ and it annihilates the states with one quantum in the pair $(1,i), (2,i)$. (ii) and (iii) follow by expanding the product. □

Fixed (e2-17, exact for (i), and for (ii), (iii) at $\zeta = 0.3, 1, 2$ to relative 10^{-9}); the lift of the copy boost is checked against $\varepsilon\otimes G$ on all twelve generators. This is pair emergence across the constituents. The copy boost is a metric connection between two binding frames but not a gauge field (Proposition 9.4 of [2]; Corollary 2.4 of [1]). Each emerging pair has one quantum in each constituent, with normalised persistence $\cosh^6\zeta/\cosh^3 2\zeta = (1+\tanh^2\zeta)^{-3}$.

Proposition 2.7 (Lorentz-invariant states)**.** *(i) At $n = 1$ no nonzero state is $\mathfrak{l}_{\text{int}}$-invariant. The two projectors $\frac{1}{2}(\mathbf{1}\pm\omega)$ of Proposition 26.4 of [1] are the projectors onto odd and even N. The paired-mode projector of Example 2.6 of [1] is the projector onto $N \geq 2$.*

(ii) At $n = 2$ the states invariant under the diagonal $\mathfrak{l}_{\text{int}}$ form a space of dimension 8. Its weights in the sectors $N = 0, \ldots, 6$ (the traces of the invariant projector restricted to them) are $0, \frac{1}{2}, 2, 3, 2, \frac{1}{2}, 0$. It contains no state with $N = 0$ or $N = 6$, and its states of definite N form a space of dimension 2, at $N = 3$, on which $N_i^{(1)} + N_i^{(2)} = 1$ for $i = 1, 2, 3$.

(iii) Under the spectator action no nonzero state of S_2 is invariant. The projector $\Pi_0 = \frac{1}{2}(\mathbf{1}+H)$ of Theorem 26.6 of [1] commutes with $(-1)^N$ and not with N, and its compression to every sector $\{N = k\}$ has trace equal to half the dimension of that sector.

Proof. (i) $S\otimes\mathbb{C}$ contains only spin $\frac{1}{2}$. ω is $-(-1)^N$ (Lemma 2.4(i)). With $\mathsf{P}=\sum_i e_ie_{i+3}=\sum_i(1-2N_i)$ and $\mathcal{V}=\omega$, the projector $\frac{1}{2}\mathbf{1}-\frac{1}{4}(\mathsf{P}+\mathcal{V})$ of Example 2.6 vanishes exactly where at most one mode is occupied, by the table of signs given there. (ii) and (iii) are exact kernel computations; the weights are traces of the orthogonal projector onto the kernel. $H=E_{(1,2)}E_{(1,3)}E_{(2,1)}E_{(2,2)}$ is a blade of even grade, so it commutes with the volume element $(-1)^N$. It shares one index with each of the pole operators of the modes $(1,2)$, $(1,3)$, $(2,1)$, $(2,2)$ and none with the other two, so it anticommutes with those four and is a signed permutation of the Fock basis that flips exactly those four occupation numbers. Hence $H|0\rangle$ has $N=4$, so H and Π_0 do not commute with N, and H has zero diagonal, so the compression of Π_0 to every sector has trace half its dimension. □

Fixed (e2-18). At one constituent the number of quanta is not Lorentz invariant (Proposition 1.8), while its parity is: the Lorentz-invariant projectors of Proposition 26.4 of [1] are the parity projectors, and the paired-mode projector, a function of N that is not a function of the parity, is not Lorentz invariant. The Fock vacuum is not Lorentz invariant at either rung. At the second rung the Lorentz scalars of the diagonal action are spread over all numbers from 1 to 5; the two scalars of definite number carry one quantum in each mode pair i, shared between the constituents. The Lorentz-invariant vacuum of Theorem 26.6 of [1] is not a function of N. It selects half of every number sector, and so it is not a Fock vacuum.

2.5 A Lorentz-invariant number at even rungs

The Fock number is the lift of the involution K_n (Lemma 2.4(i)), and K_n commutes with no Lorentz algebra of the tower except through the pole boost. Any other η_n-antisymmetric involution defines a number in the same way. The question is whether one commutes with a Lorentz action.

Theorem 2.8 (An invariant number of quanta)**.** *(i) Let $X\in\mathfrak{so}(3n,3n)$ with $X^2=\mathbf{1}$. Its eigenspaces $V_\pm$ are totally null of dimension $3n$, and $N_X:=\frac{3n}{2}\mathbf{1}+\sigma(X)$ has spectrum $0,\dots,3n$ with multiplicities $\binom{3n}{k}$. If X commutes with a Lie algebra $\mathfrak{l}\subset\mathfrak{so}(3n,3n)$, then N_X commutes with $\sigma(\mathfrak{l})$.*

(ii) Under the common binding action (the diagonal $\mathfrak{l}_G$, $(k_I,k_B,k_S)=(0,n,0)$ in Theorem 10.8 of [1]) such an X exists iff n is even. At $n=2$ the only ones are $\pm\varepsilon\otimes G$, with $V_\pm=\{X^{(2)}=\pm GX^{(1)}\}$, the null subspaces of Proposition 8.4(ii) of [2]. At $n=4$, $(\varepsilon\oplus\varepsilon)\otimes G$ is one; it is not unique, since each of the three pairings of the four constituents gives one.

(iii) At $n=2$ none exists under the other four actions: the commutants, of dimensions 7, 1, 28 and 15 for $(2,0,0)$, $(1,1,0)$, $(1,0,1)$ and $(0,1,1)$, consist of compact elements for the first two and of elements vanishing on the Lorentz-carrying subspace of constituent 1 for the last two. At $n=3$ the action $(0,2,1)$ admits $(\varepsilon\otimes G)\oplus K$.

(iv) At $n=2$, with $N_K:=N_{\varepsilon\otimes G}$: the diagonal-$\mathfrak{l}_G$-invariant states form a space of dimension 4, with weights $1,2,1$ at $N_K=0,3,6$. The vacuum $|0_K\rangle$ of N_K is $\mathfrak{l}_G$-invariant, has weights $\frac{1}{8},\frac{3}{8},\frac{3}{8},\frac{1}{8}$ at $N=0,2,4,6$, and equals the limit of $U_\zeta|0\rangle/\|U_\zeta|0\rangle\|$ as $\zeta\to-\infty$. Every vector of $V_\pm$ has components of equal length in the two constituents.

Proof. (i) For $v,w\in V_+$, $\eta(v,w)=\eta(Xv,w)=-\eta(v,Xw)=-\eta(v,w)$, so V_+ is totally null, and likewise V_-; as $V_n=V_+\oplus V_-$ and null subspaces of $\mathbb{R}^{3n,3n}$ have dimension at most $3n$, both have dimension $3n$. Two pairs of complementary maximal null subspaces are conjugate under $O(\eta_n)$ (Witt's theorem), so $X=gK_ng^{-1}$, and a lift $\hat g$ of g to the Pin group gives $\sigma(X)=\hat g\sigma(K_n)\hat g^{-1}$. The spectrum of $\sigma(K_n)=N-\frac{3n}{2}$ is that of Lemma 2.4. Finally σ is a Lie algebra homomorphism. (ii) The commutant of the diagonal $\mathfrak{l}_G$ in $\mathfrak{so}(3n,3n)$ is $\{A\otimes\mathbf{1}_6+B\otimes G:A,B\in\mathfrak{so}(n)\}$, which acts on $z\in\mathbb{C}^n\otimes\mathbb{C}^3$ as $(A+iB)\otimes\mathbf{1}_3$ (Proposition 9.1(i) of [2]). $X^2=\mathbf{1}$ requires

$(A+iB)^2 = \mathbf{1}$, so $A+iB$ is invertible; but a complex antisymmetric matrix of odd size has $\det(A+iB) = \det(A+iB)^\top = -\det(A+iB) = 0$. For even n, $A+iB = i\varepsilon \oplus \cdots \oplus i\varepsilon$ squares to $\mathbf{1}$. At $n=2$ the solutions of $(a\,\varepsilon \otimes \mathbf{1}_6 + b\,\varepsilon \otimes G)^2 = \mathbf{1}$ are $a=0$, $b=\pm 1$. (iii) and (iv) are computed. For the limit, $|0_K\rangle$ is the eigenvector of $\sigma(\varepsilon \otimes G)$ for its lowest eigenvalue -3, and $\langle 0|0_K\rangle \neq 0$. On $V_\pm$, $|X^{(2)}| = |GX^{(1)}| = |X^{(1)}|$. □

Part (i) is inherited, the correspondence between polarisations and Fock structures [3, 12]; the spectrum and the commutation are fixed in the realisation at $n=2$ (e2-19) and on S_4, of dimension 4096, where $\sigma((\varepsilon \oplus \varepsilon) \otimes G)$ was checked to lift its vector and to commute with the six lifted diagonal $\mathfrak{l}_G$ generators in sparse integer arithmetic (e2-22). Part (ii) is fixed; the parity argument is general, and the exact checks cover $n = 2, 3, 4$ (e2-19, e2-21). Part (iii) is fixed and re-verifies the dimensions of Theorem 10.8 of [1] (e2-21). Part (iv) is fixed (e2-20; the limit to 10^{-9} at $\zeta = -12$).

So a Lorentz-invariant number of quanta exists, under the common binding action, exactly at even rungs. At the second rung it is unique up to $N_K \mapsto 6 - N_K$. Its quanta are not quanta of one constituent: $V_\pm$ is the graph of $\pm G$ from constituent 1 to constituent 2, and N_K does not commute with $N^{(1)}$, since $\sigma(\varepsilon \otimes G)$ changes $N^{(1)}$ and $N^{(2)}$ by one each (Proposition 2.6). Its vacuum, read in the Fock structure of the constituents, is a condensate of pairs with one quantum in each constituent, the infinite copy boost of the Fock vacuum (fixed). Read as a particle number, N_K counts quanta that are Lorentz invariant under the binding action and belong to two constituents at once (interpretive). The parity pattern can be set beside Proposition 1.4(iv): under the diagonal interior action an imaginary unit is selected exactly at odd rungs, and under the diagonal binding action an invariant number exists exactly at even rungs. The two statements concern different Lorentz actions, and the juxtaposition is a reading.

process	sense	result	class
$S \to S_n$, common interior action	module	none for even n; 160 maps at $n=3$	inherited (selection rule); fixed
$S \to S \otimes E$, $S \otimes \Lambda^2 E$; $S \to S \otimes V^{(2)}$ (mixed)	module	Clifford multiplication, 8 maps each	inherited counts; fixed; mixed action constructed
free particle of mass $M \to$ two	mass shell	only if $M \geq m_1 + m_2$	inherited
graded composite $\to$ free pair	mass shell	$E_+ < E_1 + E_2$ iff $m_1 m_2 \neq 0$	fixed; binding interpretive
one null field $\to$ massive pair	field	no covariant map; closure moves nothing	obstructed (restated)
two null fields $\to$ massive pair	field	$M_{\text{pair}} = \lvert z_1 \cdot z_2 \rvert$	inherited (restated)
boosts ($\mathfrak{l}_{\text{int}}$, $\mathfrak{l}_G$, copy)	quanta	pairs, fermionic Bogoliubov amplitudes	fixed; form inherited
compact, outside Cent(G_n) (S_k, J)	quanta	pairs, unitarily	fixed
compact, inside Cent(G_n) (paired rotations)	quanta	N conserved	fixed
copy rotation	quanta	one quantum hops; Rabi at unequal rates	fixed; two-level inherited; i constructed
N_X under $(0, n, 0)$	quanta	exists iff n even; vacuum a pair condensate	fixed; particle reading interpretive
dynamical pair-creating agent	—	—	open

Table 4: Processes that take one to two, by sense of "particle".

statement	class	checks
conventions: tower, ungraded copy, common bivectors, CAR, Fock vacuum	restated (RE Props. 7.1, 7.2)	e2-1
Proposition 2.1(i): no vertex $S \leftrightarrow S_2$; $(-1)^n$; 160 at $n = 3$	inherited; fixed in realisation	e2-2, e2-6
Proposition 2.1(ii), (iii): counts $8, 8, 24, 8, 32$; Clifford multiplication	inherited counts; fixed	e2-3, e2-4
Proposition 2.1(iv): spectator 64; mixed 16 and 8; $\mathfrak{l}_G$ Casimir 3	fixed; mixed action constructed	e2-5
Proposition 2.2(i): free-pair threshold	inherited	e2-7
Proposition 2.2(ii), (iii): $(H_g^2 - a)^2 = 4b$; $E_+ < E_1 + E_2$ iff $m_1 m_2 \neq 0$	fixed; binding energy interpretive	e2-8
Corollary 2.3: one null field, two null fields, closure	restated (RE Prop. 8.5, Thm 8.6, Cor. 8.3)	e2-9
Lemma 2.4(i), (ii): $N = \frac{3n}{2} + \sigma(K_n)$; $\mathrm{Cent}(K_n) \oplus \mathrm{Anti}(K_n)$	restated (Proposition 1.6); inherited	e2-10, e2-11
Lemma 2.4(iii), (iv): compact and number-conserving = paired rotations; outside $\mathrm{Cent}(K_n)$ the vacuum acquires a pair	fixed	e2-12, e2-13, e2-26
Proposition 2.5: Bogoliubov amplitudes; pairs from S_1 and J	fixed; form inherited; Unruh analogy interpretive	e2-14 to e2-16
Proposition 2.6: copy boost, pairs across constituents	fixed	e2-17
Proposition 2.7: invariant states at $n = 1, 2$; Prop. 26.4, Thm 26.6	fixed (comparison with ES17)	e2-18
Theorem 2.8: invariant number iff n even; N_K at $n = 2$	fixed; (i) inherited; reading interpretive	e2-19 to e2-22
Proposition 2.9: hopping, ladder, Rabi transfer	fixed; Rabi inherited; external i constructed	e2-23 to e2-25
Section 2.7: dynamical agent of pair creation	open	—

Table 5: Claims of Section 2, their classes and the checks printed by `verify/e2_processes.py`.

2.6 The copy rotation moves one quantum

Proposition 2.9 (Hopping and Rabi transfer)**.** *(i)* $\hat{C} = \sigma(\varepsilon \otimes \mathbf{1}_6) = \sum_i (a^+_{2,i} a^-_{1,i} - a^+_{1,i} a^-_{2,i})$. *It commutes with* N *and not with* $N^{(1)}$*;* $\hat{C} a^+_{1,i}|0\rangle = a^+_{2,i}|0\rangle$ *and* $\hat{C} a^+_{2,i}|0\rangle = -a^+_{1,i}|0\rangle$*, and* $e^{\theta \hat{C}} a^+_{1,1}|0\rangle = \cos\theta\, a^+_{1,1}|0\rangle + \sin\theta\, a^+_{2,1}|0\rangle$*.* $e^{\theta \hat{C}}$ *has no matrix element between different values of* N*, and* $e^{2\pi \hat{C}} = \mathbf{1}$*.*

(ii) At $n = 3$ *the copy rotations commute with* K_3*, and their lifts with* N*; the copy boosts anticommute with* K_3*.*

(iii) With rates κ_1, κ_2 *and coupling* β*, let* $\mathsf{H} = \kappa_1 N^{(1)} + \kappa_2 N^{(2)} + i\beta \hat{C}$*, with an external* i*.* H *is Hermitian and commutes with* N*. On the sector* $N = 1$*, in the basis* $a^+_{1,i}|0\rangle$*,* $a^+_{2,i}|0\rangle$*, it is* $\left(\begin{smallmatrix} \kappa_1 & -i\beta \\ i\beta & \kappa_2 \end{smallmatrix}\right) \otimes \mathbf{1}_3$*, whose eigenvalues are the lines* $\omega_\pm$ *of Theorem 7.12(ii) of [2]. From* $a^+_{1,1}|0\rangle$ *the probability of finding the quantum in constituent 2 is* $s \sin^2(\Omega t/2)$*, with* $s = 4\beta^2/((\kappa_2 - \kappa_1)^2 + 4\beta^2)$ *and* $\Omega = ((\kappa_2 - \kappa_1)^2 + 4\beta^2)^{1/2}$*.*

Proof. (i) $\varepsilon \otimes \mathbf{1}_6$ commutes with K_2 and maps the vector of each mode of constituent 1 to the same mode of constituent 2, so its lift lies in $L_+ \otimes L_-$ and is a hopping term; the coefficients are computed. The rest follows from $[\hat{C}, a^\pm_{1,i}] = a^\pm_{2,i}$ and $[\hat{C}, a^\pm_{2,i}] = -a^\pm_{1,i}$, and from $\hat{C}$ being a sum of six commuting terms of square $-\frac{1}{4}$ (Proposition 7.3(ii) of [2]). (ii) $GK = -KG$. (iii) $\hat{C}$ is real antisymmetric, so $i\beta\hat{C}$ is Hermitian. On the sector $N = 1$, $\hat{C}$ acts as $\varepsilon \otimes \mathbf{1}_3$, and the two-level formula follows. □

Fixed (e2-23 to e2-25; the transfer probability at $(\kappa_1, \kappa_2, \beta) = (1, 3, 0.4)$ to 2×10^{-16}, with weight below 10^{-24} outside $N = 1$, and complete transfer at equal rates at $t = \pi/(2\beta)$). The two-level formula is inherited [17]. The external i is constructed, as in Proposition 8.11(i) of [2];

on fields the same matrix acts with G_2 as the imaginary unit and gives the per-copy rest energies of Section 8.4 of [2]. Read as one and two: one quantum can be shared between constituents, in any proportion at equal rates and up to the fraction s at unequal rates, but it is never doubled, because $\hat{C}$ commutes with N. At equal rates the coupling is removable (Theorem 7.12(iii) of [2]), so which constituent carries the quantum is gauge dependent if the copy rotation is gauged, as the per-copy rest energy is (Proposition 8.9 of [2]). At unequal rates the partial transfer is physical. Up the ladder the copy rotations $\mathfrak{so}(n)$ hop quanta among n constituents and the copy boosts create pairs across them (e2-24).

2.7 Answer: which processes take one to two

Table 4 lists the processes. Read together they say the following. In the sense of Lorentz modules, one fermion never becomes two under the common interior action. It becomes a fermion and a boson, by Clifford multiplication, and under the mixed action the boson can be the second constituent read as a field strength (fixed; the mixed action constructed). In the sense of mass shells, the graded composite of two masses lies below the free-pair threshold at every momentum, by an amount controlled by $m_1 m_2$, so the tower's own composite is the bound side of the threshold (fixed; binding reading interpretive). In the sense of fields, one null field becomes a massive pair only together with a second field (restated). In the sense of quanta, no element of $\mathfrak{so}(3n, 3n)$ changes N by an odd number, so one quantum never becomes two. The vacuum becomes partly a pair under every compact or Euclidean-symmetric element outside $\mathrm{Cent}(K_n)$: the boosts of $\mathfrak{l}_{\mathrm{int}}$ other than the pole boost, the boosts of $\mathfrak{l}_G$, the copy boost, and the compact generators outside $\mathrm{Cent}(G_n)$, including the spatial rotations of the interior four-plane and the charge J (fixed). The compact binding-compatible generators, and the copy rotation among them, never create quanta; the copy rotation moves one (fixed). The number of Fock quanta is therefore not Lorentz invariant under any action of the tower, since every action contains, on some constituent, the boosts of $\mathfrak{l}_{\mathrm{int}}$ or of $\mathfrak{l}_G$, and a Lorentz-invariant number exists only for the common binding action at even rungs, where its quanta belong to two constituents at once (fixed). Whether a dynamical mechanism of the arena supplies the non-binding-compatible or non-compact generator that creates a pair is open.

Up the ladder. Rung by rung the following changes, each fixed. Under the common interior action $e^{2\pi S_3} = (-1)^n$, so a fermion has vertices into odd rungs only. $\mathfrak{so}(3n, 3n)$ splits into $9n^2$ number-conserving and $9n^2 - 3n$ pair-creating directions, and the compact number-conserving ones are the $\frac{3}{2}n(3n-1)$ paired rotations. The copy rotations $\mathfrak{so}(n)$ hop quanta among the constituents and the copy boosts create pairs across them. An invariant number under the common binding action exists exactly at even n, uniquely up to $N_K \mapsto 6 - N_K$ at $n = 2$ and in several ways at $n = 4$.

3 Up the ladder

Sections 1 and 2 treat the second rung. This section treats the rung $\mathrm{Cl}(3n, 3n)$ for every n and the step from n to $n+1$. The question is the same as at $n = 2$: in what precise sense, and under which added structures, does rung n carry n particles rather than one particle with a larger internal space, and what changes from one rung to the next. The answers, with their classes, are these. (1) $\mathrm{Cl}(3n, 3n) = M_{8^n}(\mathbb{R})$ at every rung. The step embeds rung n in rung $n+1$ with multiplicity eight and adds three Fock modes, one frame and one copy direction; the commuting copy of the new constituent is its generators dressed by the volume element of rung n (fixed; Section 3.1). (2) For n equal masses the graded rest Hamiltonian is one clock of mass $\sqrt{n}\,m$, and the ungraded one is n clocks of mass m, with spectrum $m(n-2j)$ of multiplicity $\binom{n}{j}4^n$ (fixed, partly restated from Proposition 1.2; Section 3.2). (3) A rotation through 2π of the diagonal

interior Lorentz action is $(-1)^n$ on S_n, and one of the diagonal field-strength action is $+\mathbf{1}$. The interior composite is a fermion exactly at odd n, and the field-strength composite has integer spin at every rung (restated from Proposition 10.9 of [1], with the second half fixed; Section 3.3). (4) On vectors the copy rotations of all six directions span the commutant of the diagonal $\mathfrak{so}(3,3)$, and the copy group $O(n)$ has the diagonal $\mathfrak{so}(3,3)$ as its commutant. On S_n the Casimirs satisfy $C_{\mathrm{cop}} + C_{\mathrm{diag}} = \frac{3}{4}n(n+4)$. This is Howe's skew duality $(O(n), \mathfrak{so}(6))$ on the spinor module of $\mathfrak{so}(6n)$, and at $n = 2, 3$ the joint decomposition has the predicted multiplicity-free form (fixed; the duality inherited; Section 3.4). (5) The Koszul swaps realise the permutation matrices of that $O(n)$, twisted by the sign character. Their symmetric and antisymmetric sectors are mirror images, of equal dimension and never empty. The plain permutations of the common complex space give $\Lambda^n_{\mathbb{C}}$ of the Dirac module, which vanishes for $n \geq 5$ (fixed; the exterior powers inherited; Section 3.5). (6) The algebra singles out no rung, and its invariant forms single out only the parity of n, through a period of four. Odd rungs are singled out by the interior Lorentz action, even rungs by the binding action, and $n = 3$ by the projector P_* (restated and fixed; Section 3.6). (7) Going up the ladder enlarges one particle; it becomes the addition of a particle only through the constructed structures of Section 1, now needed at every rung (interpretive, drawn from the fixed statements; Section 3.7). Identifiers e3-n refer to the `PASS` lines of `verify/e3_ladder.py` (Table 8).

Notation. As in Section 1: $S = \mathbb{R}^8$, $S_n = S^{\otimes n}$, $E_{(k,a)} = \omega^{\otimes(k-1)} \otimes e_a \otimes \mathbf{1}^{\otimes(n-k)}$, and $V_n = \mathbb{R}^n \otimes V = \mathbb{R}^{3n,3n}$ with $\eta_n = \mathbf{1}_n \otimes \eta$ and $G_n = \mathbf{1}_n \otimes G$. $\Gamma^{(k)} = E_{(k,1)} \cdots E_{(k,6)} = \mathbf{1} \otimes \cdots \otimes \omega \otimes \cdots \otimes \mathbf{1}$ is the volume element of constituent k, and $\Gamma_n = \Gamma^{(1)} \cdots \Gamma^{(n)}$. The ungraded generators are $u_{(k,a)} = \mathbf{1}^{\otimes(k-1)} \otimes e_a \otimes \mathbf{1}^{\otimes(n-k)}$, $J^{(k)} = E_{(k,2)}E_{(k,3)}$, and $a^{\pm}_{k,i} = \frac{1}{2}(E_{(k,i)} \pm E_{(k,i+3)})$, $N = \sum_{k,i} a^+_{k,i} a^-_{k,i}$ (Proposition 7.2 of [2]); $|0\rangle$ is the first basis vector of S_n, the Fock vacuum. The diagonal generators are $\Delta_{ab} = \frac{1}{2}\sum_k E_{(k,a)}E_{(k,b)}$. All fifteen span the *diagonal* $\mathfrak{so}(3,3)$; those with $a, b \in \{1,4,5,6\}$ span the diagonal $\mathfrak{l}_{\mathrm{int}}$; and the paired rotations $\Delta_{ij} - \Delta_{i+3,j+3}$ and paired boosts $\Delta_{i,j+3} - \Delta_{j,i+3}$ span the diagonal $\mathfrak{l}_G$ (Theorem 12.1 of [1]). The copy rotations are $C^{kl} = \frac{1}{2}\sum_a \eta_{aa} E_{(k,a)}E_{(l,a)}$ for $k < l$, with $C^{12} = C_b$ of Proposition 7.3(ii) of [2], and M_{kl} are their interior parts (Section 10.1 of [1]). The Casimirs are $C_{\mathrm{diag}} = -\sum_{a<b} \eta^{aa}\eta^{bb}\Delta^2_{ab}$ over all six directions, C_Δ the same over the interior four (Section 10.2 of [1]), $C_G = -\sum_i L_i^2 + \sum_i K_i^2$ for the paired rotations L_i and boosts K_i of $\mathfrak{l}_G$, $C_{\mathrm{cop}} = -\sum_{k<l}(C^{kl})^2$ and $C^{\mathrm{int}}_{\mathrm{cop}} = -\sum_{k<l} M^2_{kl}$. A Lorentz Casimir takes the value $2[j_L(j_L+1) + j_R(j_R+1)]$ on (j_L, j_R). We write $\mathfrak{S}_n$ for the symmetric group, to keep S_n for the module.

3.1 The rung and the step

Proposition 3.1 (The rung and the step)**.** *(i) Every blade of* $\mathrm{Cl}(3n,3n)$ *other than* $\pm\mathbf{1}$ *is traceless on* S_n*. Hence the* 64^n *blades are linearly independent and* $\mathrm{Cl}(3n,3n) \cong M_{8^n}(\mathbb{R})$*, with* S_n *its unique irreducible module. The volume element is* $E_{(1,1)} \cdots E_{(n,6)} = \Gamma_n = \omega^{\otimes n}$*, and* $(-1)^N = (-1)^n \Gamma_n$*.*

(ii) $E^{(n+1)}_{(k,a)} = E^{(n)}_{(k,a)} \otimes \mathbf{1}$ *for* $k \leq n$ *and* $E_{(n+1,a)} = \Gamma_n \otimes e_a$*. So* $\mathrm{Cl}(3n,3n) \otimes \mathbf{1} \subset \mathrm{Cl}(3n+3, 3n+3)$*, and* S_{n+1} *restricted to* $\mathrm{Cl}(3n,3n)$ *is eight copies of* S_n*. The commutant of* $\mathrm{Cl}(3n,3n) \otimes \mathbf{1}$ *is* $\mathbf{1} \otimes M_8(\mathbb{R})$*, generated by the ungraded generators* $\mathbf{1} \otimes e_a = (\Gamma_n \otimes \mathbf{1})E_{(n+1,a)}$ *of the new constituent.*

(iii) $N_{n+1} = N_n \otimes \mathbf{1} + \mathbf{1} \otimes N_1$*,* $\operatorname{spec} N_n = \{0, \ldots, 3n\}$ *with multiplicities* $\binom{3n}{j}$*, and* $[N, E_{(k,i)}] = E_{(k,i+3)}$*,* $[N, E_{(k,i+3)}] = E_{(k,i)}$*. Each step adds three modes.*

(iv) Every diagonal bivector satisfies $\sum_{k\leq n+1} E_{(k,a)}E_{(k,b)} = \left(\sum_{k\leq n} E_{(k,a)}E_{(k,b)}\right) \otimes \mathbf{1} + \mathbf{1} \otimes e_a e_b$*. Hence* $S_{n+1} \cong S_n \otimes S$ *and* $S_n \cong S^{\otimes n}$ *as modules of the diagonal* $\mathfrak{so}(3,3)$*, and so of the diagonal* $\mathfrak{l}_{\mathrm{int}}$ *and* $\mathfrak{l}_G$*.*

Proof. (i) Moving the generators of each constituent to their own factor gives $e_I = \pm \bigotimes_j e_{I_j} \omega^{p_j}$ with $p_j = |I_{j+1}| + \cdots + |I_n| \bmod 2$, and the trace is multiplicative. On S, $\operatorname{tr}(e_J \omega^p) \neq 0$ only for $(J,p) = (\emptyset, 0)$ and for J the full index set with $p = 1$. The last factor has $p_n = 0$, so a nonzero trace needs $I_n = \emptyset$; then $p_{n-1} = 0$, and so on, down to $I = \emptyset$. Trace orthogonality gives independence, and $64^n = (8^n)^2$. The volume element is $\prod_k \Gamma^{(k)}$ because $\prod_a E_{(k,a)} = (\omega^{\otimes(k-1)})^6 \otimes \omega \otimes \mathbf{1}$. On one constituent $(-1)^N = -\omega$ (Proposition 7.2 of [2]), and $N = \sum_k N^{(k)}$. (ii) The formulas are equation (2) of [1] at rungs n and $n+1$. The commutant of $M_{8^n}(\mathbb{R}) \otimes \mathbf{1}$ in $M_{8^{n+1}}(\mathbb{R})$ is $\mathbf{1} \otimes M_8(\mathbb{R})$, and $\Gamma_n^2 = \mathbf{1}$. (iii) $N = \sum_{k,i} \frac{1}{2}(\mathbf{1} - E_{(k,i)} E_{(k,i+3)})$, the terms are commuting diagonal involutions with traceless products, and the commutators are those of Proposition 7.2 of [2]. (iv) $E_{(n+1,a)} E_{(n+1,b)} = \Gamma_n^2 \otimes e_a e_b = \mathbf{1} \otimes e_a e_b$. □

Part (i) is fixed in the realisation (e3-1a, exhaustive over all 64^n blades at $n \leq 4$). That $\mathrm{Cl}(p,q) \cong M_{2^{(p+q)/2}}(\mathbb{R})$ for $p - q \equiv 0 \pmod 8$ is inherited [6]: the real structure of the algebra is the same at every rung, so it singles out none. Parts (ii) and (iii) restate Propositions 7.1 and 7.2 of [2] and Proposition 1.1(ii),(v) at the last factor (e3-1b, with the commutant exhaustive over blades at $n = 1, 2$; e3-1c, spectrum also at $n = 5$). Part (iv) is fixed (e3-6c, $n \leq 3$). Under (iv) every statement about the diagonal action is a statement about tensor powers of the one-constituent module, and the step is tensoring with S.

3.2 Mass up the ladder: one clock or n clocks

Put $H_g = m \sum_k E_{(k,1)}$, built from the vectors of the rung, and $H_u = m \sum_k u_{(k,1)}$, built from the commuting copies.

Proposition 3.2 (Mass up the ladder)**.** *(i) The $E_{(k,1)}$ are pairwise anticommuting traceless involutions. So $H_g^2 = nm^2\mathbf{1}$ and $\operatorname{spec} H_g = \{\pm\sqrt{n}\, m\}$, each of multiplicity $8^n/2$. Every $J^{(k)}$ commutes with H_g, and $Z_g = J^{(1)} H_g$ satisfies $Z_g^2 = -nm^2\mathbf{1}$: the rest flow $e^{-\lambda Z_g}$ is one clock of angular frequency $\sqrt{n}\, m$.*

(ii) The ungraded generators are $u_{(k,a)} = \Gamma^{(1)} \cdots \Gamma^{(k-1)} E_{(k,a)}$, with no further sign. The $u_{(k,1)}$ commute, square to $\mathbf{1}$, and every product over a nonempty subset is traceless. So $\operatorname{spec} H_u = \{m(n-2j) : j = 0, \ldots, n\}$, the value $m(n-2j)$ with multiplicity $\binom{n}{j} 4^n$.

(iii) $T_u^{(k)} = J^{(k)} u_{(k,1)} = \mathbf{1} \otimes \cdots \otimes T \otimes \cdots \otimes \mathbf{1}$ are n commuting complex structures: n clocks of angular frequency m.

(iv) $H_g^{(n+1)} = H_g^{(n)} \otimes \mathbf{1} + \Gamma_n \otimes m e_1$ with anticommuting summands, and $H_u^{(n+1)} = H_u^{(n)} \otimes \mathbf{1} + \mathbf{1} \otimes m e_1$ with commuting summands. The graded mass grows in quadrature, $\sqrt{n}\, m \to \sqrt{n+1}\, m$, and the ungraded energies add.

Proof. (i) Graded generators of distinct constituents anticommute, and each $E_{(k,1)}$ is traceless because $\operatorname{tr} e_1 = 0$. $J^{(k)}$ is a bivector sharing no index with any $E_{(l,1)}$, so it commutes with each (equation (3) of [1]), and $(J^{(k)})^2 = -\mathbf{1}$. (ii) $\Gamma^{(1)} \cdots \Gamma^{(k-1)} = \omega^{\otimes(k-1)} \otimes \mathbf{1}$ and $\omega^2 = \mathbf{1}$, so the product is $\mathbf{1}^{\otimes(k-1)} \otimes e_a \otimes \mathbf{1}$. Products of the $u_{(k,1)}$ are tensor products of factors e_1 and $\mathbf{1}$, traceless unless all factors are $\mathbf{1}$. Commuting involutions whose nonempty products are traceless have all 2^n joint signs with equal multiplicity $8^n/2^n = 4^n$, and j minus signs give $m(n-2j)$. (iii) $J e_1 = e_2 e_3 e_1 = T$ on the kth factor. (iv) Proposition 3.1(ii); Γ_n anticommutes with every generator of $\mathrm{Cl}(3n, 3n)$. □

At general masses (i) and (ii) are Proposition 1.2(vi), restated; the multiplicities, the clocks (iii) and the step (iv) are fixed (e3-2a and e3-2b, exact over signed permutations at $n \leq 5$, eigenvalue counts in floating point at $n \leq 3$; e3-6c). At $n = 2$, (i) is the Dirac duplication of Proposition 8.11(iv) of [2]. So rung n read through its vectors carries one Dirac mass, $\sqrt{n}\, m$, along the unit vector $n^{-1/2} \sum_k E_{(k,1)}$ of $\mathbb{R}^{3n,3n}$, with one clock; read through its commuting

copies it carries n masses m with n clocks and threshold nm. The quadrature is the statement that the copies of the graded tower are directions of one momentum space (Section 1.2).

3.3 Spin parity

Let $R_n = \frac{1}{2}\sum_k E_{(k,4)}E_{(k,5)}$, the rotation about the third axis of the diagonal $\mathfrak{l}_{\text{int}}$, and $L_n = \frac{1}{2}\sum_k (E_{(k,1)}E_{(k,2)} - E_{(k,4)}E_{(k,5)})$, the paired rotation of the $(1,2)$ and $(4,5)$ planes in the diagonal $\mathfrak{l}_G$. Both generate rotations of period 2π on vectors.

Proposition 3.3 (Spin parity up the ladder)**.** *(i) R_n is a sum of n commuting terms $\frac{1}{2}b_k$ with $b_k^2 = -\mathbf{1}$. Hence $e^{2\pi R_n} = (-1)^n\mathbf{1}$, and the eigenvalues of $2R_n$ are i times integers congruent to n modulo 2. Under the diagonal $\mathfrak{l}_{\text{int}}$ the whole of S_n has half-integer spin for odd n and integer spin for even n. The Casimir spectra are*

$$\begin{aligned}
&n=1: && C_\Delta = \tfrac{3}{2}\,(8),\\
&n=2: && 0\,(8),\ 3\,(32),\ 4\,(24),\\
&n=3: && \tfrac{3}{2}\,(160),\ \tfrac{11}{2}\,(288),\ \tfrac{15}{2}\,(64),\\
&n=4: && 0\,(160),\ 3\,(1024),\ 4\,(864),\ 8\,(864),\ 9\,(1024),\ 12\,(160),
\end{aligned}$$

where the values $0, \frac{3}{2}, 3, 4, \frac{11}{2}, 8, \frac{15}{2}, 9, 12$ are those of $(0,0)$, $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$, $(\frac{1}{2},\frac{1}{2})$, $(1,0)\oplus(0,1)$, $(1,\frac{1}{2})\oplus(\frac{1}{2},1)$, $(1,1)$, $(\frac{3}{2},0)\oplus(0,\frac{3}{2})$, $(\frac{3}{2},\frac{1}{2})\oplus(\frac{1}{2},\frac{3}{2})$ and $(2,0)\oplus(0,2)$.

(ii) L_n commutes with G_n on vectors and is a sum of $2n$ commuting terms $\pm\frac{1}{2}b$ with $b^2 = -\mathbf{1}$. Hence $e^{2\pi L_n} = +\mathbf{1}$ at every rung. The diagonal $\mathfrak{l}_G$ has Casimir $C_G = 3$ on all of S; $0, 4, 8$ on $4, 24, 36$ dimensions at $n = 2$; and $3, 9, 15$ on $128, 256, 128$ at $n = 3$: the content of $(\frac{1}{2},\frac{1}{2})^{\otimes n}$, integer spin at every rung.

(iii) The central element $-\mathbf{1}$ of $\mathrm{Spin}(3,3)$, which is $e^{\pi e_4 e_5} = e^{\pi e_1 e_2}$ on S, acts on S_n through the diagonal $\mathrm{Spin}(3,3)$ as $(-1)^n$. The rotation through 2π of $\mathfrak{l}_{\text{int}}$ is this element, and that of $\mathfrak{l}_G$ is not.

(iv) For the actions of Theorem 10.8 of [1], with $\mathfrak{l}_{\text{int}}$ on k_I constituents, $\mathfrak{l}_G$ on k_B and trivially on the rest, the rotation through 2π is $(-1)^{k_I}$.

Proof. (i) $(E_{(k,4)}E_{(k,5)})^2 = -E_{(k,4)}^2E_{(k,5)}^2 = -\mathbf{1}$, and bivectors of distinct constituents share no index and commute. So $e^{2\pi R_n} = \prod_k(\cos\pi + b_k\sin\pi)$. Each listed value of $2[j_L(j_L+1) + j_R(j_R+1)]$ is attained by exactly one unordered pair (j_L, j_R), which identifies the modules. (ii) $e_ie_j - e_{i+3}e_{j+3}$ commutes with G (Proposition 7.8(i) of [2]). The $2n$ terms share no index, and $e^{2\pi L_n} = \prod_k(-\mathbf{1})(-\mathbf{1})$. By Proposition 3.1(iv), $S_n \otimes \mathbb{C} \cong (S\otimes\mathbb{C})^{\otimes n} = 2^n(\frac{1}{2},\frac{1}{2})^{\otimes n}$ under the diagonal $\mathfrak{l}_G$, since $S\otimes\mathbb{C} = 2(\frac{1}{2},\frac{1}{2})$ (Proposition 13.12 of [1]); the Clebsch–Gordan series gives the listed values. (iii) The diagonal $\mathrm{Spin}(3,3)$ acts on $S^{\otimes n}$ by $g \mapsto g^{\otimes n}$ (Proposition 3.1(iv)). $e^{2\pi\cdot\frac{1}{2}e_4e_5}$ is $-\mathbf{1}$, while the double rotation $e^{2\pi\cdot\frac{1}{2}(e_1e_2-e_4e_5)} = e^{\pi e_1e_2}e^{-\pi e_4e_5}$ is $+\mathbf{1}$. (iv) Products of the factors of (i) and (ii) over the respective constituents. □

Part (i) restates Proposition 10.9 and Proposition 10.10(i) of [1] (fixed there), with the element and the spectra re-verified (e3-3a, exact at $n \le 5$; e3-3c, floating-point spectra at $n \le 4$). Part (ii) extends Proposition 13.12 of [1] from $n = 1$ to every rung and is fixed (e3-3b, e3-3c). Part (iii) is fixed (e3-3b), and (iv) restates Proposition 10.9 of [1]. The parity of (i) is the Ehrenfest–Oppenheimer rule, a composite of n spin-$\frac{1}{2}$ particles is a fermion iff n is odd (inherited [16]). The rule holds equally for one particle whose internal space is the n-fold tensor product of spinors. So the spin parity does not decide between one particle and n. Under the field-strength reading it does not arise: every rung carries integer spin, $\mathrm{Spin}(3,3) \cong SL(4,\mathbb{R})$ restricted to $\mathfrak{l}_G$ integrates to $SO^+(1,3)$ on every S_n, and the diagonal $\mathfrak{l}_G$ composite is a boson at every rung.

Remark 3.4 (Comparison with Theorem 13.15 of [1])**.** Theorem 13.15 of [1] says that a copy-invariant four-vector in $\Lambda^g(\mathbb{R}^n \otimes \mathbb{R}^{1,3})$ needs n odd and g odd, $n \le g \le 3n$. The statement concerns the algebra and (i) concerns the module, and they are logically independent. Both are products over the n constituents: in (i) each constituent contributes a factor -1 to the 2π rotation, and in the theorem the copy invariant of odd grade needs the Levi-Civita symbol of $SO(n)$ with its n indices. They meet on the Dirac block. At odd n the module contains spin $\frac{1}{2}$, and at $n = 3$ the copy-singlet four-vector Q_μ of grade three builds a gauge-invariant $\mathrm{Cl}(1,3)$ on that block (Theorem 10.6 of [1]). The check e3-6b re-verifies that Q_μ commutes with the interior copy rotations M_{kl} and satisfies $[\Delta_{ab}, Q_c] = \eta_{bc}Q_a - \eta_{ac}Q_b$, and that the $n = 2$ analogue $\sum_\nu \eta_{\nu\nu}(\gamma^{(1)}_\mu\gamma^{(2)}_\nu + \gamma^{(1)}_\nu\gamma^{(2)}_\mu)$ commutes with M_{12} but is not a vector. At even n there is neither a fermion to carry gamma matrices nor a copy-invariant four-vector to build them. Under the diagonal $\mathfrak{l}_G$ neither statement has an analogue: the spin is integral at every rung, and grade one is $n[(1,0)\oplus(0,1)]$, n field strengths and no four-vector (Section 13.5 of [1]). The juxtaposition is interpretive; the facts are restated and fixed.

3.4 A dual pair up the ladder

Theorem 3.5 (The copy algebra and the diagonal $\mathfrak{so}(3,3)$)**.** *(i) On vectors, the commutant of the diagonal $\mathfrak{so}(3,3)$ in $\mathfrak{so}(3n,3n)$ is $\mathfrak{so}(n)\otimes\mathbf{1}_6$, of dimension $\binom{n}{2}$, spanned by the vector actions of the C^{kl}. The commutant of $\mathfrak{so}(n)\otimes\mathbf{1}_6$ together with the copy permutations is $\mathbf{1}_n \otimes \mathfrak{so}(3,3)$ for $n \ge 2$; that of $\mathfrak{so}(n)\otimes\mathbf{1}_6$ alone has dimension 36 at $n = 2$ and 15 for $n \ge 3$. The weights η_{aa} are forced: $\sum_a E_{(k,a)}E_{(l,a)}$ does not commute with the diagonal $\mathfrak{so}(3,3)$.*

(ii) On S_n, C^{kl} is the spin lift of $(e_k e_l^\top - e_l e_k^\top)\otimes\mathbf{1}_6$: $[C^{kl}, E_{(l,a)}] = E_{(k,a)}$ and $[C^{kl}, E_{(k,a)}] = -E_{(l,a)}$. The C^{kl} commute with the fifteen Δ_{ab} and close on $\mathfrak{so}(n)$, with $[C^{kl}, C^{lm}] = C^{km}$. Each is a sum of six commuting terms of square $-\frac{1}{4}$, so $e^{2\pi C^{kl}} = \mathbf{1}$. In Fock form

$$C^{kl} = \sum_{i=1}^{3}(a^+_{k,i}a^-_{l,i} - a^+_{l,i}a^-_{k,i}), \qquad C^{kl}a^+_{l,i}|0\rangle = a^+_{k,i}|0\rangle,$$

so the copy algebra conserves N and moves quanta between constituents.

(iii) As operators on S_n,

$$C_{\mathrm{cop}} + C_{\mathrm{diag}} = \tfrac{3}{4}\,n(n+4)\,\mathbf{1},$$

that is $\frac{15}{4}, 9, \frac{63}{4}, 24, \frac{135}{4}, \frac{81}{2}$ at $n = 1,\dots,6$. The interior identity of Theorem 10.4 of [1], $C^{\mathrm{int}}_{\mathrm{cop}} + C_\Delta = \frac{1}{2}n(n+2)\,\mathbf{1}$, holds at every n. Both constants are $nd(d+n-2)/8$, at $d = 6$ and $d = 4$.

(iv) At $n = 2$ the commutant of the diagonal $\mathfrak{so}(3,3)$ in $M_{64}(\mathbb{R})$ has dimension 14 and is spanned by the words in C^{12} and the Koszul swap τ_K, with $C^{12}\tau_K = -\tau_K C^{12}$: it is the image of the group algebra of $O(2)$. The words in C^{12} alone span 7 dimensions.

Proof. (i) $V = \mathbb{R}^{3,3}$ is absolutely irreducible under $\mathfrak{so}(3,3)$, so by Schur's lemma the commutant of $\mathbf{1}_n \otimes \mathfrak{so}(3,3)$ in $\mathrm{End}(\mathbb{R}^n \otimes V)$ is $\mathrm{End}(\mathbb{R}^n)\otimes\mathbf{1}_6$. An element $A\otimes\mathbf{1}_6$ is η_n-antisymmetric iff $A^\top = -A$. Conversely, $\mathbb{R}^n$ is absolutely irreducible under the group generated by $SO(n)$ and the permutation matrices for $n \ge 2$, and under $\mathfrak{so}(n)$ alone for $n \ge 3$; at $n = 2$ the commutant of $\mathfrak{so}(2)$ is $\mathrm{span}\{\mathbf{1}, \varepsilon\}$, which adds $\varepsilon \otimes \mathrm{Sym}_\eta(V)$, of dimension 21. The unweighted sum lifts $(e_k e_l^\top - e_l e_k^\top)\otimes\eta$, and η does not commute with the boosts. (ii) The spin lift $\sigma(X) = \frac{1}{4}\sum_{A,B} X_{AB}\eta_{BB}E_AE_B$ satisfies $[\sigma(X), E_C] = \sum_A X_{AC}E_A$ and is a Lie algebra homomorphism; the vector matrices commute by (i). The six terms $\frac{1}{2}\eta_{aa}E_{(k,a)}E_{(l,a)}$ share no index, and $(E_{(k,a)}E_{(l,a)})^2 = -\eta^2_{aa}\mathbf{1}$. For the Fock form substitute $E_{(k,i)} = a^+_{k,i} + a^-_{k,i}$, $E_{(k,i+3)} = a^+_{k,i} - a^-_{k,i}$:

n	copy label	C_{cop}	C_{diag}	$\mathfrak{so}(6)$ modules $(\lambda_1,\lambda_2,\lambda_3)$	dimensions	rank in S_n
2	$\lvert j\rvert=0$	0	9	$(1,1,1)$, $(1,1,-1)$	$10+10$	$1\cdot 20=20$
	$\lvert j\rvert=1$	1	8	$(1,1,0)$	15	$2\cdot 15=30$
	$\lvert j\rvert=2$	4	5	$(1,0,0)$	6	$2\cdot 6=12$
	$\lvert j\rvert=3$	9	0	$(0,0,0)$	1	$2\cdot 1=2$
3	$j=0$	0	$\frac{63}{4}$	$(\frac{3}{2},\frac{3}{2},\pm\frac{3}{2})$	$20+20$	$1\cdot 40=40$
	$j=1$	2	$\frac{55}{4}$	$(\frac{3}{2},\frac{3}{2},\pm\frac{1}{2})$	$36+36$	$3\cdot 72=216$
	$j=2$	6	$\frac{39}{4}$	$(\frac{3}{2},\frac{1}{2},\pm\frac{1}{2})$	$20+20$	$5\cdot 40=200$
	$j=3$	12	$\frac{15}{4}$	$(\frac{1}{2},\frac{1}{2},\pm\frac{1}{2})$	$4+4$	$7\cdot 8=56$

Table 6: Joint decomposition of S_n under the copy algebra and the diagonal $\mathfrak{so}(3,3)$ (Theorem 3.5). At $n=2$, j is the $\mathfrak{so}(2)$ weight; at $n=3$, j is the copy isospin. $C_{\mathrm{diag}}=\langle\lambda,\lambda+2\rho\rangle$ with $\rho=(2,1,0)$, so that the spinor has $\frac{15}{4}$ and the vector 5. Ranks are real dimensions of S_n, equal to complex dimensions of $S_n\otimes\mathbb{C}$. Check e3-4d.

$C^{kl}=\sum_i(a^+_{k,i}a^-_{l,i}+a^-_{k,i}a^+_{l,i})$, and $a^-_{k,i}a^+_{l,i}=-a^+_{l,i}a^-_{k,i}$ for $k\neq l$ by the CAR of Proposition 7.2 of [2]. (iii) The proof of Theorem 10.4 of [1] uses neither $d=4$ nor the signature of the interior: with $B^{kl}_{ab}=E_{(k,a)}E_{(k,b)}E_{(l,a)}E_{(l,b)}$ it gives $C_{\mathrm{diag}}=\frac{nd(d-1)}{8}-\frac{1}{2}\mathsf{S}$ and $C_{\mathrm{cop}}=\binom{n}{2}\frac{d}{4}+\frac{1}{2}\mathsf{S}$ with the same cross term $\mathsf{S}=\sum_{k<l}\sum_{a<b}\eta^{aa}\eta^{bb}B^{kl}_{ab}$. At $d=6$ the constant is $\frac{15n}{4}+\frac{3n(n-1)}{2}=\frac{3}{4}n(n+4)$. (iv) Computed: the nullity of $\sum_X \mathrm{ad}_X^{\top}\mathrm{ad}_X$ over the fifteen generators, and the rank of the words. □

Part (i) is fixed (e3-4a, exact over $\mathbb{Q}$ at $n\leq 5$); at $n=3$ its interior analogue is Theorem 10.2 of [1]. Part (ii) restates Proposition 7.3(ii) of [2] at $n=2$ and Proposition 9.6(i) of [2], and is fixed at $n\leq 4$ (e3-4b, e3-4f); the Fock form at $n=2$ is Proposition 2.9(i) with the opposite orientation. Part (iii) is fixed: the constant $\frac{3}{4}n(n+4)$ is new, and the identity was checked exactly as matrices on S_n at $n\leq 4$ and blade by blade in the abstract Clifford algebra at $n\leq 6$; the interior identity is restated and checked in the same two ways (e3-4c). Part (iv) is fixed (e3-4e; nullity in floating point with gap >1, rank exact). The η-weighting is where the signature enters: a copy rotation pairs two generators of the same square, so it is compact (Theorem 10.2 of [1]), and $C_{\mathrm{cop}}\geq 0$.

Howe's skew duality. By Proposition 7.2 of [2], $S_n\otimes\mathbb{C}$ is the exterior algebra on the $3n$ creation operators $a^+_{k,i}$, that is $\Lambda(\mathbb{C}^n\otimes\mathbb{C}^3)$ with k the copy index and i the mode index. The complexified diagonal $\mathfrak{so}(3,3)$ is spanned by the quadratics $\sum_k x_k y_k$ with $x,y\in\{a^{\pm}_{\cdot,i}\}$, the $O(n)$-invariant contractions over the copy index, which form, up to multiples of $\mathbf{1}$ and by the substitution used in (ii), $\Lambda^2\mathbb{C}^3\oplus\mathfrak{gl}(3)\oplus\Lambda^2\mathbb{C}^3\cong\mathfrak{so}(6,\mathbb{C})$. By (ii) the copy rotations are the $\mathfrak{so}(n)\subset\mathfrak{gl}(n)$ acting on the copy index. This is the setting of Howe's skew $(O(n),\mathfrak{so}(2m))$ duality on $\Lambda(\mathbb{C}^n\otimes\mathbb{C}^m)$, here with $m=3$ [18]: the two actions generate mutual commutants, and the module decomposes without multiplicity as $\bigoplus_\lambda\rho_\lambda\otimes\pi_{\lambda'}$, with ρ_λ an irreducible representation of $O(n)$ and $\pi_{\lambda'}$ one of $\mathfrak{so}(6)$ determined by it. The theorem, and the linear relation between the two Casimirs, are inherited; (iii) is its constant and (iv) its commutant at $n=2$ in the tower. The $O(n)$ contains more than $SO(n)$: at $n=2$ the reflection is τ_K, and Section 3.5 shows that the Koszul swaps realise the permutation matrices at every rung.

Proposition 3.6 (Joint decomposition)**.** *The joint spectrum of $(C_{\mathrm{cop}},C_{\mathrm{diag}})$ on S_2 and S_3 is that of Table 6. Every rank is the dimension of the copy module times the sum of the dimensions of the listed $\mathfrak{so}(6)$ modules, and every $\mathfrak{so}(6)$ Casimir equals $\frac{3}{4}n(n+4)-C_{\mathrm{cop}}$. At $n=4$, $C_{\mathrm{cop}}=0,3,4,8,9,12,15,16,19,24$ on $70,560,504,810,1024,200,320,450,144,14$ dimensions, the values $2[j_+(j_++1)+j_-(j_-+1)]$ of the $\mathfrak{so}(4)=\mathfrak{su}(2)\oplus\mathfrak{su}(2)$ modules $(0,0)$, $(\frac{1}{2},\frac{1}{2})$, $(1,0)\oplus(0,1)$, $(1,1)$, $(\frac{3}{2},\frac{1}{2})\oplus(\frac{1}{2},\frac{3}{2})$, $(2,0)\oplus(0,2)$, $(\frac{3}{2},\frac{3}{2})$, $(2,1)\oplus(1,2)$, $(\frac{5}{2},\frac{1}{2})\oplus(\frac{1}{2},\frac{5}{2})$, $(3,0)\oplus(0,3)$, each paired with $\mathfrak{so}(6)$ modules of Casimir $24-C_{\mathrm{cop}}$ whose dimensions reproduce the ranks.*

Proof. At $n = 2, 3$ the minimal polynomial of C_{cop} is checked exactly and the ranks are traces of its Lagrange projectors, in integer arithmetic. The $\mathfrak{so}(6)$ dimensions are the Weyl dimension formula of D_3. At $n = 4$, C_{cop} commutes with N (by (ii)) and is diagonalised on each of the thirteen number sectors. □

This is fixed (e3-4d), and the $n = 3$ copy spectrum restates Proposition 9.6(ii) of [2]. The joint spectra lie on the line of Theorem 3.5(iii). Copy type and $\mathfrak{so}(3,3)$ type determine each other, as j_{cop} and Lorentz type do in Corollary 10.5 of [1]. Two features recur up the ladder. The sector of largest copy Casimir carries the smallest $\mathfrak{so}(3,3)$ module: the trivial module at $n = 2, 4$ and the half-spinors $\mathbf{4}, \bar{\mathbf{4}}$ of one constituent at $n = 3$. And the copy isospin is integral at $n = 2, 3$, while at $n = 4$ the modules of $\mathfrak{so}(4) = \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ include doublets of each factor, always with $j_+ + j_-$ integral, as single-valuedness requires (Theorem 3.5(ii)).

3.5 The symmetric group

Let $\tau_k = \mathbf{1}^{\otimes(k-1)} \otimes \tau_K \otimes \mathbf{1}^{\otimes(n-k-1)}$, with $\tau_K = D\,P$ the Koszul swap of Proposition 7.3(iii) of [2]. For a surplus direction $s \in \{2, 3\}$ let $\Theta_s = E_{(1,s)}E_{(2,s)}\cdots E_{(n,s)}$, and let $C_3 = \sum_{a<b,\,c<d,\,e<f} \varepsilon_{abcdef} L^{ab}L^{cd}L^{ef}$ be the cubic invariant of the diagonal $\mathfrak{so}(3,3)$, with $L^{ab} = \eta^{aa}\eta^{bb}\Delta_{ab}$.

Proposition 3.7 (The Koszul symmetric group)**.** *(i) $\tau_k^2 = \mathbf{1}$, $(\tau_k\tau_{k+1})^3 = \mathbf{1}$ and $\tau_k\tau_l = \tau_l\tau_k$ for $|k - l| > 1$, so the τ_k define an action of $\mathfrak{S}_n$ on S_n. Moreover $\tau_k E_{(j,a)}\tau_k = E_{(s_k j,a)}$ and $\tau_k = \frac{1}{8}\prod_a (E_{(k,a)} - E_{(k+1,a)})$: τ_k is the lift to $\mathrm{Pin}(3n, 3n)$ of the transposition matrix of copies $k, k+1$, a product of six reflections. It commutes with the diagonal $\mathfrak{so}(3,3)$, with N, with H_g and with C_{cop}, and conjugates C^{kl} as the permutation does. On the vacuum $\tau_k|0\rangle = -|0\rangle$.*

(ii) At $n = 3$ the $\mathfrak{S}_3$-isotypic parts have the content

	$C_\Delta = \frac{3}{2}$	$\frac{11}{2}$	$\frac{15}{2}$	$C_{\mathrm{cop}} = 0$	2	6	12
trivial	24	48	16	20	36	20	12
sign	24	48	16	20	36	20	12
standard	112	192	32	0	144	160	32

(real dimensions; the standard rows are twice the multiplicity of the two-dimensional representation). The Dirac block, $C_\Delta = \frac{3}{2}$, contains 24 totally symmetric and 24 totally antisymmetric dimensions.

(iii) Θ_s is invertible, anticommutes with every τ_k, commutes with the diagonal $\mathfrak{l}_{\mathrm{int}}$ and with every C^{kl}, and anticommutes with C_3. It maps the Koszul-symmetric part isomorphically onto the Koszul-antisymmetric part, as modules of $\mathfrak{l}_{\mathrm{int}} \oplus \mathfrak{so}(n)$, and exchanges the two chiralities of $\mathfrak{so}(6)$. At $n = 3$ the eigenvalues of $8C_3$ are $-630, -210, -126$ on the symmetric part of the copy sectors $j = 0, 1, 2$ and $+630, +210, +126$ on the antisymmetric part.

(iv) The Koszul-symmetric part and the Koszul-antisymmetric part of S_n have the same dimension, $\sum_k \binom{n-k+3}{3}\binom{4}{k}$, which is $8, 32, 88, 192, 360$ at $n = 1, \ldots, 5$. Neither is ever zero.

Proof. (i) The relations are checked. $(E_{(k,a)} - E_{(k+1,a)})^2 = 2\eta_{aa}$, and the reflection in this vector exchanges $E_{(k,a)}$ and $E_{(k+1,a)}$; the product over the six directions acts on vectors as the transposition matrix P, because the sign $(-1)^6$ of the twisted adjoint action is $+1$. Hence $\tau_k\sigma(X)\tau_k^{-1} = \sigma(PXP^{-1})$ for every $X \in \mathfrak{so}(\eta_n)$ and $\tau_k v\tau_k^{-1} = Pv$ for every vector. P commutes with $\mathbf{1}_n \otimes \mathfrak{so}(3,3)$ and with K_n, and fixes $\sum_k E_{(k,1)}$, so τ_k commutes with the Δ_{ab}, with $N = \frac{3n}{2} + \sigma(K_n)$ (Lemma 2.4(i)) and with H_g; it permutes the C^{kl} up to sign and so fixes C_{cop}.

(ii) Traces of the isotypic projectors $\frac{\dim\rho}{6}\sum_\sigma \chi_\rho(\sigma)\tau_\sigma$ times the spectral projectors of C_Δ and C_{cop}, in exact integer arithmetic. (iii) Θ_s is a blade. Against $E_{(k,a)} - E_{(k+1,a)}$ it gives the sign $(-1)^n$ for $a \neq s$ and $(-1)^{n-1}$ for $a = s$, so against τ_k the sign is $(-1)^{6n-1} = -1$. The interior bivectors share no index with Θ_s; the terms of C^{kl} share none or two; a diagonal bivector Δ_{sb} shares one, so $\mathrm{Ad}\,\Theta_s$ acts on the diagonal $\mathfrak{so}(3,3)$ as the reflection of the direction s, an outer automorphism, which reverses C_3. (iv) τ_σ permutes the basis tensors of $S^{\otimes n}$ with the Koszul sign of ω-parity, and ω has four even and four odd basis vectors. The image of the symmetriser is spanned by orbit sums, and an orbit contributes iff no odd basis vector repeats, so its dimension is the number of monomials of degree n in four even and four odd variables, $\sum_k \binom{n-k+3}{3}\binom{4}{k}$; for the antisymmetriser exchange the roles of even and odd, which gives the same number. □

Part (i) restates Proposition 7.3(iii) of [2] at $n = 2$ and is fixed at every rung (e3-5a at $n = 3, 4$; e3-5e, relations and vacuum at $n \leq 5$). Parts (ii)–(iv) are fixed (e3-5b; e3-5c, exact at $n \leq 4$, the C_3 eigenvalues in floating point to 10^{-6}; e3-5e). Since τ_k and the natural action $\Lambda(s_k)$ of the transposition on $\Lambda(\mathbb{C}^n \otimes \mathbb{C}^3)$ implement the same automorphism of $\mathrm{Cl}(3n, 3n)$, they differ by a scalar, and the vacuum fixes it: $\tau_\sigma = \mathrm{sgn}(\sigma)\Lambda(\sigma)$. The Koszul swaps are therefore the permutation matrices of Howe's $O(n)$, twisted by the sign character. Part (iii) generalises the mirror statement of Proposition 1.10(ii) to every rung: the Koszul sign does not sort the module by Lorentz type or by copy type, and its two sectors carry the same content up to the reflection of one direction.

Proposition 3.8 (The ungraded n-particle space)**.** *Let P_σ permute the tensor factors of S_n without signs, and let $W_n = \{J^{(1)} = \cdots = J^{(n)}\}$, of dimension $2 \cdot 4^n$, with $J := J^{(1)}|_{W_n}$ as imaginary unit, so that $W_n \cong S \otimes_{\mathbb{C}} \cdots \otimes_{\mathbb{C}} S = (\mathbb{C}^4)^{\otimes_{\mathbb{C}} n}$ (Proposition 1.4(iv)).*

(i) $P_\sigma J^{(k)} P_\sigma^{-1} = J^{(\sigma k)}$, so P_σ preserves W_n and is $\mathbb{C}$-linear there.

(ii) The totally antisymmetric part of W_n under P_σ is $\Lambda^n_{\mathbb{C}}\mathbb{C}^4$, of real dimension $2\binom{4}{n} = 8, 12, 8, 2, 0$ at $n = 1, \ldots, 5$, and the totally symmetric part is $\mathrm{Sym}^n_{\mathbb{C}}\mathbb{C}^4$, of real dimension $2\binom{n+3}{3} = 8, 20, 40, 70, 112$.

(iii) Under the diagonal $\mathfrak{l}_{\mathrm{int}}$, $\Lambda^2_{\mathbb{C}}\mathbb{C}^4$ has $C_\Delta = 0$ on 4 and 3 on 8 dimensions, $2(0,0) \oplus (\frac{1}{2}, \frac{1}{2})$; $\Lambda^3_{\mathbb{C}}\mathbb{C}^4$ has $C_\Delta = \frac{3}{2}$ on all 8, a Dirac spinor; and $\Lambda^4_{\mathbb{C}}\mathbb{C}^4$ has $C_\Delta = 0$, a scalar.

Proof. (i) $P_\sigma(A_1 \otimes \cdots \otimes A_n)P_\sigma^{-1}$ permutes the factors. (ii) Exact traces of $\frac{1}{n!}\sum_\sigma \mathrm{sgn}(\sigma)P_\sigma$ and of $\frac{1}{n!}\sum_\sigma P_\sigma$, composed with the projector $2^{1-n}\prod_k(\mathbf{1} - J^{(k)}J^{(k+1)})$ onto W_n, as signed permutations. (iii) The projector onto the antisymmetric part, as an exact integer matrix, satisfies the stated polynomial identities with C_Δ. Since $\mathbb{C}^4 = (\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$, $\Lambda^2 = \Lambda^2(\frac{1}{2}, 0) \oplus (\frac{1}{2}, 0) \otimes (0, \frac{1}{2}) \oplus \Lambda^2(0, \frac{1}{2})$, $\Lambda^3\mathbb{C}^4 \cong (\mathbb{C}^4)^* \otimes \Lambda^4\mathbb{C}^4$ and $\Lambda^4\mathbb{C}^4 = \det$. □

The representation theory is inherited [7]; its realisation on W_n is fixed (e3-5d). With the plain exchange and antisymmetry, rung n on W_n carries n identical Dirac fermions in one momentum mode, and the Pauli principle holds: at most four, since a Dirac spinor has four states, so there is no antisymmetric state at $n \geq 5$ (inherited [10]). The two exchanges therefore differ at every rung, not only at $n = 2$ (Proposition 1.9). The Koszul exchange is the one the algebra supplies, since it permutes the vectors of $\mathrm{Cl}(3n, 3n)$. Its antisymmetric sector has dimension 88 at $n = 3$ and contains all three Lorentz types, and at $n = 5$ it has dimension 360 where the plain one is empty. It is not a statistics label. The plain exchange is a statistics label, but only on the constructed space W_n, with the constructed generators $u_{(k,a)}$ as gamma matrices.

3.6 The ladder table

Table 7 collects the rung-by-rung data. The Lie algebra rows are fixed (e3-6a, exact over $\mathbb{Q}$ at $n \leq 5$) and restate Theorem 6.1, Theorem 10.8 and Proposition 13.14 of [1]; the row of

structure	$n=1$	2	3	4	5	source
$\dim V_n$, $\dim S_n$	6, 8	12, 64	18, 512	24, 8^4	30, 8^5	—
$\mathrm{Cl}(3n,3n)$		$M_{8^n}(\mathbb{R})$ at every rung				e3-1a ($n \le 4$)
Fock modes	3	6	9	12	15	e3-1c
spectator centraliser $\mathfrak{so}(3n-1,3n-3)$	1	28	91	190	325	ES17 Thm 6.1; e3-6a
commutant of diagonal $\mathfrak{l}_{\mathrm{int}}$, $\mathfrak{so}(n)\oplus\mathfrak{so}(2n)$	1	7	18	34	55	ES17 Thm 10.8; e3-6a
commutant of diagonal $\mathfrak{l}_G$, $\mathfrak{so}(n,\mathbb{C})$	0	2	6	12	20	ES17 Prop. 13.14; e3-6a
its compact part, $\mathfrak{so}(n)$	0	1	3	6	10	e3-6a
rotor algebra $\mathfrak{u}(n)$ (in $\mathfrak{so}(6n)$)	1	4	9	16	25	ES17 Prop. 13.14; e3-6a
copy algebra, commutant of diagonal $\mathfrak{so}(3,3)$	0	1	3	6	10	e3-4a
$C_{\mathrm{cop}} + C_{\mathrm{diag}}$	$\frac{15}{4}$	9	$\frac{63}{4}$	24	$\frac{135}{4}$	e3-4c
graded rest mass	m	$\sqrt{2}\,m$	$\sqrt{3}\,m$	$2m$	$\sqrt{5}\,m$	e3-2a
ungraded rest energies $\lvert m(n-2j)\rvert$	m	$0, 2m$	$m, 3m$	$0, 2m, 4m$	$m, 3m, 5m$	e3-2b
2π rotation, diagonal $\mathfrak{l}_{\mathrm{int}}$	$-\mathbf{1}$	$+\mathbf{1}$	$-\mathbf{1}$	$+\mathbf{1}$	$-\mathbf{1}$	e3-3a
2π rotation, diagonal $\mathfrak{l}_G$	$+\mathbf{1}$	$+\mathbf{1}$	$+\mathbf{1}$	$+\mathbf{1}$	$+\mathbf{1}$	e3-3b
copy-invariant four-vector	yes	no	yes	no	yes	ES17 Thm 13.15; e3-6b
selected imaginary unit (diagonal $\mathfrak{l}_{\mathrm{int}}$)	yes	no	yes	no	yes	Proposition 1.4(iv)
invariant number of quanta (diagonal $\mathfrak{l}_G$)	no	yes	no	yes	no	Theorem 2.8
invariant Grassmann mass kernel	T_1	plane	S_3	none	T_5	Theorem 6.11(ii)
copy isospin	—	integral	integral	$j_\pm \in \frac{1}{2}\mathbb{Z}$	—	RE Prop. 9.6; e3-4d
projector P_*	—	—	yes	—	—	ES17 Thm 8.1
$\Lambda^n_{\mathbb{C}}$ of the Dirac module on W_n	Dirac	$2(0,0)\oplus(\frac{1}{2},\frac{1}{2})$	Dirac	$(0,0)$	0	e3-5d
Koszul-antisymmetric part of S_n	8	32	88	192	360	e3-5e

Table 7: The ladder. Dimensions of Lie algebras are real. "Plane" and "none": at $n \equiv 2 \pmod 4$ the invariant Grassmann kernels form a plane, at $n \equiv 0$ there is none; T_n and S_n there are the temporal and spatial volumes of Theorem 6.11, not the spinor module. The copy isospin is that of the six-direction copy algebra.

copy-invariant four-vectors restates Theorem 13.15 of [1], with the case $n = 3$ and the absence at $n = 2$ re-verified (e3-6b). Read as a whole, the table says which structures single out which rungs.

No rung is singled out by the algebra. $\mathrm{Cl}(3n,3n) = M_{8^n}(\mathbb{R})$, its module is S_n, and every Lie algebra row is a polynomial in n. The first structure with a period in n is the symmetry of the invariant bilinear forms on S_n, of period four (Theorem 6.11(ii)). It is Chevalley's pattern for spinors of $\mathbb{R}^{3n,3n}$, the dimension $6n$ running through the residues $6, 4, 2, 0$ modulo 8 with period four in n (inherited [3]). Through it the $\mathfrak{so}(3n,3n)$-invariant Grassmann mass kernel is unique exactly at odd n. That is the only selection the algebra makes by itself, and it does not distinguish $n = 3$ from the other odd rungs.

Odd rungs are singled out by the interior Lorentz action. Under the diagonal $\mathfrak{l}_{\mathrm{int}}$, odd n is where the composite is a fermion (Proposition 3.3(i)), where an imaginary unit is selected (Proposition 1.4(iv)), where copy-invariant four-vectors and the Dirac block exist (Theorem 13.15 and Proposition 10.10 of [1]). *Even rungs are singled out by the binding action*: only there does a Lorentz-invariant number of quanta exist under the diagonal $\mathfrak{l}_G$ (Theorem 2.8). *The third rung is singled out by added projectors and by the low-rank coincidences of the diagonal algebra*: the chiral Pati–Salam projector P_* is constructed at $n = 3$ (Theorem 8.1 of [1]), the diagonal connection algebra there is $\mathfrak{so}(3) \oplus \mathfrak{so}(6) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(4)$ with colour $\mathfrak{su}(3)$ inside $\mathfrak{so}(6)$, and

the copy isospin is integral (Proposition 10.10(iii) of [1]; Proposition 9.6 of [2]). At $n = 5$ the Dirac block carries one Standard-Model family per copy multiplet (Proposition 10.10(iv) of [1]). Apart from the period of the forms, every one of these selections comes from a Lorentz action, a projector or a gauge-algebra identification added to the rung. None comes from the ladder.

3.7 The step from n to $n+1$, and what going up the ladder means

What the step preserves. By Proposition 3.1(ii) the step keeps rung n as the subalgebra $\mathrm{Cl}(3n, 3n) \otimes \mathbf{1}$, with all of its operators. Under the spectator action the interior four-plane stays in the first factor, so the tower is one Lorentz frame with nested internal centralisers (Proposition 6.3 of [1]). The diagonal module is tensored with S (Proposition 3.1(iv)), and the dual pair $(O(n), \mathfrak{so}(3,3))$ sits inside $(O(n+1), \mathfrak{so}(3,3))$.

What the step adds. Three Fock modes, $N_{n+1} = N_n \otimes \mathbf{1} + \mathbf{1} \otimes N_1$ (Proposition 3.1(iii)). One frame: $V_{n+1} = V_n \oplus V$, which under the diagonal $\mathfrak{l}_G$ is one more field strength $(1,0) \oplus (0,1)$ and under the diagonal $\mathfrak{l}_{\mathrm{int}}$ one more potential $(\frac{1}{2}, \frac{1}{2})$ with two scalars (Section 13.5 of [1]). One copy direction: the copy space $\mathbb{R}^n$ becomes $\mathbb{R}^{n+1}$, the copy algebra gains the n rotations $C^{k,n+1}$, and the Casimir constant grows by $\frac{3}{4}(2n+5)$ (Theorem 3.5(iii); e3-6c). The new constituent has its own commuting copy, $(\Gamma_n \otimes \mathbf{1}) E_{(n+1,a)}$.

What the step changes. The graded mass grows in quadrature and the ungraded energies add (Proposition 3.2(iv)). The spin parity under the diagonal $\mathfrak{l}_{\mathrm{int}}$ flips, and that under the diagonal $\mathfrak{l}_G$ does not (Proposition 3.3). Every parity-dependent selection of Section 3.6 switches on or off.

One particle or several. Read together, the statements of this section answer the question in four sentences. (1) As given by its vectors, every rung carries one particle: one Dirac mass $\sqrt{n}\, m$ along a unit vector of $\mathbb{R}^{3n,3n}$ with one clock, and copies that are the components of a vector of the copy group $O(n)$. That group is the Howe dual of the diagonal $\mathfrak{so}(3,3)$, its type is locked to the $\mathfrak{so}(3,3)$ type by $C_{\mathrm{cop}} + C_{\mathrm{diag}} = \frac{3}{4} n(n+4)$, and its Lie algebra moves quanta between constituents without creating any. So the copy index behaves as a flavour index of one particle, and going up the ladder adds a flavour direction and three modes to that particle (fixed statements, read interpretively; Proposition 3.2 and Theorem 3.5). (2) The Lorentz content does not decide. The parity $(-1)^n$ under the interior action is the parity of n spin-$\frac{1}{2}$ particles and equally that of one particle in an n-fold tensor product, and under the field-strength action every rung is bosonic (fixed; inherited rule; Proposition 3.3). (3) A rung carries n particles in the sense of ordinary quantum mechanics only with the structures of Section 1, now needed at every rung: the commuting copies $u_{(k,a)} = \Gamma^{(1)} \cdots \Gamma^{(k-1)} E_{(k,a)}$ (fixed as an algebra), the additive Hamiltonian H_u with its n clocks, the common complex space W_n with $J|_{W_n}$, and the plain exchange with antisymmetry (constructed; the unit selected up to sign at odd n only). With them the rung carries n identical Dirac fermions in one mode, with the Pauli bound $n \leq 4$ (fixed; inherited; Proposition 3.8). (4) The exchange the algebra supplies is the Koszul $\mathfrak{S}_n$, the permutation matrices of $O(n)$ up to sign, and it is not a statistics label: its two sectors are mirror images and never empty (fixed; Proposition 3.7).

Which process goes up the ladder. Within a rung, the copy algebra hops quanta among the n constituents and conserves N (Theorem 3.5(ii); Proposition 2.9). Between rungs there is no process in the algebra: rung $n+1$ is not an element or a state of rung n, and the embedding $\mathrm{Cl}(3n, 3n) \otimes \mathbf{1}$ makes rung n the part of rung $n+1$ that does not see the last constituent. In the two-particle reading the step is the tensor embedding $\psi \mapsto \psi \otimes \phi_0$, which needs a state ϕ_0 that nothing selects (constructed, as at $n = 2$). Whether a dynamical mechanism selects W_n, the additive Hamiltonian and a rung is open.

statement	class	checks
Proposition 3.1(i): $\mathrm{Cl}(3n,3n) = M_{8^n}(\mathbb{R})$, volume $\omega^{\otimes n}$, $(-1)^N = (-1)^n\Gamma_n$	fixed in the realisation; classification inherited	e3-1a
Proposition 3.1(ii), (iii): step formulas, eight copies, commutant $\mathbf{1} \otimes M_8(\mathbb{R})$; Fock modes	restated (RE Props. 7.1, 7.2; Proposition 1.1)	e3-1b, e3-1c
Proposition 3.1(iv): diagonal bivectors additive, $S_{n+1} = S_n \otimes S$	fixed	e3-6c
Proposition 3.2(i), (ii): $\pm\sqrt{n}\,m$; $m(n-2j)$ with $\binom{n}{j}4^n$; exact dressing	restated (Proposition 1.2(vi), RE Prop. 8.11(iv)); multiplicities fixed	e3-2a, e3-2b
Proposition 3.2(iii), (iv): one clock against n clocks; the step	fixed	e3-2a, e3-2b, e3-6c
Proposition 3.3(i), (iv): $e^{2\pi R_n} = (-1)^n$; $(-1)^{k_I}$; C_Δ spectra	restated (ES17 Props. 10.9, 10.10(i)); spectra fixed	e3-3a, e3-3c
Proposition 3.3(ii), (iii): $e^{2\pi L_n} = +\mathbf{1}$; $(\frac{1}{2},\frac{1}{2})^{\otimes n}$; central element	fixed (extends ES17 Prop. 13.12)	e3-3b, e3-3c
Remark 3.4: spin parity against copy-invariant four-vectors	restated (ES17 Thms 10.6, 13.15); comparison interpretive	e3-6b
Theorem 3.5(i): mutual commutants on vectors; η-weighting forced	fixed	e3-4a
Theorem 3.5(ii): spin lifts, $\mathfrak{so}(n)$, single-valued, Fock form	fixed; restated at $n = 2, 3$ (RE Props. 7.3, 9.6)	e3-4b, e3-4f
Theorem 3.5(iii): $C_{\mathrm{cop}} + C_{\mathrm{diag}} = \frac{3}{4}n(n+4)$; interior identity	fixed (constant new); interior restated (ES17 Thm 10.4)	e3-4c
Theorem 3.5(iv): commutant at $n = 2$ is the image of $\mathbb{R}[O(2)]$	fixed	e3-4e
Howe's skew duality $(O(n), \mathfrak{so}(6))$ on $\Lambda(\mathbb{C}^n \otimes \mathbb{C}^3)$	inherited [18]	e3-4f
Proposition 3.6, Table 6: joint decomposition at $n = 2, 3, 4$	fixed; $n = 3$ copy spectrum restated (RE Prop. 9.6)	e3-4d
Proposition 3.7(i): Koszul $\mathfrak{S}_n$, Pin lift, commutation, vacuum sign	fixed; restated at $n = 2$ (RE Prop. 7.3(iii))	e3-5a, e3-5e
Proposition 3.7(ii)–(iv): $\mathfrak{S}_3$ content; mirror Θ_s; sector dimensions	fixed	e3-5b, e3-5c, e3-5e
Proposition 3.8: W_n, $\Lambda^n_{\mathbb{C}}\mathbb{C}^4$, $\mathrm{Sym}^n_{\mathbb{C}}\mathbb{C}^4$, Lorentz content	fixed in the realisation; inherited (exterior powers, Pauli)	e3-5d
Table 7: the ladder $n = 1, \ldots, 5$	fixed; restated (ES17 Thms 6.1, 8.1, 10.8, 13.15, Prop. 13.14; RE Prop. 9.6)	e3-6a, e3-6b, as listed
Section 3.6: no rung selected by the algebra; odd, even, third rung	interpretive (drawn from fixed statements); period four inherited	—
Section 3.7: step; one particle or several; process	fixed parts as cited; reading interpretive; tensor embedding constructed; mechanism open	e3-1b, e3-1c, e3-6c

Table 8: Claims of Section 3, their classes and their checks, which are `PASS` lines of `verify/e3_ladder.py`. "Restated": fixed in [1], [2] or Section 1, re-verified here.

4 The b-field up the ladder

Section 9 of [2] studied the field that acts between the constituents of one rung, the copy field $b = B \otimes \mathbf{1}_6$, and settled what it does to rates and lines on two and three constituents. This section follows that field up the ladder, $n = 2, 3, 4, \ldots$, and reads it in the Fock picture of Sections 1.4 and 2.4. The field always acts between the constituents of one rung $\mathrm{Cl}(3n, 3n)$; going up the ladder means increasing n.

The answers, with their classes, are these. (1) Every compact binding-compatible field is the second quantisation of a one-quantum Hamiltonian. It conserves the number N of quanta, moves quanta between the constituents, and its many-quantum lines are sums of one-quantum lines. The copy boosts create pairs, one quantum in each of the two constituents they join (fixed; Section 4.1). (2) On n constituents with distinct rates and all couplings nonzero, the invariants of the copy field are the $|B_{kl}|$ and the signs of the products of B around the cycles of the complete graph. Exactly $(n-1)(n-2)/2$ of these signs are independent, and all of them can be taken time-odd. The Jarlskog-type determinant $\det[H, K]$ has time parity $(-1)^n$, so it detects time reversal only at odd n (fixed; Section 4.2). (3) At $n = 4$ the copy group is $SO(4)$, and each of its two $\mathfrak{su}(2)$ factors has half-integer isospin exactly on the states with an odd number of quanta. The copy inversion acts as $(-1)^N$. Above $n = 4$ the same holds for the $\mathfrak{so}(4)$ of any four constituents, with the quanta in those four. At no rung does a spinor of the copy algebra occur (fixed; Section 4.3). (4) For rates of multiplicities $n_1 + \cdots + n_r = n$ the removable part of the copy $\mathfrak{so}(n)$ is $\bigoplus_i \mathfrak{so}(n_i)$, and $\sum_{i<j} n_i n_j$ directions are physical (fixed; the Higgs-like reading is interpretive; Section 4.4). (5) For a rotor with rates (κ_k), a frame bracket between copies i, j and k is admitted only if $\kappa_k = \kappa_i + \kappa_j$. A harmonic ladder of rates $\kappa_k = k\kappa_1$ therefore admits non-abelian, nilpotent frames. From $n = 4$ on the Jacobi identity among the frame brackets is a quadratic condition (fixed; the reading is constructed; Section 4.5). Section 4.6 says what the field can and cannot do for the emergence of particles. Identifiers e4-n refer to the `PASS` lines of `verify/e4_bfield_ladder.py` (Table 11).

Notation. As in Section 9 of [2]: $V_n = \mathbb{R}^n \otimes \mathbb{R}^6$, η_n, $G_n = \mathbf{1}_n \otimes G$, $z = (z_1, \ldots, z_n) \in \mathbb{C}^{3n}$ with $z_k = \mathbf{B}_k + i\mathbf{E}_k$, copy rotations $C^{kl} = (E_{lk} - E_{kl}) \otimes \mathbf{1}_6$, copy boosts $(E_{lk} - E_{kl}) \otimes G$, copy field $b = B \otimes \mathbf{1}_6$ with $B \in \mathfrak{so}(n)$, rates $K = \mathrm{diag}(\kappa_1, \ldots, \kappa_n)$, $\tilde{K} = K \otimes \mathbf{1}_3$, and, for a compact binding-compatible b acting on z as the real antisymmetric R, the frame Hamiltonian $H = \tilde{K} - iR$ of Theorem 9.3 of [2]. σ is the spin lift to $S_n = (\mathbb{R}^8)^{\otimes n}$, with $[\sigma(X), v] = Xv$ for $v \in V_n$, and $\hat{C}^{kl} = \sigma(C^{kl})$. The $3n$ modes are $p = (k, i)$, $k \le n$, $i \le 3$, with $a^{\pm}_{k,i} = \frac{1}{2}(E_{(k,i)} \pm E_{(k,i+3)})$ (Proposition 7.2 of [2]). $N_{k,i} = a^+_{k,i} a^-_{k,i}$, $N^{(k)} = \sum_i N_{k,i}$ is the number in constituent k, $N_{[i]} = \sum_k N_{k,i}$ the number in direction i, and $N = \sum_k N^{(k)}$. For a complex $3n \times 3n$ matrix M we write $\mathcal{N}(M) = \sum_{p,q} M_{pq}\, a^+_p a^-_q$. The first basis vector of the realisation is the Fock vacuum $|0\rangle$ (e4-1a).

4.1 Quanta between constituents: transfer, not creation

Theorem 1.7 shows, at every rung, that the compact binding-compatible fields are exactly the binding-compatible fields that conserve N, and Propositions 2.6 and 2.9 give the hop and the pair at $n = 2$. The theorem below adds the second-quantised form, which says how the quanta move, and the bookkeeping by constituent and by direction at $n = 3$.

Theorem 4.1 (The field between constituents, second quantised)**.** *(i) Let X be a paired rotation, acting on z as the real antisymmetric $R \in \mathfrak{so}(3n)$. Then $\sigma(X) = \mathcal{N}(R)$. Hence $\sigma(X)$ commutes with N, and $\sigma(X)\, a^+_q |0\rangle = \sum_p R_{pq}\, a^+_p |0\rangle$.*

(ii) For the copy field $b = B \otimes \mathbf{1}_6$ and rates K, the operator $\mathsf{H} = \sum_k \kappa_k N^{(k)} - i\sigma(b)$ on $S_n \otimes \mathbb{C}$ equals $\mathcal{N}(H)$, $H = (K - iB) \otimes \mathbf{1}_3$. It is Hermitian and commutes with N. Its block on

$\{N=1\}$, *in the basis* $a_p^+|0\rangle$, *is* H, *and its eigenvalues on* $\{N=m\}$ *are the sums of* m *eigenvalues of* H *with distinct indices.*

(iii) *At* $n=3$, $\hat{C}^{kl}=\sum_i(a^+_{l,i}a^-_{k,i}-a^+_{k,i}a^-_{l,i})$. *It commutes with* N, *with* $N^{(k)}+N^{(l)}$, *with* $N^{(m)}$ *for* $m\notin\{k,l\}$ *and with every* $N_{[i]}$, *and not with* $N^{(k)}$. *The three lifts close on* $\mathfrak{so}(3)$ *as on vectors,* $[\hat{C}^{12},\hat{C}^{13}]=\hat{C}^{23}$.

(iv) *At* $n=3$, $\sigma((E_{lk}-E_{kl})\otimes G)=-\sum_i(a^+_{k,i}a^+_{l,i}+a^-_{l,i}a^-_{k,i})$. *It joins only sectors whose* N *differs by* 2, *in both directions, and it commutes with* $N^{(k)}-N^{(l)}$ *and with* $N^{(m)}$, $m\notin\{k,l\}$: *every pair it creates has one quantum in constituent* k *and one in constituent* l.

(v) *At* $n=3$ *the quarter turn* $U=e^{\pi\hat{C}^{12}/2}=\frac{1}{8}\prod_a(\mathbf{1}+2t_a)$, *with* $t_a=-\frac{1}{2}\eta_{aa}E_{(1,a)}E_{(2,a)}$ *the six commuting terms of* $\hat{C}^{12}$, *is orthogonal, commutes with* N *and* $N^{(3)}$, *and satisfies* $UN^{(1)}U^{-1}=N^{(2)}$. *For* $\kappa_1\neq\kappa_2$ *it changes the rate operator* $\sum_k\kappa_kN^{(k)}$ *and keeps its spectrum.*

Proof. (i) A paired rotation acts in the same way on the temporal directions and on their partners, so $X(2a_q^\pm)=\sum_p R_{pq}\,2a_p^\pm$, and $[\sigma(X),a_q^\pm]=\sum_p R_{pq}a_p^\pm$. The CAR give $[\mathcal{N}(R),a_q^+]=\sum_p R_{pq}a_p^+$ and $[\mathcal{N}(R),a_q^-]=-\sum_p R_{qp}a_p^-$, which is the same because $R^\top=-R$. So $\sigma(X)-\mathcal{N}(R)$ commutes with every generator, and S_n is irreducible, so it is a multiple of $\mathbf{1}$. Both operators are traceless: $\sigma(X)$ is a sum of products of two anticommuting generators, and $\mathrm{Tr}(a_p^+a_q^-)=0$ for $p\neq q$, while $R_{pp}=0$. Hence they are equal. Each term $a_p^+a_q^-$ conserves N, and $a_q^-|0\rangle=0$ gives the action on one quantum. (ii) $N^{(k)}=\mathcal{N}(E_{kk}\otimes\mathbf{1}_3)$, and by (i) $\sigma(b)=\mathcal{N}(B\otimes\mathbf{1}_3)$, so $\mathsf{H}=\mathcal{N}(H)$. $N^{(k)}$ is real diagonal and $\sigma(b)$ real antisymmetric, so H is Hermitian. For $H=WDW^\dagger$ with W unitary, $b_r^\pm=\sum_p W^{(\pm)}_{pr}a_p^\pm$ ($W^{(+)}=W$, $W^{(-)}=\bar{W}$) satisfy the CAR and $\mathsf{H}=\sum_r d_r\,b_r^+b_r^-$; its eigenvectors are $b^+_{r_1}\cdots b^+_{r_m}|0\rangle$ with $r_1<\cdots<r_m$. (iii) is (i) for $R=(E_{lk}-E_{kl})\otimes\mathbf{1}_3$. Each term moves one quantum between (k,i) and (l,i). (iv) A copy boost anticommutes with K_3, so its lift is a combination of a^+a^+ and a^-a^- terms (Lemma 2.4(ii)). It maps the plane of mode (k,i) to that of mode (l,i), which fixes the modes paired, and the coefficients are computed. (v) $t_a^2=-\frac{1}{4}$, so $e^{\theta\hat{C}^{12}}=\prod_a(\cos\frac{\theta}{2}+2\sin\frac{\theta}{2}\,t_a)$. At $\theta=\frac{\pi}{2}$, $\mathrm{Ad}\,U$ maps $E_{(1,a)}$ to $\pm E_{(2,a)}$ and $E_{(2,a)}$ to $\mp E_{(1,a)}$, and fixes $E_{(3,a)}$. $N^{(k)}$ is a function of the products $E_{(k,i)}E_{(k,i+3)}$, so $UN^{(1)}U^{-1}=N^{(2)}$, and conjugation keeps spectra. □

The theorem is fixed. Part (i) is the standard realisation of $\mathfrak{so}(3n)$ on the exterior algebra by one-body operators [3, 9], inherited, and its realisation in the tower is checked exactly for all 15 and 36 basis elements at $n=2,3$ (e4-1b) and for the copy rotations at $n=4$ (e4-3a). Part (ii) is checked exactly on $\{N=1\}$ at $n=3$, rates $(1,2,4)$ and $(B_{12},B_{23},B_{31})=(\frac{1}{2},\frac{1}{3},\frac{1}{5})$, and the additivity on $\{N=2\}$ and $\{N=3\}$ to 10^{-9} (e4-1e). Parts (iii) to (v) are exact (e4-1c, e4-1d, e4-1f); in (iv) a random paired boost of $\mathfrak{so}(9,\mathbb{C})$ also changes N by exactly ±2. The external i in H is constructed, as in Proposition 2.9(iii). Under $z\mapsto\sum_p z_pa_p^+|0\rangle$ the one-quantum sector of $e^{-i\mathsf{H}t}$ is the frame flow $z(t)=e^{-iHt}z(0)$ of Theorem 9.3(i) of [2].

Corollary 4.2 (Transfer, not creation)**.** *Let* b *be a compact element of* $\mathrm{Cent}(G_n)\cap\mathfrak{so}(3n,3n)$, *the class of gauge-admissible fields between the constituents (Theorem 9.3(ii) of [2]; Corollary 2.4 of [1]).*

(i) $e^{t\sigma(b)}$ *conserves* N *and acts on* $\{N=m\}$ *as the* m*-th exterior power of* e^{tR}. *It moves quanta between modes and constituents, and never creates or annihilates one.*

(ii) *The binding-compatible elements that change* N *are the paired boosts, among them the copy boosts. Each of their terms creates or annihilates one pair, and a copy boost puts one quantum in each of the two constituents it joins.*

(iii) Let $\hat{U}$ lift a compact η_n-isometry commuting with G_n, which acts on z as a unitary W. A change of frame by $\hat{U}$ takes the rate operator $\mathcal{N}(\tilde{K})$ to $\hat{U}\mathcal{N}(\tilde{K})\hat{U}^{-1} = \mathcal{N}(W\tilde{K}W^{-1})$. This operator commutes with N and has the same spectrum on every sector $\{N = m\}$, the sums of m rates. The numbers $N^{(k)}$ are not kept.

Proof. (i) The proof of Theorem 4.1(i) holds at every n, and $\mathcal{N}(R)$ is the derivation of the exterior algebra induced by R, so its exponential is $\Lambda(e^{tR})$. (ii) is Theorem 1.7(ii), with the pairing of Theorem 4.1(iv) at $n = 3$ and of Proposition 2.6(iii) at $n = 2$. (iii) $\operatorname{Ad}\hat{U}$ acts on vectors as the isometry, hence on the a_p^+ as W, which gives the formula. A compact isometry commuting with G_n commutes with $K_n = -\eta_n G_n$, so $\hat{U}$ commutes with N, and conjugation keeps spectra. This is the Fock form of Theorem 9.3(iii) of [2], and Theorem 4.1(v) exhibits a frame change that moves $N^{(1)}$ to $N^{(2)}$. □

The corollary is fixed. It answers the first question precisely. The gauge-admissible field between constituents transfers quanta and never creates them. It does not change a rate either (Theorem 9.3(iii) of [2]). Which constituent carries a quantum depends on the frame when the copy rotation is gauged, as Proposition 2.9 found at $n = 2$. How many quanta there are, and the multiset of their rates, does not. Pair creation across constituents needs a copy boost, which is a metric connection and not a gauge field (Proposition 9.4 of [2]). Because H is one-body, its many-quantum lines carry no interaction energy. The field mixes quanta; it does not bind them.

4.2 Time-odd invariants on n constituents

Theorem 9.5 of [2] found on three constituents one invariant that two do not carry, the sign of $J = B_{12}B_{23}B_{31}$, odd under interior time reversal. For a cycle $\gamma = (k_1, \dots, k_L)$ of distinct copy indices write $\Pi_\gamma B = B_{k_1k_2}B_{k_2k_3}\cdots B_{k_Lk_1}$; reversing the orientation multiplies it by $(-1)^L$.

Theorem 4.3 (The copy field on n constituents)**.** *Let $n \geq 3$, let the rates be distinct, and let $b = B \otimes \mathbf{1}_6$ with every $B_{kl} \neq 0$.*

(i) The endomorphisms of V_n commuting with G_n, the diagonal $\mathfrak{l}_G$ and $K \otimes G$ are $\sum_k (a_k + b_k G)$ on copy k, and the η_n-isometries among them are the 2^n copy reflections $\operatorname{diag}(s_1, \dots, s_n) \otimes \mathbf{1}_6$, $s_k = \pm 1$, under which $B_{kl} \mapsto s_k s_l B_{kl}$.

(ii) Two fields with the same $|B_{kl}|$ are related by a copy reflection iff $\Pi_\gamma B$ has the same sign for both on every cycle γ. The signs of the $c_n = \frac{1}{2}(n-1)(n-2)$ triangles $B_{1k}B_{kl}B_{l1}$, $2 \leq k < l \leq n$, are independent and determine all the others. The invariants of the field are the $\frac{1}{2}n(n-1)$ magnitudes and these c_n signs.

(iii) Interior time reversal, $(K, B) \mapsto (K, -B)$ (Theorem 9.5(iv) of [2]), multiplies $\Pi_\gamma B$ by $(-1)^L$. It flips all c_n triangle signs, and no copy reflection undoes it.

(iv) The triangle invariants are spectral. With P_k the spectral projectors of K, polynomials in K, and $H = K - iB$ on the copy index, $\operatorname{tr}(P_k H P_l H P_m H) = iB_{kl}B_{lm}B_{mk}$ for every triangle (k, l, m).

Proof. (i) The commutant of G_n and the diagonal $\mathfrak{l}_G$ is $\{A \otimes \mathbf{1} + A' \otimes G\}$ with $A, A' \in \mathfrak{gl}(n)$ (Proposition 9.2 of [2]). $[A \otimes \mathbf{1} + A' \otimes G, K \otimes G] = [A, K] \otimes G - [A', K] \otimes \mathbf{1}$, so A and A' commute with K and are diagonal. The isometry condition per copy is $a_k b_k = 0$, $a_k^2 - b_k^2 = 1$, as in the proof of Theorem 9.5(iii) of [2]. (ii) Encode the signs of B_{kl}, $k < l$, as $\sigma \in \mathbb{Z}_2^E$, E the edges of the complete graph K_n. A reflection adds δs, $(\delta s)_{kl} = s_k + s_l$, so the orbits are the cosets of $\operatorname{im}\delta$. $\ker\delta$ is the constants because K_n is connected, so $\operatorname{rank}\delta = n - 1$ and there are $2^{|E|-n+1} = 2^{c_n}$ orbits. The functions of σ that are constant on cosets are the characters $\sigma \mapsto \sum_{e\in\gamma}\sigma_e$ with γ in the annihilator of $\operatorname{im}\delta$, the edge sets of even degree at every vertex, which form the cycle space [22]. Its dimension is $|E| - n + 1 = c_n$, and the fundamental cycles of the star at vertex 1, the

triangles $(1, k, l)$, form a basis. A character pairs perfectly with the cosets, so the triangle signs separate the orbits. The oriented product $\Pi_\gamma B$ differs from the unoriented one by a sign fixed by γ. (iii) $-B$ adds the all-ones vector to σ, and its character on γ is $L \bmod 2$. A triangle is odd, so the all-ones vector is not in $\operatorname{im}\delta$. (iv) $P_k H P_l = H_{kl} E_{kl}$ and $H_{kl} = -iB_{kl}$ for $k \neq l$, and $P_k = \prod_{j\neq k}(K - \kappa_j)/(\kappa_k - \kappa_j)$ for distinct rates. □

The theorem is fixed. Part (i) is exact at $n = 4$, rates $(1, 2, 3, 5)$ (e4-2a), and at $n = 3$ it is Theorem 9.5(iii) of [2]. Parts (ii) and (iii) are checked exhaustively on all sign patterns at $n = 3, 4, 5, 6$, with $c_n = 1, 3, 6, 10$ and $\operatorname{rank}\delta = n-1$ over $\mathbb{Z}_2$ (e4-2b, e4-2c). Part (iv) is symbolic at $n = 4$, where the product of the four triangle invariants is a product of squares, because every edge of K_4 lies on two triangles (e4-2d). The count c_n is the cycle rank of K_n, inherited from graph theory. It is also the number of rephasing-invariant phases of n families of quarks [20, 21] (inherited). Here the phases are fixed at $\pm\pi/2$ because the field is real (Section 9.5 of [2]), so they reduce to signs. On n constituents the copy field carries c_n time-odd signs, its Wilson loops around the triangles.

Remark 4.4 (Couplings that vanish)**.** For a general coupling graph Γ_B, with $|E'|$ nonzero couplings and $c(\Gamma_B)$ connected components, the same argument gives $2^{|E'|-n+c(\Gamma_B)}$ classes of sign patterns, so the number of independent cycle signs is the cycle rank $|E'| - n + c(\Gamma_B)$ (exhaustive over all 64 graphs at $n = 4$, rank over $\mathbb{Z}_2$ for all graphs at $n = 5, 6$; e4-2f). A chain of constituents carries no sign. A ring of n constituents carries one sign, odd under time reversal iff n is odd, in agreement with the bipartite criterion of Theorem 9.5(iv) of [2]. This is fixed.

Proposition 4.5 (The Jarlskog-type determinant up the ladder)**.** *On the copy index* $[H, K] = iD$ *with* $D_{kl} = (\kappa_k - \kappa_l)B_{kl}$*, a real symmetric matrix with zero diagonal. Hence* $\det[H, K] = i^n \det D$*, and* $\det D$ *is the sum over derangements* π *of* $(-1)^{n-\#\mathrm{cycles}(\pi)} \prod_k D_{k\pi(k)}$*. Under* $B \mapsto -B$ *it changes by* $(-1)^n$*.*

(i) $n = 3$*:* $\det[H, K] = 2iJ(\kappa_1 - \kappa_2)(\kappa_1 - \kappa_3)(\kappa_2 - \kappa_3)$*.*

(ii) $n = 4$*:* $\det[H, K] = D_{12}^2 D_{34}^2 + D_{13}^2 D_{24}^2 + D_{14}^2 D_{23}^2 - 2(D_{12}D_{23}D_{34}D_{41} + D_{12}D_{24}D_{43}D_{31} + D_{13}D_{32}D_{24}D_{41})$*. It is even under time reversal and contains no triangle.*

(iii) $n = 5$*:* $\det[H, K]$ *is odd under time reversal; every cycle cover contains a* 5*-cycle or a triangle.*

Proof. $[K - iB, K] = -i[B, K]$ and $[B, K]_{kl} = -(\kappa_k - \kappa_l)B_{kl}$. D is a product of two antisymmetric factors entrywise, hence symmetric. The Leibniz expansion of a matrix with zero diagonal runs over derangements, and $-D$ multiplies each term by $(-1)^n$. At $n = 4$ the derangements are the three double transpositions and the six 4-cycles. At $n = 5$ they are the 24 five-cycles and the 20 products of a transposition and a 3-cycle. □

The proposition is fixed. Part (i) restates Theorem 9.5(ii) of [2], and the identities are symbolic at $n = 3, 4, 5$, with the cycle-cover sum checked at $n = 4, 5$ (e4-2e). The determinant measures cycle covers. A 2-cycle contributes the magnitude $(\kappa_k - \kappa_l)^2 B_{kl}^2$, and a cycle of length $L \geq 3$ contributes twice the product of the rate differences around it times $\Pi_\gamma B$. At $n = 4$ the 4-cycle products are time-even: $B_{12}B_{23}B_{34}B_{41} = -B_{12}B_{23}B_{31} \cdot B_{13}B_{34}B_{41}/B_{13}^2$ is fixed by the product of two triangle signs. So $\det[H, K]$ cannot tell (K, B) from $(K, -B)$ at $n = 4$, although the field is time-odd. The time-odd invariants there are the triangle traces of Theorem 4.3(iv). Jarlskog's determinant [19] measures time reversal on three constituents and, among the rungs, only at odd n; the triangle traces measure it at every $n \geq 3$.

4.3 Copy isospin at four constituents

On two and three constituents the copy isospin is integral (Proposition 9.6 of [2]), and Remark 9.8(d) of [2] notes that this does not extend to four (d7-6d). We determine the copy isospin content of S_4, of dimension 4096. Write

$$I_1^{\pm} = \tfrac{1}{2}(\hat{C}^{23} \pm \hat{C}^{14}), \qquad I_2^{\pm} = \tfrac{1}{2}(\hat{C}^{31} \pm \hat{C}^{24}), \qquad I_3^{\pm} = \tfrac{1}{2}(\hat{C}^{12} \pm \hat{C}^{34})$$

(the self-dual and anti-self-dual combinations; $J_a^{\pm}$ in the script), and let $j_{\pm}(j_{\pm}+1)$ be the eigenvalues of the Casimirs $-\sum_a (I_a^{\pm})^2$.

Theorem 4.6 (Copy isospin on four constituents)**.** *Let $n = 4$.*

(i) Each $\hat{C}^{kl}$ equals $\sum_i (a^+_{l,i}a^-_{k,i} - a^+_{k,i}a^-_{l,i})$, commutes with every $N_{[i]}$, and is a sum of six commuting terms of square $-\frac{1}{4}$, so $e^{2\pi\hat{C}^{kl}} = +\mathbf{1}$. The six close on $\mathfrak{so}(4)$, and the copy group acting on S_4 is $SO(4)$.

(ii) $[I_a^+, I_b^-] = 0$ and $[I_1^{\pm}, I_2^{\pm}] = I_3^{\pm}$ cyclically, on vectors and on S_4. On vectors $4(I_a^{\pm})^2 = -\mathbf{1}$, so the copy index is the module $(\frac{1}{2},\frac{1}{2})$. On S_4, $e^{2\pi I_3^+} = e^{2\pi I_3^-} = (-1)^N$. Hence $I_3^+ = \frac{1}{2}(\hat{C}^{12} + \hat{C}^{34})$ has eigenvalues $i\mu$ with μ half-integral exactly on the sectors of odd N. In particular $e^{2\pi I_3^+} a^+_{1,1}|0\rangle = -a^+_{1,1}|0\rangle$.

(iii) As an $\mathfrak{so}(4)$-module, $S_4 \otimes \mathbb{C} \cong (\Lambda\mathbb{C}^4)^{\otimes 3}$ with $\Lambda\mathbb{C}^4 = 2(0,0) \oplus 2(\frac{1}{2},\frac{1}{2}) \oplus (1,0) \oplus (0,1)$. The modules (j_+, j_-) and their multiplicities are those of Table 9. Always $j_+ - j_- \in \mathbb{Z}$. The modules with half-integral j_+ fill exactly the odd-N sectors, 2048 dimensions. The one-quantum sector is $(\frac{1}{2},\frac{1}{2}) \otimes \mathbb{R}^3$.

even N: (j_+, j_-)	$(0,0)$	$(1,0)$	$(0,1)$	$(1,1)$	$(2,0)$	$(0,2)$	$(2,1)$	$(1,2)$	$(3,0)$	$(0,3)$
multiplicity	70	84	84	90	20	20	15	15	1	1
odd N: (j_+, j_-)	$(\frac{1}{2},\frac{1}{2})$	$(\frac{3}{2},\frac{1}{2})$	$(\frac{1}{2},\frac{3}{2})$	$(\frac{3}{2},\frac{3}{2})$	$(\frac{5}{2},\frac{1}{2})$	$(\frac{1}{2},\frac{5}{2})$				
multiplicity	140	64	64	20	6	6				

Table 9: The copy isospin content of $S_4 \otimes \mathbb{C}$ under $\mathfrak{so}(4) = \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-$. Each half has dimension 2048 (e4-3c).

Proof. (i) The one-body form is Theorem 4.1(i), and the six terms are those of Proposition 9.6(i) of [2] with $|D| = 6$. A module of $\mathfrak{so}(4)$ on which the 2π rotation in one plane acts as $\mathbf{1}$ integrates to $SO(4)$, not to its double cover. (ii) On vectors $2I_3^{\pm} = (C^{12} \pm C^{34})$ squares to $-\mathbf{1}$, so $e^{2\pi I_3^{\pm}} = -\mathbf{1}$ on V_4. Its lift is the product of the twelve commuting factors $e^{2\pi u}$, where u runs over the terms of $I_3^{\pm}$, $u^2 = -\frac{1}{16}$ and $e^{2\pi u} = 4u$. That product is a multiple of the volume element of $\mathrm{Cl}(12,12)$, and the computation gives $(-1)^N$. On a sector where $e^{2\pi I_3^+} = -\mathbf{1}$ the eigenvalues $i\mu$ of I_3^+ satisfy $e^{2\pi i\mu} = -1$. (iii) S_4 is the exterior algebra on the modes, and the modes of direction i span a copy $\mathbb{C}_i^4$ of the copy index, so $S_4 \otimes \mathbb{C} = \Lambda(\mathbb{C}_1^4 \oplus \mathbb{C}_2^4 \oplus \mathbb{C}_3^4) = \bigotimes_i \Lambda\mathbb{C}_i^4$. By (i), $\mathfrak{so}(4)$ acts on each factor as the derivation induced by the vector module. $\Lambda^0 = \Lambda^4 = (0,0)$, $\Lambda^1 \cong \Lambda^3 = (\frac{1}{2},\frac{1}{2})$ and $\Lambda^2 = \Lambda^2_+ \oplus \Lambda^2_- = (1,0) \oplus (0,1)$ [39]. The Clebsch–Gordan rule gives the table. Odd degree in $\Lambda\mathbb{C}^4$ means half-integral (j_+, j_-), and the parity of a product is the parity of the total degree, $(-1)^N$. □

The theorem is fixed. (i) and (ii) are exact on S_4, with sparse integer matrices (e4-3a, e4-3b), and $e^{2\pi\hat{C}^{12}} = +\mathbf{1}$ also on S_5, of dimension 32768 (e4-3a). (iii) is checked on the joint spectrum of the two Casimirs, block-diagonalised by the direction numbers $N_{[i]}$, which commute with every copy rotation; the eigenvalues are integral to 10^{-8} and the multiplicities agree with the character

(e4-3c). The decomposition rule is inherited. The same rule at $n = 3$, $\Lambda\mathbb{C}^3 = 2(j{=}0) \oplus 2(j{=}1)$, gives the Casimir spectrum $j = 0, 1, 2, 3$ on $40, 216, 200, 56$ dimensions of Proposition 9.6(ii) of [2], recomputed here (e4-3d, restated).

Corollary 4.7 (No spinor of the copy algebra at any rung)**.** *For every n, $e^{2\pi\hat{C}^{kl}} = +\mathbf{1}$ on S_n, and $S_n \otimes \mathbb{C} \cong (\Lambda\mathbb{C}^n)^{\otimes 3}$ as a module of the copy $\mathfrak{so}(n)$. The copy algebra acts through $SO(n)$, and for $n \geq 3$ no spinor module of $\mathfrak{so}(n)$ occurs. At $n = 4$ the half-integral isospin is that of the vector $(\frac{1}{2}, \frac{1}{2})$ and of its odd exterior powers. At $n = 6$, where $\mathfrak{so}(6) \cong \mathfrak{su}(4)$, the fundamental* $\mathbf{4}$ *does not occur.*

Proof. The six terms of $\hat{C}^{kl}$ commute and square to $-\frac{1}{4}$ at every n, and the proof of Theorem 4.6(iii) holds with $\mathbb{C}^n$ in place of $\mathbb{C}^4$. Exterior powers of the vector module have integral weights in the orthonormal basis, and the spinor modules have half-integral ones [6]. □

The corollary is fixed, checked on S_4 and S_5 (e4-3a). It makes the doublets of Remark 9.8(d) of [2] precise. At $n = 4$ a state carries half-integral copy isospin under either $\mathfrak{su}(2)$ factor iff it has an odd number of quanta; the copy inversion $-\mathbf{1} \in SO(4)$, the rotation through π in the planes $(1, 2)$ and $(3, 4)$, acts as $(-1)^N$. This is possible without any double cover because $\mathfrak{so}(4)$ is not simple and its vector is a doublet of each factor. Up the ladder the copy algebra $\mathfrak{so}(n)$ is simple again ($n \geq 5$), so it has no $\mathfrak{su}(2)$ ideal. Its subalgebra $\mathfrak{so}(4)$ on any four constituents still has doublets: on S_5, $e^{2\pi I_3^{\pm}}$ built on the constituents $(1, 2, 3, 4)$ or $(2, 3, 4, 5)$ is $(-1)^{N'}$, with N' the number of quanta in those four constituents (e4-3e). Half-integral copy isospin therefore first appears at the fourth rung and persists above it, under the $\mathfrak{so}(4)$ of four constituents. The doublets are frames and their odd products, not spinors of a spin cover.

4.4 Rate patterns and the breaking of the copy symmetry

Proposition 4.8 (Rate patterns)**.** *Let the rates take r distinct values with multiplicities $n_1 + \cdots + n_r = n$.*

(i) The removable part of the copy $\mathfrak{so}(n)$, the part commuting with $K \otimes G$, is $\bigoplus_i \mathfrak{so}(n_i)$, block diagonal on the equal-rate constituents, of dimension $\sum_i \frac{1}{2} n_i(n_i - 1)$. The non-removable directions number $\frac{1}{2}n(n-1) - \sum_i \frac{1}{2} n_i(n_i - 1) = \sum_{i<j} n_i n_j$. Within the compact binding-compatible $\mathfrak{so}(3n)$ the removable part is $\bigoplus_i \mathfrak{so}(3n_i)$, and the non-removable directions number $9 \sum_{i<j} n_i n_j$.

(ii) The quadratic form $B \mapsto \|[K, B]\|^2$ on $\mathfrak{so}(n)$ is diagonal in the basis C^{kl} (on the copy index), with eigenvalue $2(\kappa_k - \kappa_l)^2$.

Proof. (i) $[K, E_{lk} - E_{kl}] = (\kappa_l - \kappa_k)(E_{lk} + E_{kl})$, so $X \in \mathfrak{so}(n)$ commutes with K iff $X_{kl} = 0$ whenever $\kappa_k \neq \kappa_l$, and likewise for R and $\tilde{K}$ on $\mathbb{C}^{3n}$. Removability is commutation with $K \otimes G$ (Theorem 9.3(v) of [2]). The count uses $n^2 = \sum n_i^2 + 2\sum_{i<j} n_i n_j$. (ii) The matrices $E_{lk} + E_{kl}$, $k < l$, are orthogonal, of norm squared 2. □

The proposition is fixed, exact for every partition of $n = 2, \ldots, 5$ in $\mathfrak{so}(n)$ and of $n = 2, 3, 4$ in $\mathfrak{so}(3n)$ (e4-4a), and at rates $(1, 1, 2, 5)$, where the form is $\mathrm{diag}(0, 2, 32, 2, 32, 18)$ on $C^{12}, C^{13}, C^{14}, C^{23}, C^{24}, C^{34}$ (e4-4b). At $n = 3$ it restates the dimensions $36, 18, 9$ and $3, 1, 0$ of Theorem 9.3(v) of [2]. With the factor 6 of $\mathbf{1}_6$, (ii) is the norm $6\sum_{k\neq l}(\kappa_k - \kappa_l)^2 B_{kl}^2$ of Remark 9.8(e) of [2].

Remark 4.9 (The breaking pattern)**.** The rate matrix is a symmetric tensor of the copy $\mathfrak{so}(n)$, and its stabiliser is $\bigoplus_i \mathfrak{so}(n_i)$; its stabiliser in $O(n)$ is $O(n_1) \times \cdots \times O(n_r)$, which for distinct rates is the group of copy reflections of Theorem 4.3(i). Read as in Remark 9.8(e) of [2], this is the pattern of a gauge symmetry broken by a symmetric-tensor vacuum value [23], with $\sum_{i<j} n_i n_j$ massive directions of masses proportional to $|\kappa_k - \kappa_l|$ [35, 36, 37]. Up the ladder, constituents

with equal rates form a multiplet whose internal copy rotations are removable, so which of them carries a quantum is a gauge choice. Constituents with different rates are distinguishable, and the field between them is physical. The reading is interpretive, with the caveat of [2]: b is a constant connection without kinetic term, and the rates are inputs.

4.5 A ladder of rates admits a non-abelian frame

Section 9.2 of [2] shows that a common binding action lets the rotor act by a non-central $A \in \mathfrak{u}(n)$. It also shows that the frame then admits equivariant brackets $[X,Y]_k = \sum_{i,j} \Gamma^k_{ij} \beta_0(X_i, Y_j)$, with $\beta_0(x,y) = z(x) \times z(y)$ the complex cross product and $\Gamma^k_{ij} = \Gamma^k_{ji} \in \mathbb{C}$, subject to $\Gamma^k_{ij}(\omega_k - \omega_i - \omega_j) = 0$, where A acts on copy k as $i\omega_k$. For $A = K \otimes G$ the ω_k are the rates. With a copy field, A acts on z as iH, and the condition applies to the eigenvalues of H, the lines.

Proposition 4.10 (Rotor-compatible frame brackets)**.** *(i) The real bilinear maps $V \times V \to V$ equivariant under $\mathfrak{l}_G$ form the two-dimensional space spanned by β_0 and $G\beta_0$, and all of them are antisymmetric. The equivariant antisymmetric brackets on V_n are the maps above, a space of real dimension $n^2(n+1) = 12, 36, 80$ for $n = 2, 3, 4$.*

(ii) $A = K \otimes G$ is a derivation of the bracket iff $\Gamma^k_{ij} = 0$ unless $\kappa_k = \kappa_i + \kappa_j$. The rotor-compatible brackets have real dimension 0, 0, 2 for rates $(1,1)$, $(1,3)$, $(1,2)$; 0, 4, 0, 4 for $(1,1,1)$, $(1,2,3)$, $(1,3,5)$, $(1,2,4)$; and 8 for $(1,2,3,4)$. For the harmonic ladder $\kappa_k = k$ the dimension is $2\sum_{k \le n} \lfloor k/2 \rfloor = 2, 4, 8, 12, 18$ for $n = 2, \dots, 6$.

(iii) Rates $(1,2,3)$. The rotor-compatible brackets are $[X,Y]_2 = a\,\beta_0(X_1, Y_1)$, $[X,Y]_3 = b\,(\beta_0(X_1, Y_2) + \beta_0(X_2, Y_1))$, $(a,b) \in \mathbb{C}^2$. Every one of them satisfies the Jacobi identity. For $ab \neq 0$ the frame is a nilpotent Lie algebra $V_1 \oplus V_2 \oplus V_3$, graded by the rate and generated by V_1, with lower central series of real dimensions $18, 12, 6, 0$.

(iv) Rates $(1,2,4)$. The rotor-compatible brackets are $[X,Y]_2 = a\,\beta_0(X_1, Y_1)$, $[X,Y]_3 = b\,\beta_0(X_2, Y_2)$. For $X, Y \in V_1$ and $Z \in V_2$ the Jacobiator is $ab\,\beta_0(\beta_0(X,Y), Z) \in V_3$, so the Jacobi identity holds iff $ab = 0$.

(v) Rates $(1,2,3,4)$, with $a = \Gamma^2_{11}$, $b = \Gamma^3_{12}$, $c = \Gamma^4_{13}$, $d = \Gamma^4_{22}$. For $X, Y \in V_1$ and $Z \in V_2$ the Jacobiator is $(ad - bc)\,\beta_0(\beta_0(X,Y), Z) \in V_4$, and the Jacobi identity holds on all of V_4 iff $ad = bc$.

Proof. (i) is an exact null-space computation on the 216 coefficients of a bilinear map (e4-5a); G acts as i, so $G\beta_0 = i\beta_0$, and the maps are complex bilinear. A bracket is a choice of Γ^k_{ij} for each k and each unordered pair $\{i,j\}$, $n \cdot \frac{1}{2}n(n+1)$ complex numbers. (ii) On z, A acts as $i\kappa_k$ on copy k, and the bracket is complex bilinear, so $A[X,Y]_k - [AX,Y]_k - [X,AY]_k = i(\kappa_k - \kappa_i - \kappa_j)\Gamma^k_{ij}\beta_0(X_i, Y_j)$. For $\kappa_k = k$ the pairs $i \le j$ with $i + j = k$ number $\lfloor k/2 \rfloor$. (iii) The only composite is $[[X,Y]_2, Z]_3$, so the Jacobiator is $ab \sum_{\mathrm{cyc}} \beta_0(\beta_0(X_1, Y_1), Z_1)$, which vanishes by the Jacobi identity of the complex cross product, the bracket of $\mathfrak{so}(3,\mathbb{C})$. $\beta_0(V_1, V_1) = V_2$ and $\beta_0(V_1, V_2) = V_3$ for $ab \neq 0$. The Jacobiator is trilinear and alternating, and inputs of rates r_1, r_2, r_3 give an output of rate $r_1 + r_2 + r_3$, so each of its components is fixed by inputs whose rates add to the target rate. (iv) The only composite is $[[X,Y]_2, Z]_3 = ab\,\beta_0(\beta_0(X_1, Y_1), Z_2)$, with inputs of rates $1, 1, 2$. For $X, Y \in V_1$ and $Z \in V_2$ the other two cyclic terms vanish, because $V_1 \times V_2$ brackets to zero. For $ab = 0$ there is no composite. (v) The V_3-component is as in (iii) and vanishes. For $X, Y \in V_1$ and $Z \in V_2$, $[[X,Y], Z]_4 = ad\,\beta_0(\beta_0(X,Y), Z)$, while $[[Y,Z], X]_4 + [[Z,X], Y]_4 = bc\,(\beta_0(\beta_0(Y,Z), X) + \beta_0(\beta_0(Z,X), Y)) = -bc\,\beta_0(\beta_0(X,Y), Z)$ by the Jacobi identity of β_0. □

The proposition is fixed. (i) restates the space of Section 9.2 of [2] at $n = 2$ (e4-5a). (ii) is exact, block by block, for the listed rates and for the harmonic ladder up to $n = 4$; for $n = 5, 6$ the

count is the number of resonant blocks (e4-5b). At $n = 2$ it restates the bracket of $\mathrm{diag}(1,2) \otimes G$ of Section 9.2 of [2]. (iii) to (v) are checked on random samples to 10^{-10}: antisymmetry, $\mathfrak{l}_G$-equivariance, the rotor as a derivation and the Jacobi identity; at $(1,2,4)$ and $(1,2,3,4)$ the samples off the stated conditions fail (e4-5c, e4-5d, e4-5e). That every rotor-compatible frame is nilpotent when all rates are positive is inherited: A is then a nonsingular derivation, and a Lie algebra with a nonsingular derivation is nilpotent [24].

The reading is constructed. A harmonic ladder of rates admits a non-abelian frame, from $n = 3$ on and already for $\mathrm{diag}(1,2)$ at $n = 2$. Its bracket joins two frames whose rates add into the frame of the summed rate, so the frame algebra is graded by rate. With three constituents on the harmonic ladder the Lie frame brackets form the linear space $\mathbb{C}^2$. With four the Jacobi identity cuts out the cone $ad = bc$ in $\mathbb{C}^4$, and on a longer ladder the same condition recurs, because the V_4-component of the Jacobiator involves only the blocks with target rate at most 4. The resonant pattern $(1,2,4)$ admits a non-abelian frame only with one of its two blocks, $ab = 0$. Nothing selects such a rotor. The paper's closure takes $A = G_n$ and an abelian frame (Theorem 17.7 and Proposition 17.18 of [1]). The non-central rotor, its rate ladder and the bracket are admitted, not selected. The resonance condition $\omega_k = \omega_i + \omega_j$ is that of three-wave mixing and sum-frequency generation [25], a parallel only (interpretive). Read with $mc^2 = \hbar\kappa$, the bracket composes frames whose rest energies add (interpretive).

4.6 Answer: what the field between constituents can and cannot do

Remark 4.11 (What the field does up the ladder)**.** (a) *It moves quanta and keeps their number* (fixed). A gauge-admissible field between the constituents acts as a one-body operator. It hops quanta between constituents and mixes them, and its N-quantum lines are sums of one-quantum lines (Theorem 4.1 and Corollary 4.2).

(b) *It does not take one to two* (fixed). It creates no quantum and changes no rate. Pair emergence across constituents needs a copy boost, which is not a gauge field (Corollary 4.2(ii); Proposition 9.4 of [2]). Because it is one-body, it binds no quanta into composites. If two particles are to emerge from one, the process lies outside the compact field between constituents.

(c) *It carries time-odd invariants that grow with the rung* (fixed). From $n = 3$ on it carries $\frac{1}{2}(n-1)(n-2)$ independent triangle signs, each odd under interior time reversal. The triangle traces measure them at every n, Jarlskog's determinant only at odd n (Theorem 4.3 and Proposition 4.5).

(d) *It acts on doublets from the fourth rung on* (fixed). At $n = 4$ the copy isospin is half-integral exactly on odd N, and above it under the $\mathfrak{so}(4)$ of any four constituents, on an odd number of quanta in them. At every rung the copy algebra acts through $SO(n)$, so no spinor of it occurs (Theorem 4.6 and Corollary 4.7).

(e) *It is physical exactly between unequal rates* (fixed). The physical directions number $\sum_{i<j} n_i n_j$; equal-rate constituents stay gauge-equivalent (Proposition 4.8). The reading as symmetry breaking by the rate matrix is interpretive (Remark 4.9).

(f) *It admits non-abelian frames on a harmonic ladder of rates* (constructed). The rotor must be non-central and its rates resonant, and nothing in the closure selects either (Proposition 4.10).

Up the ladder, then, the field between constituents distinguishes constituents by their rates, carries more time-odd structure, and from $n = 4$ on acts on doublets. It never changes the number of quanta that the rung carries. In the sense of Section 1.4, a rung carries N quanta, and the compact b-field keeps N fixed. It decides where the quanta are, not how many there are. This reading is interpretive, drawn from the fixed statements above.

n	copy algebra, group on S_n	physical directions, distinct rates	independent signs c_n	half-integral copy isospin	rotor brackets, rates $1, \dots, n$
2	$\mathfrak{so}(2)$, $SO(2)$	1	0	no; charges $i\{-3, \dots, 3\}$	2
3	$\mathfrak{so}(3)$, $SO(3)$	3	1	no; $j = 0, \dots, 3$	4
4	$\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, $SO(4)$	6	3	yes, exactly on odd N	8, Jacobi: $ad = bc$
5	$\mathfrak{so}(5)$, $SO(5)$	10	6	under the $\mathfrak{so}(4)$ of four constituents; no spinor of $\mathfrak{so}(5)$	12
6	$\mathfrak{so}(6) \cong \mathfrak{su}(4)$, $SO(6)$	15	10	as for $n = 5$; no $\mathbf{4}$ of $\mathfrak{su}(4)$	18

Table 10: The field between constituents, rung by rung. The $n = 2, 3$ isospin entries restate Proposition 9.6 of [2]. The others follow from Theorems 4.3 and 4.6, Corollary 4.7, and Propositions 4.8 and 4.10; the rotor column counts real dimensions.

statement	class	checks
Theorem 4.1(i): paired rotations are one-body operators $\mathcal{N}(R)$, conserve N	fixed (structure inherited, realisation fixed)	e4-1a, e4-1b
Theorem 4.1(ii): $\mathsf{H} = \mathcal{N}(H)$; $N = 1$ block is H; additive lines	fixed; external i constructed	e4-1e
Theorem 4.1(iii), (iv): copy rotations and copy boosts at $n = 3$ by constituent and direction	fixed	e4-1c, e4-1d
Theorem 4.1(v): quarter turn moves $N^{(1)}$ to $N^{(2)}$, keeps the rate spectrum	fixed	e4-1f
Corollary 4.2: transfer, not creation; rates gauge invariant	fixed (partly restated: Theorem 1.7; RE Thm 9.3(iii))	e4-1b, e4-1d, e4-3a
Theorem 4.3(i): residual group is the copy reflections	fixed (restated at $n = 3$: RE Thm 9.5(iii))	e4-2a
Theorem 4.3(ii), (iii): invariants are $\lvert B_{kl}\rvert$ and c_n triangle signs; all time-odd	fixed (cycle space inherited)	e4-2b, e4-2c
Theorem 4.3(iv): triangle traces	fixed	e4-2d
Remark 4.4: cycle rank of the coupling graph	fixed	e4-2f
Proposition 4.5: $\det[H, K] = i^n \det D$, parity $(-1)^n$, $n = 3, 4, 5$	fixed ($n = 3$ restated: RE Thm 9.5(ii))	e4-2e
parallel with the $(n-1)(n-2)/2$ phases of n quark families	inherited	—
Theorem 4.6(i), (ii): $SO(4)$, $(\frac{1}{2}, \frac{1}{2})$, $e^{2\pi I_3^{\pm}} = (-1)^N$	fixed	e4-3a, e4-3b
Theorem 4.6(iii), Table 9: $(\Lambda\mathbb{C}^4)^{\otimes 3}$, doublets on odd N	fixed (decomposition rule inherited)	e4-3c
$n = 3$ Casimir spectrum $40, 216, 200, 56$	restated (RE Prop. 9.6(ii))	e4-3d
Corollary 4.7: no spinor of the copy algebra at any rung; doublets of the $\mathfrak{so}(4)$ of four constituents at $n = 5$	fixed	e4-3a, e4-3e
Proposition 4.8: removable $\bigoplus \mathfrak{so}(n_i)$, $\sum_{i<j} n_i n_j$ physical; mass form	fixed ($n = 3$ restated: RE Thm 9.3(v))	e4-4a, e4-4b
Remark 4.9: breaking by the rate matrix	interpretive	—
Proposition 4.10(i), (ii): equivariant brackets, resonance condition, dimensions	fixed ($n = 2$ restated: RE Sec. 9.2)	e4-5a, e4-5b
Proposition 4.10(iii)–(v): Jacobi at $(1, 2, 3)$, $(1, 2, 4)$, $(1, 2, 3, 4)$; nilpotency	fixed; nilpotency inherited	e4-5c, e4-5d, e4-5e
Section 4.5: a harmonic ladder admits a non-abelian frame	constructed; three-wave parallel interpretive	e4-5b, e4-5c
Remark 4.11, Table 10: what the field can and cannot do	(a)–(e) fixed, (f) constructed; summary interpretive	—

Table 11: Claims of Section 4, their classes and their checks, which are `PASS` lines of `verify/e4_bfield_ladder.py`.

5 The closure at two constituents

Section 17 of [1] closes the force law $\dot{p} = \kappa G^{\top} p$ into the Lie–Poisson system of $\mathfrak{s}_n = \mathbb{R} \ltimes_{G_n} V_n$, in which the rate is the angular velocity of a rotor. It treats one constituent, several constituents driven by one rotor (Proposition 17.18 there), and three constituents (Theorems 17.19–17.24 there): the closure symmetry, the absence of a common Lorentz algebra, the part of the Pati–Salam algebra that commutes with the binding root, what survives each projector, its chirality and anomalies, and the freedom left in the inertia. The case $n = 2$ is not treated there. Parts of it are in [2]: the closure symmetry $\mathfrak{so}(6, \mathbb{C})$ (Proposition 7.8), the balance surface (Theorem 8.1), the balanced pairs (Proposition 8.4) and the pair invariant M_{pair} (Theorem 8.6). This section supplies the $n = 2$ analogue of every item, with the five Lorentz actions at $n = 2$ in place of the two projectors, which exist only at $n = 3$, and asks whether $n = 2$ is a closure rung in its own right. The answers, with their classes, are these. (1) The closure symmetry $\mathfrak{so}(6, \mathbb{C})$ is the complex orthogonal algebra of $q_{\text{tot}} = z_1 \cdot z_1 + z_2 \cdot z_2$. It mixes the two field strengths, keeps no quadratic invariant except q_{tot}, and does not preserve the pair invariant: on each of its orbits M_{pair} takes every value in $[\frac{1}{2}|q_{\text{tot}}|, \infty)$ (fixed; Section 5.1). (2) There is no common Lorentz algebra, and under $\mathfrak{l}_G^{(1)}$ the spinor module carries no Weyl spinor (fixed, partly restated; Section 5.2). (3) There is no Pati–Salam algebra at $n = 2$ (obstructed). The binding root leaves an $\mathfrak{su}(2)$ of the spectator gauge algebra, compact, non-abelian and kinetically admissible, which acts on the Weyl block with integer isospin and vector-like; of the diagonal connection algebra it leaves the copy rotation; of the mixed rung it leaves nothing (fixed; Section 5.3). (4) The inertia freedom is $(2, 0)$ under the closure symmetry, as at $n = 1$ and $n = 3$, and it is $(14, 10)$ or $(18, 18)$ under the binding-compatible gauge parts (fixed, partly restated; Section 5.4). (5) The balance surface is connected and of codimension two. It is transversally neutral, since the imbalance is a constant of motion; the rate is stable, but the bounded frame orbits are not, and they need $h(0) \neq 0$. With equal inertias a massive balanced pair is never an exchange eigenstate of its momenta (fixed; Section 5.5). (6) Table 12 compares $n = 1, 2, 3$, and Section 5.6 answers the question. Identifiers e5-n refer to the `PASS` lines of `verify/e5_closure_two.py` (Table 13).

Notation. As in Section 8 of [2]. $V_2 = V \oplus V$ has the basis $e_a^{(1)}$, $e_a^{(2)}$, the metric $\eta_2 = \eta \oplus \eta$ and the binding root $G_2 = G \oplus G$. The temporal directions are $e_1^{(k)}, e_2^{(k)}, e_3^{(k)}$, each with its partner $e_{i+3}^{(k)}$, and the paired rotations and boosts $\mathrm{rot}(i, j)$, $\mathrm{bst}(i, j)$ of temporal $i < j$ are those of Section 17.6 of [1]. A vector X of V_2 gives $Z = (z_1, z_2) \in \mathbb{C}^6$, $z_k = z(X^{(k)})$, on which G_2 acts as i; $q_k = z_k \cdot z_k$, $q_{\text{tot}} = q_1 + q_2$, $Q_{kl} = z_k \cdot z_l$ is the complex Gram matrix with singular values $\sigma_1 \geq \sigma_2$, and $M_{\text{pair}} = \frac{1}{2}(\sigma_1 + \sigma_2)$ is the rest energy of the pair (Theorem 8.6 of [2]). $C_v = \varepsilon \otimes \mathbf{1}_6$ is the copy rotation and $\varepsilon \otimes G$ the copy boost, $\varepsilon = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. $\mathfrak{l}_G^{(k)}$ and $\mathfrak{l}_{\text{int}}^{(k)}$ are the frame and interior Lorentz algebras of constituent k, and $\mathfrak{l}_G^{\Delta}$ is the diagonal $\mathfrak{l}_G$. On $S_2 = \mathbb{R}^{64}$ the tower generators are $E_{(1,a)} = e_a \otimes \mathbf{1}$, $E_{(2,a)} = \omega \otimes e_a$, the spin lift of $M \in \mathfrak{so}(6, 6)$ is $S(M) = \frac{1}{4} \sum_{a,b} (M\eta_2)_{ab} E_a E_b$, and $\Gamma_{12} = E_{(1,1)} \cdots E_{(2,6)}$. In the closure the per-constituent quantities are those of the momentum fields $\pi^{(k)} = \eta^{-1} p^{(k)}$, $w_k = w(p^{(k)}) = q(z(\pi^{(k)}))$ (equation (30) of [1]); Lorentz-class inertias are $N_k = (a_k \eta^{-1} + b_k G \eta^{-1})/m_k$, $c_k = (a_k + i b_k)/m_k$, $F(p) = \sum_k c_k w_k$ and $R_{\text{eff}} e^{i\varphi_{\text{eff}}} = F(p(0))$. The *kinetic condition* is that of Proposition 5.5 of [1], used in Theorem 17.22 there with the kernels compressed by the projector: $X^{\top} K^{\mu} + K^{\mu} X = 0$ for $\mu = 0, \ldots, 3$, with the kernels $K^{\mu} = -\gamma^0 \gamma^{\mu}$ of Proposition 4.5 there. At $\mu = 0$ it says that X is antisymmetric, and at $\mu = a$ that X commutes with the boosts $\gamma^0 \gamma^a$.

5.1 The closure symmetry and the invariants of a pair

Proposition 5.1 (The closure symmetry at two constituents)**.** *$G_2^{\top} \eta_2 G_2 = -\eta_2$ and $G_2^2 = -\mathbf{1}$. The centraliser of G_2 in $\mathfrak{so}(6, 6)$ has dimension 30 and is spanned by the 15 paired rotations and the 15 paired boosts. In the basis of the six temporal directions followed by their partners every*

element is $X = \left(\begin{smallmatrix} R & S \\ -S & R \end{smallmatrix}\right)$ *with* R, S *antisymmetric, and* X *acts on* Z *as the complex antisymmetric matrix* $R - iS$*. The map* $X \mapsto R - iS$ *is a Lie isomorphism onto* $\mathfrak{so}(6,\mathbb{C})$*, the algebra of the complex orthogonal group of* q_{tot}*, and the compact part of the centraliser is* $\mathfrak{so}(6)$*, the paired rotations.*

This restates Proposition 7.8(i) of [2] (fixed there), the $n = 2$ member of Theorem 17.19 of [1]; the bracket identity was checked on all 435 pairs of basis elements (e5-1a). By Proposition 7.8(ii) of [2], $\mathfrak{so}(6,\mathbb{C}) = \mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)} \oplus (\mathbb{C}^3 \otimes \mathbb{C}^3)$, and the cross part maps z_1 into z_2: the closure symmetry does not keep the two field strengths apart.

Proposition 5.2 (Quadratic invariants of a pair)**.** *The holomorphic quadratic polynomials in* (z_1, z_2) *invariant under (a)* $\mathfrak{so}(6,\mathbb{C})$ *are the multiples of* q_{tot}*; (b)* $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$ *form* $\mathrm{span}_{\mathbb{C}}\{q_1, q_2\}$*; (c)* $\mathfrak{l}_G^{\Delta}$ *form* $\mathrm{span}_{\mathbb{C}}\{q_1, q_2, z_1 \cdot z_2\}$*. In each case the invariant real quadratic forms on* V_2 *have real dimension* 2*,* 4*,* 6 *and are spanned by the real and imaginary parts of these.*

Proof. $\mathfrak{l}_G \cong \mathfrak{so}(3,\mathbb{C})$ acts on $\mathbb{C}^3$ by the vector representation. By the first fundamental theorem for the orthogonal group [28], the polynomial invariants of $SO(3,\mathbb{C})$ acting diagonally on (z_1, z_2) are generated by the inner products $z_k \cdot z_l$ (and $\det(z_1, z_2, \cdot)$, which needs three vectors), which gives (c). Under $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$ a cross term $z_1 \cdot Az_2$ is invariant under $z_1 \mapsto \Lambda z_1$ with z_2 fixed only if $A = 0$, which gives (b). $\mathbb{C}^6$ is irreducible under $\mathfrak{so}(6,\mathbb{C})$, with q_{tot} its only invariant form, which gives (a). A real invariant form that is not the real part of a holomorphic one would be Hermitian, $h(Xu, v) + h(u, Xv) = 0$, and applied to X and iX this forces $h(Xu, v) = 0$ for all X, so $h = 0$. □

Inherited (classical invariant theory), with the real dimensions computed as exact null spaces (e5-1b).

Theorem 5.3 (The closure symmetry and the pair invariant)**.** *(i)* $q(z_1 + z_2) = q_1 + q_2 + 2\, z_1 \cdot z_2$*,* $\det Q = q_1 q_2 - (z_1 \cdot z_2)^2$*, and*

$$M_{\mathrm{pair}}^2 = \tfrac{1}{4}(|q_1|^2 + |q_2|^2 + 2|z_1 \cdot z_2|^2) + \tfrac{1}{2}|q_1 q_2 - (z_1 \cdot z_2)^2|.$$

So M_{pair}*,* $z_1 \cdot z_2$ *and* $q(z_1 + z_2)$ *are Lorentz invariants of the pair under* $\mathfrak{l}_G^{\Delta}$*. None of them is invariant under* $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$*: a boost of constituent* 2 *alone fixes* q_1, q_2 *and changes* $z_1 \cdot z_2$ *and* M_{pair}*. Under the closure symmetry only* $q_{\mathrm{tot}} = \mathrm{tr}\, Q$ *is invariant.*

(ii) M_{pair} *is invariant under* $\mathfrak{l}_G^{\Delta}$ *and the copy rotation* C_v*, and the subalgebra of* $\mathfrak{so}(6,\mathbb{C})$ *that leaves it invariant is* $\mathfrak{l}_G^{\Delta} \oplus \mathbb{R}C_v$*, of real dimension* 7*.*

(iii) The closure symmetry does not preserve M_{pair}*. The copy boost has the eigenvector* (e_1, ie_1) *with eigenvalue* 1*, and maps the balanced pair* (e_1, ie_1)*, with* $q_{\mathrm{tot}} = 0$ *and* $M_{\mathrm{pair}} = 1$*, to* $e^t(e_1, ie_1)$*, with* $M_{\mathrm{pair}} = e^{2t}$*. The compact rotation* $R_\varphi^{(1)} \in SO(6)$ *of Example 8.8 of [2] takes* $((1, i, 0), 0)$ *from* $M_{\mathrm{pair}} = 0$ *to* $M_{\mathrm{pair}} = \sin\varphi$*.*

(iv) For each $c \in \mathbb{C}$ *the set* $\{Z \neq 0 : \ q_{\mathrm{tot}}(Z) = c\}$ *is a single orbit of* $SO(6,\mathbb{C})$*, and on it* M_{pair} *takes every value in* $[\frac{1}{2}|c|, \infty)$*. The bound* $M_{\mathrm{pair}} \geq \frac{1}{2}|q_{\mathrm{tot}}|$ *of Theorem 8.6(iii) of [2] is attained at* $(z, 0)$*, and the copy boost carries* $(z, 0)$ *to* $(\cosh t\, z,\ i \sinh t\, z)$*, with* $q_{\mathrm{tot}} = q(z)$ *and* $M_{\mathrm{pair}} = \frac{1}{2}|q(z)| \cosh 2t$*.*

Proof. (i) The first two identities are expansions, and with them the formula is $M_{\mathrm{pair}}^2 = \frac{1}{4}\|Q\|_F^2 + \frac{1}{2}|\det Q|$, which follows from Theorem 8.6(ii) of [2]. A diagonal Lorentz transformation Λ maps $Q = Z^\top Z$ to $Z^\top \Lambda^\top \Lambda Z = Q$; one acting on constituent 2 alone maps $z_1 \cdot z_2$ to $z_1 \cdot \Lambda z_2$. The last statement is Proposition 5.2(a). (ii) C_v maps Q to $RQR^\top$ with R real orthogonal, which keeps the singular values (Proposition 8.9(ii) of [2]). That nothing else in $\mathfrak{so}(6,\mathbb{C})$ keeps M_{pair}

is computed: the derivatives of M_{pair}^2 along the 30 basis fields at 40 random pairs form a matrix of numerical rank 23. (iii) The copy boost acts on the copy index as $\left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$ (Proposition 7.8(iii) of [2]), and M_{pair} is homogeneous of degree two. At (e_1, ie_1), $Q = vv^\top$ with $v = (1, i)$, so $\sigma_1 = 2$ and $\sigma_2 = 0$. The numbers for $R_\varphi^{(1)}$ restate Example 8.8 of [2]. (iv) Transitivity on the level sets of a nondegenerate quadratic form is Witt's theorem [29]. At $(z, 0)$, $Q = \mathrm{diag}(q(z), 0)$. Since $(\varepsilon \otimes G)^2 = \mathbf{1}$, $e^{t\,\varepsilon\otimes G} = \cosh t + \sinh t\,\varepsilon \otimes G$, and the image has $Q = q(z)\,vv^\top$ with $v = (\cosh t, i \sinh t)$, so $\sigma_1 = |q(z)|\,|v|^2 = |q(z)| \cosh 2t$ and $\sigma_2 = 0$. The orbit is connected and M_{pair} is continuous, so it takes every value between its infimum and its unbounded values; for $c = 0$ use (iii). □

Items (i) and (iv) are fixed, with (i) partly restated from Theorem 8.6 of [2]; (ii) is fixed, with the stabiliser computed numerically; (iii) is fixed, with the second example restated. Checks: the identities symbolically, the formula against the singular values at 30 random pairs to 10^{-12} (e5-1e); invariance to 10^{-11}, and the rank 23 with a singular-value gap above 10^8 (e5-1c); the orbits to 10^{-12} (e5-1d). So the closure symmetry mixes the two field strengths and does not preserve the pair invariant. The closure symmetry sees one field strength in $\mathbb{C}^6$ and its square q_{tot}. To split it into two field strengths that carry a pair invariant takes the seven-dimensional stabiliser $\mathfrak{l}_G^{\Delta} \oplus \mathbb{R}C_v$, a diagonal Lorentz action together with the copy rotation, and the closure selects neither. With equal inertias its dynamics agrees: the frame energy is $\frac{1}{2}\,\mathrm{Re}(c\,q_{\mathrm{tot}}(\pi))$ (Proposition 17.18 of [1]; Proposition 5.10(i) with $c_1 = c_2 = c$), and its compact symmetry carries every balanced pair to one null constituent and an empty copy (Proposition 8.9(iii) of [2]). In that closure "two constituents" is a label that the dynamics does not see (interpretive).

5.2 No common Lorentz algebra

Proposition 5.4 (No common Lorentz algebra at two constituents)**.** *(i)* $\mathrm{Cent}(G_2) \cap \mathfrak{l}_{\mathrm{int}}^{(1)} = 0$. *The elements of* $\mathfrak{so}(6,6)$ *commuting with both* G_2 *and* $\mathfrak{l}_{\mathrm{int}}^{(1)}$ *form* $\mathfrak{l}_G^{(2)} \cong \mathfrak{so}(3,\mathbb{C})$*, of dimension* 6*. At rung* n *they form the* $\mathfrak{so}(3n-3,\mathbb{C})$ *of constituents* $2,\dots,n$*, of dimension* $3(n-1)(3n-4)$*:* 0*,* 6 *and* 30 *at* $n = 1, 2, 3$.

(ii) The bivectors of constituent 1 *are* $e_a e_b \otimes \mathbf{1}$*, so* $S_n \otimes \mathbb{C}$ *carries* 8^{n-1} *copies of the content of* $S \otimes \mathbb{C}$*. Under* $\mathfrak{l}_G^{(1)}$*,* $S_2 \otimes \mathbb{C} = 16\,(\frac{1}{2}, \frac{1}{2})$*: no Weyl spinor (*$128\,(\frac{1}{2}, \frac{1}{2})$ *at* $n = 3$*). Under* $\mathfrak{l}_{\mathrm{int}}^{(1)}$*,* $S_2 \otimes \mathbb{C} = 16\,(\frac{1}{2}, 0) \oplus 16\,(0, \frac{1}{2})$.

(iii) Under $\mathfrak{l}_G^{(2)}$*,* $S_2 \otimes \mathbb{C} = 16\,(\frac{1}{2}, \frac{1}{2})$*, and under* $\mathfrak{l}_G^{\Delta}$*,* $S_2 \otimes \mathbb{C} = 4[(0,0) \oplus (1,0) \oplus (0,1) \oplus (1,1)]$*: integer spin only.*

Proof. (i) An element of $\mathfrak{l}_{\mathrm{int}}^{(1)}$ lives on constituent 1, where $\mathrm{Cent}(G) \cap \mathfrak{so}(3,3) = \mathfrak{l}_G \cong \mathfrak{so}(3,\mathbb{C})$ (Theorem 12.1 of [1]). It fixes the surplus pair, and an element of $\mathfrak{so}(3,\mathbb{C})$ that kills $z = (0,1,0)$ and $z = (0,0,1)$ is an antisymmetric matrix with two zero columns, hence zero. Under $\mathfrak{l}_{\mathrm{int}}^{(1)}$, V_2 is the interior four-plane of constituent 1, irreducible and nontrivial, plus eight Lorentz-trivial directions: the surplus pair $F = \langle e_2^{(1)}, e_3^{(1)} \rangle$ and all of constituent 2. By Schur's lemma an equivariant $X \in \mathfrak{so}(6,6)$ vanishes on the four-plane and preserves the eight trivial directions. If X also commutes with G_2, then for $v \in F$, $G_2 v$ lies in the interior plane $\langle e_5^{(1)}, e_6^{(1)} \rangle$, so $G_2 X v = X G_2 v = 0$ and $Xv = 0$. Hence X vanishes on constituent 1 and is an element of $\mathfrak{so}(3,3)$ of constituent 2 commuting with G, that is of $\mathfrak{l}_G^{(2)}$ (Theorem 12.1 of [1]). The same argument at rung n leaves $\mathrm{Cent}(G_{n-1})$ of constituents $2,\dots,n$, which is $\mathfrak{so}(3n-3,\mathbb{C})$ by Proposition 7.8(i) of [2]. (ii) $E_{(1,a)}E_{(1,b)} = e_a e_b \otimes \mathbf{1}$, and at one constituent $S \otimes \mathbb{C} = 2\,(\frac{1}{2}, \frac{1}{2})$ under $\mathfrak{l}_G$ and $2\,(\frac{1}{2}, 0) \oplus 2\,(0, \frac{1}{2})$ under $\mathfrak{l}_{\mathrm{int}}$ (Section 13 of [1]). (iii) The bivectors of constituent 2 are $\mathbf{1} \otimes e_a e_b$, so $\mathfrak{l}_G^{\Delta}$ acts on $(S \otimes \mathbb{C}) \otimes_{\mathbb{C}} (S \otimes \mathbb{C})$, and $(\frac{1}{2}, \frac{1}{2}) \otimes (\frac{1}{2}, \frac{1}{2}) = (0,0) \oplus (1,0) \oplus (0,1) \oplus (1,1)$. □

Part (i) restates Proposition 7.8(iv) and Proposition 9.1(iii) of [2] (fixed there), the dimension 30 restates Theorem 17.20 of [1], and the ladder formula is fixed (e5-2a). Parts (ii) and (iii) are

fixed at $n = 2$, restated at $n = 3$, and the Clebsch–Gordan rule is inherited (Casimir spectra to 10^{-9}, computed on S_2 and on one constituent; e5-2b, e5-2c). No nonzero Lorentz generator leaves both the closure and the spinor structure of matter invariant, at $n = 2$ as at $n = 1$ and $n = 3$. What differs is what commutes with both: at $n = 2$ it is a whole second Lorentz algebra, the frame algebra of the spectator constituent. That is the material of the mixed rung, which pairs $\mathfrak{l}^{(1)}_{\text{int}}$ with $\mathfrak{l}^{(2)}_{G}$ (Proposition 5.8).

5.3 Gauge algebras at two constituents and the binding root

Proposition 5.5 (No Pati–Salam algebra at two constituents)**.** *The compact parts of the candidate gauge algebras $\mathfrak{g}(k_I, k_B, k_S)$ of the five Lorentz actions at $n = 2$ (Theorem 10.8 of [1]; Table 8 of [2]) have dimensions* 13 *(spectator, $\mathfrak{so}(5) \oplus \mathfrak{so}(3)$),* 6 *(binding and spectator, $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$),* 7 *(diagonal interior, $\mathfrak{so}(2) \oplus \mathfrak{so}(4)$),* 1 *(mixed) and* 1 *(binding on both). None contains the Pati–Salam algebra $\mathfrak{so}(4) \oplus \mathfrak{so}(6)$ [30], of dimension* 21*, or its colour factor $\mathfrak{su}(4)$, of dimension* 15*.*

Obstructed, by a dimension count; the five dimensions are re-verified (e5-3a, e5-3e, e5-3h, e5-3i). At $n = 3$ the compact spectator algebra $\mathfrak{so}(8) \oplus \mathfrak{so}(6)$, of dimension 43, already contains $\mathfrak{ps}_U$ (Section 10.7 of [1]). The analogue at $n = 2$ of Theorems 17.21–17.23 of [1] is therefore the intersection of each compact candidate with $\mathrm{Cent}(G_2)$.

Proposition 5.6 (The spectator algebra and the binding root)**.** *Let the Lorentz algebra act by $\mathfrak{l}^{(1)}_{\text{int}}$, with constituent* 2 *a spectator.*

(i) $\mathfrak{g}(1, 0, 1) = \mathrm{Cent}(\mathfrak{l}^{(1)}_{\text{int}}) \cap \mathfrak{so}(6, 6) \cong \mathfrak{so}(5, 3)$ *has dimension* 28*. Its compact part is $\mathfrak{so}(5) \oplus \mathfrak{so}(3)$, with $\mathfrak{so}(5)$ on the positive Lorentz-trivial span $P = \langle e^{(1)}_2, e^{(1)}_3, e^{(2)}_1, e^{(2)}_2, e^{(2)}_3 \rangle$ and $\mathfrak{so}(3)$ on $N = \langle e^{(2)}_4, e^{(2)}_5, e^{(2)}_6 \rangle$.* $\mathfrak{g}(1, 0, 1) \cap \mathrm{Cent}(G_2) = \mathfrak{l}^{(2)}_{G}$*. Neither $\mathfrak{so}(5)$ nor $\mathfrak{so}(3)$ meets $\mathrm{Cent}(G_2)$, and with $E_2 = \langle e^{(2)}_1, e^{(2)}_2, e^{(2)}_3 \rangle$,*

$$(\mathfrak{so}(5) \oplus \mathfrak{so}(3)) \cap \mathrm{Cent}(G_2) = \{A + G_2 A G_2^{-1} : \ A \in \mathfrak{so}(E_2)\} \cong \mathfrak{su}(2),$$

the three paired rotations of constituent 2*: compact and non-abelian.*

(ii) Among the spin lifts of the 30 *elements of* $\mathrm{Cent}(G_2)$*, those satisfying the kinetic condition with the kernels of constituent* 1 *are exactly the paired rotations of constituent* 2*. The paired boosts of constituent* 2 *fail it.*

(iii) Let w be the product of the eight Lorentz-trivial generators. Then $w^2 = -\mathbf{1}$, w commutes with $\mathfrak{l}^{(1)}_{\text{int}}$ and with the spin lift of $\mathfrak{so}(5, 3)$, and $\Gamma = w\,\Omega^{(1)}$, with $\Omega^{(1)} = E_{(1,1)}E_{(1,4)}E_{(1,5)}E_{(1,6)}$ the interior volume element of constituent 1, is a chirality commuting with both. With w as i, each chirality half is a Weyl block $(\frac{1}{2}, 0) \otimes (\mathbf{4}, \mathbf{2})$, respectively $(0, \frac{1}{2}) \otimes (\mathbf{4}, \mathbf{2})$, of $\mathfrak{l}^{(1)}_{\text{int}} \oplus \mathfrak{so}(5) \oplus \mathfrak{so}(3)$ (Proposition 10.12 of [1]). Under the binding-compatible $\mathfrak{su}(2)$,

$$L = (\tfrac{1}{2}, 0) \otimes (2 \cdot \mathbf{1} \oplus 2 \cdot \mathbf{3}), \qquad R = (0, \tfrac{1}{2}) \otimes (2 \cdot \mathbf{1} \oplus 2 \cdot \mathbf{3}) :$$

integer isospin, and the same real representation on both halves, so the action is vector-like. With charges $h = -w\,S(X)$, $\mathcal{A}(h) = \mathcal{A}(h^3) = 0$, where $\mathcal{A}(f) = \frac{1}{4}[\mathrm{Tr}(fP_L) - \mathrm{Tr}(fP_R)]$ as in Theorem 17.23 of [1]. There is no doublet, hence no global $SU(2)$ anomaly [26].

Proof. (i) Let $A \in \mathfrak{so}(P)$ and $B \in \mathfrak{so}(N)$ with $[A + B, G_2] = 0$, and follow the proof of Theorem 17.21 of [1]. G_2 maps $F = \langle e^{(1)}_2, e^{(1)}_3 \rangle$ into the interior plane $\langle e^{(1)}_5, e^{(1)}_6 \rangle$, on which A and B vanish, so $G_2 A v = (A + B) G_2 v = 0$ and $Av = 0$ for $v \in F$. For $v \in E_2$, $Bv = 0$ and $G_2 v \in N$, so $G_2 A v = B G_2 v \in N$ and $Av \in G_2^{-1} N = E_2$. Hence $A \in \mathfrak{so}(E_2)$ and $B = G_2 A G_2^{-1}$ on N; conversely every such pair commutes with G_2. Taking $B = 0$ or $A = 0$ shows that neither summand

meets Cent(G_2). The rest is Proposition 5.4(i). (ii) At $\mu = 0$ the condition is antisymmetry. The spin lift of a paired boost is a sum of boost bivectors, which are symmetric (Theorem 2.3 of [1]). The spin lift of a paired rotation of constituent 2 is antisymmetric and commutes with every bivector of constituent 1, in particular with $\gamma^0\gamma^a$. A general element of Cent(G_2) was tested by an exact null space. (iii) $\mathfrak{so}(5) \supset \mathfrak{so}(E_2) \oplus \mathfrak{so}(F)$ restricts the spinor $\mathbf{4}$ to two isospin doublets, and $\mathfrak{so}(3)$ acts on $\mathbf{2}$ as a doublet, so the diagonal $\mathfrak{su}(2)$ sees $(\mathbf{2} \oplus \mathbf{2}) \otimes \mathbf{2} = 2 \cdot (\mathbf{1} \oplus \mathbf{3})$. Every representation of $\mathfrak{su}(2)$ has weights symmetric under $m \mapsto -m$, so every odd trace of h vanishes on each half separately; integer isospin excludes doublets, and with them the global anomaly. □

Part (i) is fixed: it is the structure of Theorem 17.21 of [1] at $n = 2$, with $U = P$ and $d = \dim E_2 = 3$. Part (ii) is fixed, and in (iii) the Weyl blocks restate Proposition 10.12 of [1], the branching is inherited, and the content and the anomaly traces are fixed. Checks: exact Lie algebra (e5-3a), exact null space of the kinetic condition (e5-3d), Casimirs to 10^{-9} and integer traces (e5-3b, e5-3c); the odd traces vanish as well for the $\mathfrak{so}(3)$ on N, which acts by doublets (e5-3j). The binding root therefore turns the isospin doublets of the spectator algebra, the "doublets, no colour, real" of Section 10.8 of [1], into isospin 0 and 1. No doublet survives at $n = 2$, as for the copy isospin of Proposition 9.6 of [2]. At $n = 3$ the doublets survive in the $(\frac{1}{2}, \frac{1}{2})_{\pm 1}$ of $\mathfrak{so}(4)_\Delta$ on P_* (Theorem 17.23(ii) of [1]).

Proposition 5.7 (The diagonal connection algebra and the binding root)**.** *Let the Lorentz algebra act by the diagonal* $\mathfrak{l}_{\text{int}}$*.*

(i) Cent(*diagonal* $\mathfrak{l}_{\text{int}}$) $\cap\, \mathfrak{so}(6,6) = \mathfrak{so}(2) \oplus \mathfrak{so}(4)$ *has dimension* 7 *and is compact. It meets* Cent(G_2) *in* $\mathbb{R}C_v$*, the copy rotation of all six directions: the copy rotation of the interior four-planes plus that of the surplus pairs. This algebra is abelian and compact, and* $S(C_v) = -C_b$ *with* $C_b = \frac{1}{2}\sum_a \eta_{aa} E_{(1,a)}E_{(2,a)}$*.*

(ii) There is no fermion under this action at $n = 2$ *(Proposition 10.9 of [1]) and no copy-invariant gamma matrix (Theorem 13.15 of [1]), so the kinetic condition has no canonical kernel. With the summed kernels* $K_S^\mu = \sum_k \gamma_0^{(k)}\gamma_\mu^{(k)}$ *it holds for* C_b*. With the kernels* $f_0 f_\mu$ *of the centre-of-mass generators* $f_a = (E_{(1,a)} + E_{(2,a)})/\sqrt{2}$ *(Proposition 7.3 of [2]) it fails.*

(iii) $\Gamma_{12}^2 = \mathbf{1}$*, and* Γ_{12} *commutes with the diagonal* $\mathfrak{l}_{\text{int}}$ *and with* C_b*. With copy charge* q *defined by* $C_b = iq$*, the content of* $S_2 \otimes \mathbb{C}$ *is*

$\Gamma_{12} = -1$	$(0,0)_{\pm 3} \oplus (0,0)_{\pm 1} \oplus 2\,(\frac{1}{2},\frac{1}{2})_{\pm 1} \oplus (1,0)_{\pm 1} \oplus (0,1)_{\pm 1}$
$\Gamma_{12} = +1$	$2\,(0,0)_{\pm 2} \oplus (\frac{1}{2},\frac{1}{2})_{\pm 2} \oplus 2\,(\frac{1}{2},\frac{1}{2})_0 \oplus 2\,(1,0)_0 \oplus 2\,(0,1)_0$

Spin is integral, $\Gamma_{12} = -1$ *carries the odd and* $\Gamma_{12} = +1$ *the even copy charges, and each half is invariant under* $(q, (j_L, j_R)) \mapsto (-q, (j_R, j_L))$*: self-conjugate, hence vector-like.* $\mathrm{Tr}(C_b^k P_\pm) = 0$ *for every odd* k*.*

Proof. (i) Under the diagonal $\mathfrak{l}_{\text{int}}$, $V_2 = 2V \oplus 4 \cdot \mathbf{1}$, and Schur's lemma gives $\mathfrak{so}(2) \oplus \mathfrak{so}(4)$, the $n = 2$ case of Theorem 10.2 of [1]. Within it, an element $aM + B$, with M the copy rotation of the interior four-planes and B in the surplus $\mathfrak{so}(4)$, commutes with G_2 exactly when B is a times the copy rotation of the surplus pairs, because G_2 maps each surplus pair into the interior plane of the same constituent. This is Proposition 9.1(ii) of [2], and the lift is Proposition 7.3(ii) there. (ii) C_b commutes with the diagonal boosts and is antisymmetric, while $f_0 f_\mu$ contains the cross terms $E_{(1,1)}E_{(2,\mu)}$ ($\gamma^0 = e_1$), with which it does not commute. (iii) Computed. □

Part (i) restates Proposition 9.1(ii) and Proposition 7.3 of [2] (fixed there; e5-3e), (ii) is fixed (exact; e5-3f), and (iii) is fixed (eigenvalues to 10^{-9}, traces exact; e5-3g). Under the diagonal action the binding root leaves an abelian, compact, Lorentz-invariant copy charge on matter

that has no fermions; the row "$\mathfrak{so}(2) \oplus \mathfrak{so}(4)$, no fermions" of Section 10.8 of [1] becomes "$\mathfrak{so}(2)$, no fermions".

Proposition 5.8 (The mixed rung and the binding root)**.** *Let* $\mathfrak{l}_{\mathrm{mix}}$ *pair* $\mathfrak{l}_{\mathrm{int}}^{(1)}$ *with* $\mathfrak{l}_G^{(2)}$ *generator by generator (Proposition 13.16 of [1]).*

(i) $\mathfrak{l}_{\mathrm{mix}}$ *satisfies the* $\mathfrak{so}(1,3)$ *brackets, its centraliser in* $\mathfrak{so}(6,6)$ *is* $\mathbb{R}J_1$*,* $J_1 = e_2^{(1)} e_3^{(1)}$ *acting on vectors (Proposition 13.17 of [1]), and* $\mathbb{R}J_1 \cap \mathrm{Cent}(G_2) = 0$*.* G_2 *does not commute with* $\mathfrak{l}_{\mathrm{mix}}$*.*

(ii) The $\mathfrak{l}_{\mathrm{mix}}$*-equivariant endomorphisms of* V_2 *form a seven-dimensional algebra: the scalars on the four-vector,* $\mathrm{span}\{\mathbf{1}, G\}$ *on the field strength of constituent* 2*, and* $\mathfrak{gl}(2)$ *on the two Lorentz-trivial directions. Its Euclidean-antisymmetric part, the generators that rotate without dilating, is* $\mathrm{span}\{G^{(2)}, J_1\}$*, with* $G^{(2)} = 0 \oplus G$*, and* $[J_1, G^{(2)}] = 0$*.*

Proof. (i) $\mathbb{R}^{6,6} = V \oplus \Lambda^2 V \oplus 2{\cdot}\mathbf{1}$ under $\mathfrak{l}_{\mathrm{mix}}$ (Proposition 13.16 of [1]). J_1 rotates the surplus pair of constituent 1, and G_2 maps that pair into the interior four-plane, which J_1 fixes, so $[J_1, G_2] \neq 0$. No nonzero element of $\mathfrak{l}_{\mathrm{int}}^{(1)}$ commutes with G_2 (Proposition 5.4(i)), so G_2 does not commute with $\mathfrak{l}_{\mathrm{mix}}$. (ii) Schur's lemma on the three isotypic pieces, with $\mathrm{Hom}_{\mathfrak{l}_G}(V, V) = \mathrm{span}\{\mathbf{1}, G\}$ (Proposition 17.4 of [1]). □

Part (i) restates Propositions 13.16 and 13.17 of [1] and is fixed in what it adds; (ii) is fixed (exact; e5-3h). The binding root of the closure is not covariant under the mixed action and leaves nothing of its algebra. The only covariant rotor generators that rotate without dilating are the combinations of $G^{(2)}$ and J_1. With the rotor $G^{(2)}$, which binds constituent 2 alone, all of $\mathbb{R}J_1$ survives (constructed: admitted, not selected). For completeness, under the two binding actions the binding root leaves $\mathfrak{l}_G^{(2)}$ of $\mathfrak{g}(0,1,1) = \mathfrak{so}(3,3)^{(2)}$, with compact part the $\mathfrak{su}(2)$ of Proposition 5.6, and all of $\mathfrak{g}(0,2,0) = \mathfrak{so}(2,\mathbb{C})$, with compact part $\mathbb{R}C_v$ (restating Proposition 7.8(iii) and Table 8 of [2]; e5-3i). Neither action carries a Dirac operator, so the kinetic condition does not apply.

5.4 Inertia freedom at two constituents

Theorem 5.9 (Inertia freedom at two constituents)**.** *The real dimensions of the symmetric forms* N *on* V_2^**, invariant under a symmetry (*$rN + Nr^{\top} = 0$*, Definition 17.10 of [1]), in the Lorentz class* $\{N, G_2\} = 0$ *and in the duality class* $[N, G_2] = 0$*, are:*

symmetry	*Lorentz class*	*duality class*
$\mathrm{Cent}(G_2) = \mathfrak{so}(6,\mathbb{C})$	2	0
$\mathfrak{l}_G^{\Delta} \oplus \mathbb{R}C_v$ *(the stabiliser of* M_{pair}*)*	2	0
$\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$	4	0
$\mathfrak{l}_G^{\Delta}$	6	0
$\mathfrak{l}_G^{(2)}$	14	9
$\mathfrak{su}(2)$*, the binding-compatible spectator part (Proposition 5.6)*	14	10
$\mathbb{R}C_v$*, the binding-compatible diagonal part (Proposition 5.7)*	18	18
none	42	36

Under the full closure symmetry $\mathfrak{so}(3n,\mathbb{C})$ *the freedom is* $(2,0)$ *at* $n = 1, 2, 3$*, spanned by* η_n^{-1} *and* $G_n \eta_n^{-1}$*; with no symmetry it is* $(3n(3n+1), 9n^2)$*, that is* $(12, 9)$*,* $(42, 36)$ *and* $(90, 81)$*.*

Proof. Split N into blocks between constituents. Under $\mathfrak{l}_G^{(k)}$ the invariant symmetric forms of one constituent are $\mathrm{span}\{\eta^{-1}, G\eta^{-1}\}$, of Lorentz class (Proposition 17.4 and Theorem 17.11(ii),(iii) of [1]); an invariant cross block vanishes when only one constituent is acted on, and is a complex multiple of the invariant bilinear form under $\mathfrak{l}_G^{\Delta}$. This gives $2 \cdot 2 = 4$ and $2 \cdot 3 = 6$, the real and

imaginary parts of the products $z_k \cdot z_l$ of Proposition 5.2, and $\mathfrak{so}(6,\mathbb{C})$ keeps only q_{tot}, which gives 2. A copy rotation acts on the copy index, and the copy-rotation-invariant complex symmetric 2×2 matrices are the multiples of $\mathbf{1}$, which gives 2 for the stabiliser. Under $\mathfrak{l}_G^{(2)}$ constituent 1 is free (12 and 9 by Theorem 17.11(i) of [1]), and its cross blocks vanish because $\mathfrak{so}(3)$ moves every vector of $\mathbb{R}^3$. Under the compact $\mathfrak{su}(2)$ alone the identity of constituent 2 is also invariant, which adds one duality form. Under $\mathbb{R}C_v$ alone, $N = \mathbf{1} \otimes A + \varepsilon \otimes B$ with A symmetric and B antisymmetric, which gives $12 + 6$ and $9 + 9$. With no symmetry the classes are the forms anticommuting and commuting with G_n. □

The first, third and fourth rows restate Section 8.2 of [2] ("four under $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$, six under $\mathfrak{l}_G^{\Delta}$ alone and two under $\mathrm{Cent}(G_2) \cap \mathfrak{so}(6,6)$"); the other rows, and the ladder, are fixed (exact ranks; e5-4a, e5-4b). Two observations follow. The seven-dimensional stabiliser of the pair invariant fixes the inertia as much as the thirty-dimensional closure symmetry does: up to one complex scale. And, as at $n = 3$, imposing only what the binding root leaves of a gauge algebra enlarges the freedom to 14 or 18 real parameters in the Lorentz class, so the gauge parts do not select the inertia.

5.5 Two constituents and one rotor

The closure on $\mathfrak{s}_2 = \mathbb{R} \ltimes_{G_2} V_2$ with $H = h^2/(2I_0) + \sum_k \frac{1}{2} p^{(k)\top} N_k p^{(k)}$ has $p^{(k)} = e^{-\theta G} p^{(k)}(0)$ for both constituents, so $w_k = e^{2i\theta} w_k(0)$ and $I_0 \ddot{\theta} = R_{\mathrm{eff}} \sin(2\theta + \varphi_{\mathrm{eff}})$. The rate is constant and nonzero iff $R_{\mathrm{eff}} = 0$ and $h(0) \neq 0$, and then every frame moves on a circle of diameter $|N_k p^{(k)}(0)|/|\kappa|$. The balance set $\mathcal{B} = \{F = 0\}$ has codimension two away from 0, its massive part is open and dense, and on it $|c_1| E_0(\pi^{(1)}) = |c_2| E_0(\pi^{(2)})$ (Theorem 8.1 of [2], restating Proposition 17.18 of [1]). With equal inertias the pairs on the null subspaces $V_\pm = \{p^{(2)} = \mp G p^{(1)}\}$ are balanced (Proposition 8.4 of [2]). The following items are new.

Proposition 5.10 (The rotor law with pair inertias)**.** *(i)* $\frac{1}{2} p^\top N p = \mathrm{Re}\big((a + ib) w(p)\big)/(2m)$ *for* $N = (a\eta^{-1} + bG\eta^{-1})/m$*, so block-diagonal Lorentz-class inertias give* $U(\theta) = \frac{1}{2} \mathrm{Re}\big(e^{2i\theta} F(p(0))\big)$*.*

(ii) The Lorentz-class forms on V_2^* *invariant under* $\mathfrak{l}_G^{\Delta}$ *are exactly the real and imaginary parts of the three products* $z_k \cdot z_l$ *of the momentum fields. For such an inertia* $U = \frac{1}{2} \mathrm{Re}\,\mathrm{tr}\big(C\, Q(\pi)\big)$ *with* C *complex symmetric,* $Q(\pi(\tau)) = e^{2i\theta} Q(\pi(0))$*, and*

$$I_0 \ddot{\theta} = R \sin(2\theta + \varphi), \qquad R e^{i\varphi} = \mathrm{tr}\big(C\, Q(\pi(0))\big).$$

The balance condition is $\mathrm{tr}(CQ(\pi)) = 0$*, and* $C = \mathrm{diag}(c_1, c_2)$ *recovers (i).*

Proof. (i) is Theorem 17.11(ii) of [1] per constituent. (ii) The first statement is Theorem 5.9 (row $\mathfrak{l}_G^{\Delta}$) with Proposition 5.2(c). Every $z(\pi^{(k)})$ turns as $e^{i\theta}$, so $Q \mapsto e^{2i\theta} Q$, $U(\theta) = \frac{1}{2} R \cos(2\theta + \varphi)$, and $\dot{h} = -dU/d\theta$. □

Part (i) restates Theorem 17.11(ii) of [1] (symbolic). Part (ii) restates the remark in Section 8.2 of [2], which states the form and the balance condition; the rotor law with cross terms is fixed here. Checks: exact span, and the law along a flow with a random C to 10^{-9} (e5-5a).

Proposition 5.11 (The balance surface at two constituents)**.** *Let* $c_1, c_2 \neq 0$*.*

(i) In the data $(w_1, w_2) \in \mathbb{C}^2$ *the balance condition* $c_1 w_1 + c_2 w_2 = 0$ *is a complex line, of real codimension two. In the momentum data* $\mathbb{R}^{12}$*,* $\mathcal{B}$ *has codimension two away from* 0*. If exactly one coefficient vanishes, say* $c_2 = 0$*, then* $\mathcal{B} = \{w_1 = 0\} \times \mathbb{R}^6$ *still has codimension two, but forces constituent* 1 *to be null; if both vanish,* $\mathcal{B}$ *is everything.*

(ii) $\mathcal{B} \setminus 0$ *is connected. For* $c_1 = c_2$ *it is the nonzero null cone* $\{Z \cdot Z = 0,\ Z \neq 0\}$ *of* $Z = (z(\pi^{(1)}), z(\pi^{(2)})) \in \mathbb{C}^6$*, a single orbit of* $SO(6,\mathbb{C})$*, diffeomorphic to* $\mathbb{R}_{>0} \times V_2(\mathbb{R}^6)$ *with* $V_2(\mathbb{R}^6)$ *the Stiefel manifold of orthonormal two-frames. For* $c_1 \neq c_2$ *it is the image of that cone under* $(z_1, z_2) \mapsto (z_1, \rho^{-1} z_2)$*,* $\rho^2 = c_2/c_1$*.*

(iii) The all-null part of $\mathcal{B}$ *has dimension* 8*. The massive part, on which both constituents are massive, has dimension* 10*, and* w_1 *maps it submersively onto* $\mathbb{C}^*$*, with fibre over* s *the product of the two complex spheres* $\{w(p^{(1)}) = s\}$ *and* $\{w(p^{(2)}) = -c_1 s/c_2\}$*.*

Proof. (i) is Theorem 8.1(iii) of [2]; the degenerate cases are direct. (ii) $c_1 q_1 + c_2 q_2 = c_1(z_1, \rho z_2) \cdot (z_1, \rho z_2)$. A nonzero null vector of $\mathbb{C}^6$ is $u + iv$ with $u, v \in \mathbb{R}^6$, $|u| = |v| > 0$ and $u \perp v$; $SO(6,\mathbb{C})$ is transitive on these by Witt's theorem [29], and $V_2(\mathbb{R}^6)$ is connected. (iii) The null set $\{w = 0\} \subset \mathbb{R}^6$ has dimension four (Theorem 8.1(iii) of [2]). On the massive part $w_1 \neq 0$ and $w_2 = -c_1 w_1/c_2 \neq 0$. □

Fixed, with (i) restated. Checks at 20 random points each: $\operatorname{rank} dF = 2$ for equal and unequal coefficients, orbit dimension 10, $\operatorname{rank} d(w_1, w_2) = 4$ on the all-null part, and $\operatorname{rank}(dF, dw_1) = 4$ on the massive part (e5-5f).

Theorem 5.12 (Stability of the balance surface)**.** *Let* $c_1, c_2 \neq 0$ *and let* (θ, h, p, x) *solve the two-constituent closure with* $\theta(0) = 0$*,* $\kappa_0 = h(0)/I_0$ *and* $\psi_0 = \varphi_{\text{eff}}$*.*

(i) R_{eff} *is a constant of motion for every datum. The balance surface and every level set* $\{R_{\text{eff}} = r\}$ *are invariant, so the surface is neither attracting nor repelling. A perturbation* δp *of a balanced datum has imbalance* $|dF(\delta p)| + O(|\delta p|^2)$*, and since* dF *is onto every small imbalance occurs.*

(ii) The rate is stable. With $\varepsilon = R_{\text{eff}}/(I_0 \kappa_0^2)$*,* $\kappa^2 = \kappa_0^2[1 + \varepsilon(\cos\psi_0 - \cos\psi)]$ *and* $|\kappa^2/\kappa_0^2 - 1| \leq 2\varepsilon$ *for all* τ*, and the rotor turns for all time when* $\varepsilon < \frac{1}{2}$*.*

(iii) The bounded frame orbits are not stable. In the rotating regime, during each period of the rate frame k *is displaced by* $\lambda\, e^{-\varphi_{\text{eff}} G} N_k p^{(k)}(0)$*, with* λ *of Theorem 17.16 of [1], and its mean velocity is*

$$\bar{v}_k = \tfrac{\varepsilon}{4}(1 - \varepsilon\cos\psi_0)\, e^{-\varphi_{\text{eff}} G} N_k p^{(k)}(0) + O(\varepsilon^3),$$

of first order in the imbalance. A perturbation off the surface therefore moves the frames by a distance that grows linearly in τ*, at a rate linear in the perturbation. A perturbation along the surface, or of* $h(0)$*, keeps them on circles.*

(iv) If $h(0) = 0$*, a balanced datum is an equilibrium of the rotor, and each frame moves uniformly,* $x_k = x_k(0) + \tau N_k p^{(k)}(0)$*, without bound. Off the surface, unless* $\psi_0 \in \pi\mathbb{Z}$ *(an equilibrium, at which the frames move uniformly), the rotor librates about* $\psi = \pi$ *with* $\Omega^2 = 2R_{\text{eff}}/I_0$*, amplitude* $\xi_0 \in (0, \pi)$ *given by* $\cos\xi_0 = -\cos\psi_0$*, and period* $4\mathrm{K}(m)/\Omega$*,* $m = \sin^2(\xi_0/2)$*, which diverges as* $R_{\text{eff}} \to 0$*;* K *and* E *are the complete elliptic integrals in the parameter* m *[27]. The frames move with the mean velocity* $(1 - 2\mathrm{E}(m)/\mathrm{K}(m)) e^{-\varphi_{\text{eff}} G} N_k p^{(k)}(0)$*. They drift, except at the one amplitude with* $2\mathrm{E}(m) = \mathrm{K}(m)$ *(*$m = 0.8261$*,* $\xi_0 = 130.7°$*), at which every frame returns to its initial position after each period.*

Proof. (i) $F(p(\tau)) = e^{2i\theta} F(p(0))$ because every w_k turns by $e^{2i\theta}$. (ii) Energy conservation, $H = \frac{1}{2} I_0 \kappa^2 + \frac{1}{2} R_{\text{eff}} \cos\psi$, divided by $\frac{1}{2} I_0 \kappa_0^2$; this is Corollary 17.15 of [1] with R_0/m replaced by R_{eff}. (iii) Lorentz-class N_k anticommute with G, so $\dot{x}_k = e^{2\theta G} N_k p^{(k)}(0)$ with the common angle θ, and the proof of Theorem 17.16 of [1] applies to each frame. Expanding $(1 + \varepsilon(\cos\psi_0 - \cos\psi))^{-1/2}$ to second order, $\lambda = \frac{\pi\varepsilon}{4|\kappa_0|}(1 - \frac{3}{2}\varepsilon\cos\psi_0) + O(\varepsilon^3)$ and the period is $\frac{\pi}{|\kappa_0|}(1 - \frac{1}{2}\varepsilon\cos\psi_0) + O(\varepsilon^2)$; their ratio gives $\bar{v}_k$. Along the surface $\varepsilon = 0$, and a change of $h(0)$ changes only the

radius $|N_k p^{(k)}(0)|/(2|\kappa|)$ of the circles. (iv) At $h(0) = 0$ and $R_{\text{eff}} = 0$, $\ddot{\theta} = 0$ and $\theta \equiv 0$, so $\dot{x}_k = N_k p^{(k)}(0)$. Off the surface the rotor equation is the pendulum of Proposition 4.4 of [2] with R_0/m replaced by R_{eff}. With $\psi = \pi + \xi$, $\dot{x}_k = -e^{-\varphi_{\text{eff}} G}(\cos\xi + G\sin\xi) N_k p^{(k)}(0)$. The substitution $\sin(\xi/2) = \sin(\xi_0/2)\sin t$ of that proposition gives $d\tau = dt/(\Omega\sqrt{1 - m\sin^2 t})$, so over one period $\int \sin\xi\, d\tau = (2\sqrt{m}/\Omega)\int_0^{2\pi} \sin t\, dt = 0$ and $\int \cos\xi\, d\tau = (4/\Omega)\int_0^{\pi/2}(1 - 2m\sin^2 t)(1 - m\sin^2 t)^{-1/2} dt = (4/\Omega)(2\mathrm{E}(m) - \mathrm{K}(m))$; divide by the period. □

Item (i) restates Theorem 8.1(i) of [2]; (ii) restates Corollary 17.15 of [1] for two constituents; (iii) and (iv) are fixed. The second-order term in (iii) is new. Item (iv) sharpens the closing sentence of Proposition 17.18 of [1]: on the balance surface the frames are bounded only if $h(0) \neq 0$, which is the qualifier "constant and nonzero" of Theorem 8.1(i) of [2]. Checks (DOP853, relative and absolute tolerance 10^{-12}; e5-5b to e5-5e): with equal inertias $a + ib = 1 + 0.3i$, $m = I_0 = 1$, $h(0) = 2$, and $p^{(2)} = -Gp^{(1)}$ drawn at random ($E_0(\pi^{(k)}) = 1.2156$ each), the rate stays constant to 10^{-11} over $\tau \in [0, 60]$ and both frames lie on their circles (diameter 0.942) to 10^{-9}. A generic balanced pair with unequal inertias ($m = 1, 2.5$; $a+ib = 1+0.3i,\ 0.4-0.7i$) gives the same. Off the surface ($p^{(2)}$ perturbed by 0.05 along a fixed random direction, $R_{\text{eff}} = 0.391$, $\varepsilon = 0.098$), R_{eff} and $w_1\overline{w_2}$ stay constant to 10^{-10}, the rotor obeys the pendulum law to 10^{-9}, and the displacement per period is $\lambda e^{-\varphi_{\text{eff}} G} N_k p^{(k)}(0)$ to relative 10^{-8}. For perturbations 10^{-3}, 2×10^{-3}, 4×10^{-3} the imbalance, the rate excursion and the mean frame velocity double with the perturbation (to 2%). The mean velocity equals the quadrature value to 10^{-8}, the first-order law to within 1.5ε and the second-order law to within ε^2 (relative). With $h(0) = 0$ the rotor librates with the period of (iv) to 10^{-8}, and on the balance surface it stays at rest to 10^{-12} while the frames move uniformly. Off the surface with $h(0) = 0$ the displacement after one, two and three periods equals the mean-velocity law of (iv) to relative 10^{-9} at four amplitudes, among them $\xi_0 = 130.7°$, where it vanishes (e5-5i).

Proposition 5.13 (Exchange of the constituents)**.** *Let $\sigma(p^{(1)}, p^{(2)}) = (p^{(2)}, p^{(1)})$.*

(i) σ is an η_2-isometry commuting with G_2, so it preserves the Lie–Poisson bracket of $\mathfrak{s}_2$ and commutes with the rotor; it preserves the equal-inertia Hamiltonian; as an element of $O(6,\mathbb{C})$ it has determinant -1, so it lies outside the identity component of the closure symmetry group. It exchanges V_+ and V_-.

(ii) With equal coefficients $R_{\text{eff}} = 2|c|\,|w_s|$ with $w_s = \frac{1}{2}(w_1 + w_2)$, the exchange-even part of (w_1, w_2): a pair is balanced iff (w_1, w_2) is exchange-odd, $w_2 = -w_1$.

(iii) The exchange eigenstates of the momenta, $p^{(2)} = \pm p^{(1)}$, both have $w_2 = w_1$, so $F = (c_1 + c_2)w_1$: unless $c_2 = -c_1$ they are balanced only when null. With equal inertias a massive balanced pair is never an exchange eigenstate of its momenta.

Proof. (i) σ permutes the copies; on Z it is the permutation of three pairs of coordinates, of determinant $(-1)^3$. If $p^{(2)} = -Gp^{(1)}$, then σp has second component $p^{(1)} = G(-Gp^{(1)})$. (ii) $R_{\text{eff}} = |c|\,|w_1 + w_2|$. (iii) w is quadratic, so $F = c_1 w_1 + c_2 w_1$. □

Fixed (e5-5g, e5-5j). Exchange parity of the duality invariants and exchange parity of the momenta are different things: the balanced pairs of $V_\pm$, a magnetic and an electric constituent a quarter duality turn apart (Proposition 8.4(iii) of [2]), are exchanged with each other by σ and are not exchange eigenstates. With $c_1 \neq c_2$, σ is not a symmetry of the closure.

Proposition 5.14 (What carries over from the rest closure)**.** *For one rotor on two constituents with Lorentz-class inertias:*

(i) (Theorem 4.1 of [2].) A function of p is a Casimir iff it is invariant under $p \mapsto e^{tG_2}p$. At $p \neq 0$ the leaves are two-dimensional, so there are 11 local Casimirs, against 5 at $n = 1$, and the real and imaginary parts of $\bar{z}_a z_b$, $a, b = 1, \ldots, 6$, with $z \in \mathbb{C}^6$ the pair of momentum

fields, provide them. They include E_6, E_0 and **S** *of each $\pi^{(k)}$ and the moduli $|Q_{kl}|$; no Hamiltonian moves rest energy between the constituents (Corollary 8.3 of [2]).*

(ii) (Propositions 4.3–4.8 of [2].) The pendulum, its small oscillations, the energy budget and the multiscale criterion hold with R_0/m replaced by R_{eff}, which depends on the data only through $F(p(0))$. What changes is Corollary 4.5(i) of [2]: a constant nonzero rate no longer needs a null momentum field.

(iii) (Propositions 4.10 and 4.14 of [2].) $\Gamma_{12}^2 = \mathbf{1}$, and Γ_{12} anticommutes with the twelve generators, so it implements $e^{\pi G_2} = -\mathbf{1}$ on S_2. Since $(e^{\theta G_2})^\top \eta_2 e^{\theta G_2} = \cos 2\theta\, \eta_2 + \sin 2\theta\, \eta_2 G_2$, the rotor lifts to $\mathrm{Pin}(6,6)$ only at $\theta \in \pi\mathbb{Z}$.

(iv) (Proposition 4.9 of [2].) The rotor turns both momentum fields at one rate, while the graded rest Hamiltonian of the tower has the single frequency $(m_1^2 + m_2^2)^{1/2}$ and the ungraded one the two frequencies m_1, m_2 (Proposition 1.2(v)). A closure with two rates exists, with the rotor acting by $\mathrm{diag}(\kappa_1, \kappa_2) \otimes G$, $\kappa_1 \neq \kappa_2$ (Proposition 9.2 of [2]), and is admitted, not selected.

(v) (Proposition 4.16 of [2].) Under Assumption 4.15 of [2], every leaf through the balance surface is flat, $U \equiv 0$, so the conditional ladder $\kappa_n = (n+\delta)\hbar/I_0$ holds exactly with both constituents massive, where at $n = 1$ it needs null data.

Proof. (i) The proof of Theorem 4.1 of [2] with G_2 in place of G; $zz^\dagger$ is a rank-one Hermitian matrix on $\mathbb{C}^6$, with $2 \cdot 6 - 1 = 11$ real parameters. (ii) The equations for (θ, h) are those of one constituent with $R_0 e^{i\varphi_0}/m$ replaced by $F(p(0))$. (iii) As in Proposition 4.14 of [2]: a product of all twelve anticommuting generators anticommutes with each, and $G_2^\top \eta_2 = \eta_2 G_2$, $G_2^\top \eta_2 G_2 = -\eta_2$. (iv) and (v) are direct. □

Items (i)–(iii) restate the cited results for two constituents (fixed there) and are fixed in their $n = 2$ form (Jacobian rank 11 and 5, Lie–Poisson rank 2, exact Clifford identities; e5-5h). Item (iv) restates Proposition 1.2 and Proposition 9.2 of [2] and is constructed for the two-rate closure. Item (v) is constructed and conditional on the assumption. The copy rotor of Remark 8.10 of [2], a second rotor that exists because the uniqueness argument of Theorem 17.7 of [1] fails under $\mathfrak{l}_G^\Delta$, and the two-rate rotor of (iv) are the structures of the rest closure that exist only from $n = 2$ on.

5.6 The ladder: one, two and three constituents

The new entries of Table 12 at $n = 1$ and $n = 3$ are fixed (e5-6a). At $n = 1$, $\mathrm{Cent}(\mathfrak{l}_{\mathrm{int}}) \cap \mathfrak{so}(3,3) = \mathbb{R}J$ meets $\mathrm{Cent}(G)$ in 0, so the binding root leaves no gauge algebra on one constituent. At $n = 3$ the compact spectator algebra meets $\mathrm{Cent}(G_3)$ in the 15 paired rotations of constituents 2 and 3, which contain the $\mathfrak{so}(4)_\Delta$ of Theorem 17.21 of [1], and the diagonal connection algebra meets it in the three copy rotations. At every rung the binding-compatible parts are $\mathfrak{so}(3n-3)$, the paired rotations of constituents $2, \ldots, n$, and $\mathfrak{so}(n)$, the copy rotations: the proofs of Proposition 5.6(i) and Proposition 5.7(i) apply verbatim (fixed; exact at $n = 4$, e5-6b). The balance condition $F = 0$ has codimension two at every rung, because dF has rank two at every nonzero point. At $n = 1$ it is the null cone and carries no massive datum.

Answer. *Is $n = 2$ a closure rung in its own right?* Yes, in a precise sense (interpretive, drawn from the fixed statements cited here): every structure of the three-constituent analysis of [1] except the two projectors, which exist only at $n = 3$, has a computed $n = 2$ analogue, and four of them first appear at $n = 2$. The balance surface carries massive constituents at an exactly constant rate (Theorem 8.1 of [2]). The closure symmetry mixes the constituents and does not see their pair invariant (Theorem 5.3). The closure symmetry compatible with the matter

	$n=1$	$n=2$	$n=3$
closure symmetry $\mathrm{Cent}(G_n)$	$\mathfrak{l}_G \cong \mathfrak{so}(3,\mathbb{C})$, 6	$\mathfrak{so}(6,\mathbb{C})$, 30	$\mathfrak{so}(9,\mathbb{C})$, 72
$\mathrm{Cent}(G_n) \cap \mathfrak{l}^{(1)}_{\mathrm{int}}$	0	0	0
commuting with G_n and $\mathfrak{l}^{(1)}_{\mathrm{int}}$	0	$\mathfrak{l}^{(2)}_G$, 6	$\mathfrak{so}(6,\mathbb{C})$ of 2, 3, 30
$S_n \otimes \mathbb{C}$ under $\mathfrak{l}^{(1)}_G$	$2\,(\frac{1}{2},\frac{1}{2})$	$16\,(\frac{1}{2},\frac{1}{2})$	$128\,(\frac{1}{2},\frac{1}{2})$
Pati–Salam algebra	no	no	in $\mathfrak{so}(8) \oplus \mathfrak{so}(6)$
binding-compatible compact part of the spectator algebra	0	$\mathfrak{su}(2)$, 3; vector-like, isospin $0, 1$	$\mathfrak{so}(6)$ of 2, 3, 15; contains $\mathfrak{so}(4)_\Delta$
the same for the diagonal connection algebra	0	$\mathbb{R}C_v$, 1; no fermions	copy rotations $\mathfrak{so}(3)$, 3
under the projectors	—	—	P_{fam}: $\mathfrak{u}(1)^2$; P_*: $\mathfrak{so}(4)_\Delta \oplus \mathbb{R}\,\mathrm{rot}(8,9)$, chiral
anomalies of what survives	—	none	P_{fam}: $\mathcal{A}(H^3) = -12\alpha\beta^2$; P_*: free on $\mathfrak{so}(4)_\Delta$
inertia, closure symmetry	$(2,0)$	$(2,0)$	$(2,0)$
inertia, per-constituent $\mathfrak{l}_G$	$(2,0)$	$(4,0)$	$(6,0)$
inertia, binding-compatible part	—	$(14,10)$ spectator; $(18,18)$ diagonal	$(18,13)$ P_{fam}; $(8,6)$ P_*
inertia, none	$(12,9)$	$(42,36)$	$(90,81)$
balance surface	$\{w=0\}$: null data only, dim 4	codim 2, connected; massive part dense, dim 10	codim 2, dim 16

Table 12: The closure along the ladder. Dimensions are real; the inertia entries are (Lorentz class, duality class). The $n=3$ column restates Theorems 17.19–17.24 and Section 10.7 of [1] and Theorem 8.1(iii) of [2], except the spectator and diagonal rows, which are computed here (e5-6a).

statement	class	checks
Proposition 5.1: $\mathrm{Cent}(G_2) = \mathfrak{so}(6,\mathbb{C})$, paired rotations and boosts, $R - iS$, compact part $\mathfrak{so}(6)$	restated (RE Prop. 7.8(i); ES17 Thm 17.19 at $n = 2$)	e5-1a
Proposition 5.2: quadratic invariants q_{tot}; q_1, q_2; $q_1, q_2, z_1 \cdot z_2$	inherited (invariant theory); dimensions fixed	e5-1b
Theorem 5.3(i): M_{pair} through the Gram entries; diagonal against per-constituent invariance	fixed (partly restated, RE Thm 8.6)	e5-1e
Theorem 5.3(ii): stabiliser of M_{pair} is $\mathfrak{l}_G^{\Delta} \oplus \mathbb{R}C_v$, dimension 7	fixed (numerical rank)	e5-1c
Theorem 5.3(iii),(iv): $\mathfrak{so}(6,\mathbb{C})$ does not preserve M_{pair}; range $[\frac{1}{2}\lvert c\rvert, \infty)$ on each orbit	fixed (Ex. 8.8 of RE restated; transitivity inherited)	e5-1d
Proposition 5.4: no common Lorentz algebra; $\mathfrak{l}_G^{(2)}$; $16(\frac{1}{2}, \frac{1}{2})$; integer spin under $\mathfrak{l}_G^{\Delta}$	fixed (restated: RE Props. 7.8(iv), 9.1(iii); ES17 Thm 17.20)	e5-2a to e5-2c
Proposition 5.5: no Pati–Salam algebra at $n = 2$	obstructed (dimension count)	e5-3a, 3e, 3h, 3i
Proposition 5.6: binding-compatible $\mathfrak{su}(2)$; kinetic condition; integer isospin, vector-like, anomaly free	fixed (Weyl block restated, ES17 Prop. 10.12; branching inherited)	e5-3a to e5-3d, 3j
Proposition 5.7: $\mathbb{R}C_v$; no canonical kinetic kernel; content, vector-like	fixed ((i) restated, RE Props. 9.1(ii), 7.3)	e5-3e to e5-3g
Proposition 5.8: mixed rung, nothing survives G_2; covariant rotors $G^{(2)}, J_1$	fixed (restated: ES17 Props. 13.16, 13.17); rotor $G^{(2)}$ constructed	e5-3h
binding actions: $\mathfrak{l}_G^{(2)}$ and $\mathfrak{so}(2,\mathbb{C})$ survive	restated (RE Prop. 7.8(iii), Table 8)	e5-3i
Theorem 5.9: inertia freedom at $n = 2$; $(2,0)$ at $n = 1, 2, 3$	fixed (rows 1, 3, 4 restated, RE Sec. 8.2)	e5-4a, e5-4b
Proposition 5.10: rotor law with pair inertias	fixed ((i) restated, ES17 Thm 17.11(ii))	e5-5a
Proposition 5.11: codimension two, connected, one orbit, parts	fixed ((i) restated, RE Thm 8.1(iii))	e5-5f
Theorem 5.12: neutral surface, stable rate, secular frames, $h(0) \neq 0$ needed, libration drift	fixed ((i), (ii) restated)	e5-5b to e5-5e, 5i
Proposition 5.13: exchange, $\det \sigma = -1$, balanced iff w exchange-odd	fixed	e5-5g, 5j
Proposition 5.14: Casimirs, pendulum, half turn, rates, ladder at $n = 2$	(i)–(iii) restated (RE Sec. 4), fixed at $n = 2$; (iv), (v) constructed	e5-5h
Table 12: the ladder $n = 1, 2, 3$	fixed; $n = 3$ mostly restated (ES17 Sec. 17.6)	e5-6a, 6b, 2a, 4b
Section 5.6: answer	interpretive (drawn from fixed statements)	—

Table 13: Claims of Section 5, their classes and their checks, which are `PASS` lines of the script `verify/e5_closure_two.py`.

Lorentz algebra is nonzero, and is itself a Lorentz algebra (Proposition 5.4). And the binding root leaves a compact gauge algebra, non-abelian under the spectator action (Proposition 5.6). *What* $n = 2$ *adds to* $n = 1$ is therefore a constant rate with mass, a gauge part that survives the binding root, a pair invariant and an exchange symmetry. *What it lacks*, and *what* $n = 3$ *adds*, is everything chiral. At $n = 2$ every binding-compatible gauge part acts vector-like, with zero anomaly coefficients and integer isospin, and no Pati–Salam algebra fits. At $n = 3$ the projectors P_{fam} and P_* exist, the surviving $\mathfrak{so}(4)_\Delta$ acts on inequivalent halves with doublets, the abelian charges of P_{fam} carry nonzero anomaly coefficients, and fermions exist under the diagonal action. The inertia question does not distinguish the rungs: the closure symmetry fixes the inertia up to one complex scale at $n = 1, 2, 3$ alike, and the gauge parts never do.

For the question of this note the closure gives a definite but modest answer (interpretive, drawn from the fixed statements above). The closure at $n = 2$ distinguishes two constituents only through structures it does not select. Equal inertias and the closure symmetry see one field strength in $\mathbb{C}^6$. A pair invariant needs a diagonal Lorentz action and the copy rotation. Per-constituent rest energies need the copies to be distinguished, and then nothing in the closure moves rest energy between them (Proposition 8.9 and Corollary 8.3 of [2]). What two constituents can do that one cannot is to run at an exactly constant rate with both momentum fields massive. That holds on a set of codimension two which is invariant but whose frames are secularly unstable, which is closer to a balanced pair carried by the dynamics than to two particles produced by it.

6 The b-field on the temporal triple, c on the spatial triple

An earlier assignment places the arena of c (of c^2) on the negative triple $\langle e_4, e_5, e_6\rangle$, of signature $(-,-,-)$, and the arena of the b-field on the positive triple $\langle e_1, e_2, e_3\rangle$, of signature $(+,+,+)$, with the remark that the mass is unique in $T = e_1e_2e_3$. In the conventions of [1] the positive triple is the temporal triple V^+ and carries $\mathbf{B}$, the negative triple is the spatial triple V^- and carries $\mathbf{E}$ ($X = \sum_i (B_i e_i + E_i e_{i+3})$, Theorem 12.3 of [1]), the clock is $\gamma^0 = e_1$, the spatial gamma matrices are $\gamma^a = e_{a+3}$, and c is carried by the conversion $C\colon V^+ \to V^-$, of which $C = c\,\mathbf{1}$ is the unique choice that commutes with the diagonal rotations (Theorem 3.2 of [1]; Proposition 2.7 of [2]). We therefore read the assignment as follows: the b-field takes values in the temporal rotations $\mathfrak{so}(3)_+ = \operatorname{span}\{e_2e_3, e_3e_1, e_1e_2\}$, and c^2 is a symmetric part of C. The trivector $T = \gamma^0 J$ ($J = e_2e_3$) is the Dirac mass form, the unique antisymmetric kernel invariant under all fifteen bivectors (Corollary 18.4 of [1]), kept by $\mathrm{Spin}(3,3)$ at group level (Proposition 3.5 of [2]).

The answers, with their classes, are these. (1) As a form T is invariant under all of $\mathfrak{so}(3,3)$; as an operator it commutes exactly with $\mathfrak{so}(3)_+ \oplus \mathfrak{so}(3)_-$, and its commutant is $M_4(\mathbb{C})$ (fixed; Section 6.1). (2) The temporal rotations keep the mass form and the rest flow, and the kinetic term breaks them to $\mathfrak{u}(1)_J$, in proportion to the speed of the clock. They and the Pauli–Gürsey $\mathfrak{su}(2)$ of the massless kinetic term are two diagonals of one $\mathfrak{so}(4)$ meeting in $\mathbb{R}J$ (fixed; Section 6.2). (3) A temporal b-field is half a field-strength rotation and half a counter-rotation. It keeps the rates, splits the lines into κ and $s \pm \omega/2$, $s = (\kappa^2 + \omega^2/4)^{1/2}$, cannot be removed along a worldline, and tears the binding (fixed; Section 6.3). (4) The binding root G_C preserves $\frac{1}{2}(c^2|\mathbf{B}|^2 + |\mathbf{E}|^2)$ iff $C = cR$, $R \in O(3)$. A temporal rotation keeps the symmetric factor $CC^\top$ on the spatial triple and moves only the orthogonal factor (fixed; the reading "c^2 on $(-,-,-)$, b on $(+,+,+)$" is interpretive; Section 6.4). (5) Under a temporal b-field the four-dimensional mass read by a fixed clock oscillates at fixed six-dimensional mass, unless the clock or the temporal energy is on the axis (fixed; reading interpretive; Section 6.5). (6) Between two constituents the temporal copy rotation rotates the graded masses at fixed $M = (m_1^2 + m_2^2)^{1/2}$, half of it creates quanta in pairs, and in the ungraded reading no continuous transfer at fixed spectrum exists, only the exchange of the two masses (fixed; Section 6.6). (7) At every rung the $\mathfrak{so}(3n, 3n)$-

invariant forms are spanned by the temporal and the spatial volume; their symmetry has period four in n, so the invariant Grassmann mass kernel is the temporal volume only for $n \equiv 1 \pmod 4$ (fixed; Section 6.7). Section 6.8 compares the assignment with the b-field of Sections 5 and 9 of [2]. Identifiers e6-n refer to the `PASS` lines of `verify/e6_bfield_triples.py` (Table 14).

Notation. $S = \mathbb{R}^8$ with the realisation of equation (1) of [1]; $\omega_T = T = e_1e_2e_3$, $\omega_S = e_4e_5e_6 = \gamma^1\gamma^2\gamma^3$, $\Gamma = \omega_T\omega_S = e_1 \cdots e_6$, $J = e_2e_3$, $\Omega = e_1e_4e_5e_6$ (Section 4 and Theorem 18.1 of [1]); $K^\mu = -\gamma^0\gamma^\mu$ are the kinetic kernels (Proposition 4.5 of [1]) and $\mathfrak{g}_{\text{massless}} = \text{span}\{\Omega, J, e_2\Omega, e_3\Omega\} = \mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)_{\text{PG}}$ is the massless kinetic algebra (Proposition 5.5 of [1]). The temporal generators are symmetric and the spatial ones antisymmetric (Lemma 2.1 of [1], equation (4)). $\mathfrak{so}(3)_\pm$ are the rotations of $V^\pm$, the six compact bivectors, and the nine bivectors e_ie_{a+3} are the boosts (Theorem 2.3 of [1]). A bivector Y acts on vectors by $v \mapsto Y(v) := \frac{1}{2}[Y, v]$. On $V = V^+ \oplus V^-$ we use the blocks of [2]: $G = \left(\begin{smallmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$, $G_C = \left(\begin{smallmatrix} 0 & -C^{-1} \\ C & 0 \end{smallmatrix}\right)$ (Proposition 2.7 of [2]), $N_C = \text{diag}(C^\top C, \mathbf{1})$ and $E_C(X) = \frac{1}{2}X^\top N_C X = \frac{1}{2}(|C\mathbf{B}|^2 + |\mathbf{E}|^2)$, with E_c for $C = c\,\mathbf{1}$; $\mathfrak{l}_G \oplus \mathfrak{p}_G$ is the splitting of $\mathfrak{so}(3,3)$ into the parts commuting and anticommuting with G (Proposition 5.3 of [2]). The interior shift is $S = b + \kappa G$ with phase-space generator M (equation (9) and Lemma 5.4 of [2]). Following Section 9 of [2], the *rates* are the spectrum of the η-self-adjoint part of S and the *lines* are the frequencies of S itself. On the tower $E_{(k,a)}$, N and the ungraded generators $u_a = \mathbf{1} \otimes e_a$ are as in Section 1; $C_b = -\frac{1}{2}\sum_a \eta_{aa} E_{(1,a)}E_{(2,a)}$ is the copy rotation. Units are $\hbar = c = 1$ unless c is shown. The companion note *Two readings of X* [31] is cited by its own numbers.

6.1 T as a form and as an operator

Proposition 6.1 (The two invariances of the mass form)**.** *(i)* As a form. *For every bivector Y, $Y^\top T + TY = 0$ and $Y^\top TY = -Y^2\,T$, so $(e^{sY})^\top T e^{sY} = T$ for every real s. The same holds for ω_S. The forms on S invariant under $\mathfrak{so}(3,3)$ are $\text{span}\{T, \omega_S\}$, with T antisymmetric and ω_S symmetric.*

(ii) As an operator. *T commutes with the six compact bivectors and anticommutes with the nine boosts. Hence e^{sY} commutes with T for $Y \in \mathfrak{so}(3)_+ \oplus \mathfrak{so}(3)_-$, while $e^{sY}Te^{-sY} = e^{2sY}T$ for a boost.*

(iii) The commutant. *The commutant of T in $M_8(\mathbb{R})$ is spanned by the 32 blades with an even number of spatial indices. It is $\text{Cl}(V^+) \cdot \text{Cl}^0(V^-)$, a product of two commuting subalgebras with $\text{Cl}(V^+) \cong \text{Cl}(3,0) \cong M_2(\mathbb{C})$, whose centre is $\text{span}\{\mathbf{1}, T\}$, and $\text{Cl}^0(V^-) \cong \mathbb{H}$. So it is $M_2(\mathbb{C}) \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{C})$, of dimension 32 and with centre $\text{span}\{\mathbf{1}, T\}$, and it meets $\mathfrak{so}(3,3)$ in $\mathfrak{so}(3)_+ \oplus \mathfrak{so}(3)_- \cong \mathfrak{so}(4)$. The 32 blades with an odd number of spatial indices anticommute with T.*

(iv) Ad_T *acts on V as η: on the frame it is parity.*

Proof. By the blade rule (Lemma 2.1 of [1]) a blade e_I commutes with T iff $3|I| - |I \cap \{1,2,3\}|$ is even, that is iff $|I \cap \{4,5,6\}|$ is even, and anticommutes otherwise. Conjugation by T is diagonal in the blade basis, so its commutant is spanned by the blades it contains, as in Proposition 1.1(ii). This gives (ii) and the first sentence of (iii). A temporal generator commutes with a spatial bivector, so $\text{Cl}(V^+)$ and $\text{Cl}^0(V^-)$ commute, and their products are the 8×4 commuting blades. T is central in $\text{Cl}(V^+)$; e_4e_5, e_5e_6 and their product $-e_4e_6$ square to $-\mathbf{1}$. The isomorphisms $\text{Cl}(3,0) \cong M_2(\mathbb{C})$ and $M_2(\mathbb{C}) \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{C})$ are inherited; equivalently, $T^2 = -\mathbf{1}$ makes $S \cong \mathbb{C}^4$, and the commutant of a complex structure is $\mathfrak{gl}(4, \mathbb{C})$. For (i): a compact Y is antisymmetric with $Y^2 = -\mathbf{1}$, so $Y^\top T + TY = [T, Y] = 0$ and $e^{sY} = \cos s + \sin s\, Y$; a boost is symmetric with $Y^2 = \mathbf{1}$, so $Y^\top T + TY = \{T, Y\} = 0$ and $(\cosh s + \sinh s\, Y)^\top T(\cosh s + \sinh s\, Y) = (\cosh^2 s - \sinh^2 s)T$. ω_S has the same commutation pattern. The dimension two is Theorem 6.11(i) at $n = 1$. (iv) T commutes with e_1, e_2, e_3 and anticommutes with e_4, e_5, e_6. □

Part (i) restates Corollary 18.4 of [1] and Proposition 3.5(i) of [2] (fixed there), with the symmetric partner ω_S of Section 28 of [1]. Part (iv) restates Proposition 2.11(1) of [2]. Parts (ii) and (iii) are fixed (e6-1a to e6-1c); that ω_T and ω_S are fixed by all six compact rotations is noted after Theorem 18.1 of [1]. The two invariances are one statement read twice. Because the compact bivectors are antisymmetric and the boosts symmetric, invariance of the form is commutation with the first and anticommutation with the second. As a form, T breaks nothing in $\mathfrak{so}(3,3)$. As an operator it singles out the maximal compact subalgebra, the rotations of the two triples, and among these it does not distinguish the temporal from the spatial ones. Two things distinguish T from ω_S: as a Grassmann kernel T is antisymmetric and ω_S symmetric, and as an operator $T^2 = -\mathbf{1}$ is a complex structure while $\omega_S^2 = \mathbf{1}$ gives a projector (Theorem 18.1 of [1]).

6.2 The temporal rotations on the Dirac data

The Dirac data of one constituent are the kinetic kernels K^μ, the rest Hamiltonian $H(0) = m\gamma^0 = me_1$ of the real first-order form, and the rest flow $e^{-m\lambda T}$, with $JH(0) = mT$ (Section 4.4 of [1]; Propositions 3.7 and 4.9 of [2]).

Proposition 6.2 (What the temporal rotations do to the Dirac data)**.** *Let* $Y = \alpha J + \beta\, e_3e_1 + \gamma\, e_1e_2 \in \mathfrak{so}(3)_+$.

(i) Kinetic forms. $Y^\top K^0 + K^0 Y = 0$ *and* $Y^\top K^a + K^a Y = 2\,Y(\gamma^0)\,\gamma^a$, *where* $Y(\gamma^0) = \beta e_3 - \gamma e_2$ *is the velocity of the clock. Hence* $\sum_a \|Y^\top K^a + K^a Y\|^2 = 96(\beta^2+\gamma^2)$ *(Frobenius norm), and* Y *maps the kinetic forms into their span,* $Y^\top K^\mu + K^\mu Y = \Lambda^\mu{}_\nu K^\nu$, *iff* $\beta = \gamma = 0$*:* $\mathfrak{so}(3)_+ \cap (\mathfrak{l}_{\mathrm{int}} \oplus \mathbb{R}J) = \mathbb{R}J$.

(ii) Rest Hamiltonian. $e^{\varphi e_1e_2} e_1 e^{-\varphi e_1e_2} = \cos 2\varphi\, e_1 - \sin 2\varphi\, e_2$, *and likewise with* e_1e_3 *and* e_3. *The stabiliser of* $H(0)$ *in* $\mathfrak{so}(3)_+$ *is* $\mathbb{R}J$. *A rotor of* $\mathrm{Spin}(V^+)$ *that carries* e_1 *to a clock* n *carries* J *to* $J(n) = Tn$, K^a *to* $-n\gamma^a$ *and* $H(0)$ *to* mn, *and it fixes* $K^0 = -\mathbf{1}$ *and* T. *Moreover* $nJ(n) = J(n)n = T$.

(iii) Rest flow. $e^{-m\lambda T}$ *commutes with* $\mathrm{Spin}(3)_+ \times \mathrm{Spin}(3)_-$. *The spin lift* $\frac{1}{2}e_je_k$ *of* $\mathfrak{so}(3)_+$ *has Casimir* $-\frac{3}{4}\mathbf{1}$ *on* S, *so* S *carries isospin* $\frac{1}{2}$, *and* $e^{\pi e_je_k} = -\mathbf{1}$. *The clock rotations* e_1e_2 *and* e_3e_1 *anticommute with* J *and commute with* T. *Their half turn* $\mathrm{Ad}_{e_1e_2} = \mathrm{Ad}_{e_1}\mathrm{Ad}_{e_2}$ *maps* (e_1, J, T, γ^a) *to* $(-e_1, -J, T, \gamma^a)$.

(iv) Field octants. *With the field-octant map* $\Phi\colon V \to S$ *of Theorem 4.7 of [31],* $-T\Phi = \Phi G$, *so* $e^{-m\lambda T}\Phi = \Phi\, e^{m\lambda G}$*: on the six field octants the Dirac rest flow is the duality rotation at angular frequency* m.

Proof. (i) Y is antisymmetric and commutes with each γ^a (a temporal bivector and a spatial generator), so $Y^\top K^a + K^a Y = [K^a, Y] = -[e_1, Y]\gamma^a = 2Y(e_1)\gamma^a$, and $K^0 = -\mathbf{1}$. With $v = Y(e_1)$ temporal, $(v\gamma^a)^\top(v\gamma^a) = |v|^2\mathbf{1}$, so each term has squared norm $4{\cdot}8|v|^2$. The covariance statement intersects the seven-dimensional solution space $\mathfrak{l}_{\mathrm{int}} \oplus \mathbb{R}J$ of Proposition 3.5(ii) of [2] with $\mathfrak{so}(3)_+$; it is also solved directly (e6-2a). (ii) e_1 anticommutes with e_1e_2, so $e^{\varphi e_1e_2} e_1 e^{-\varphi e_1e_2} = e^{2\varphi e_1e_2} e_1$. The stabiliser condition is $Y(e_1) = 0$. A rotor R of $\mathrm{Spin}(V^+)$ commutes with T and with every γ^a, and $J = Te_1$, so $\mathrm{Ad}_R J = Tn$ and $\mathrm{Ad}_R K^a = -n\gamma^a$. Since n commutes with T and $n^2 = \mathbf{1}$, $nJ(n) = J(n)n = T$. (iii) T commutes with the compact bivectors by Proposition 6.1(ii); the Casimir is $\frac{1}{4}\sum Y_j^2$ with $Y_j^2 = -\mathbf{1}$; the rest is the blade rule. (iv) Iterate $-T\Phi = \Phi G$. □

Part (i) is fixed (e6-2a); its covariance half restates Proposition 3.5(ii) of [2]. Part (ii) is fixed (e6-2b, exact at $\cos\varphi = \frac{3}{5}$ and at the clock $n = -\frac{7}{25}e_1 - \frac{24}{25}e_2$, where (i) holds with $Y(n)$ in place of $Y(\gamma^0)$). Its transport of the data restates Proposition 3.4 and Theorem 3.5 of [31], by which the Dirac mass form $\gamma^0 J(n) = T$ is the same for every clock while $J(n)$ turns with n.

Part (iii) is fixed (e6-2c). Part (iv) restates Proposition 4.12(i) of [31], re-verified with Φ built from the Fock vacuum (e6-2c).

So the temporal $SU(2)$ is a symmetry of the mass form and of the rest flow that the kinetic term breaks to $\mathfrak{u}(1)_J$. That reverses the usual order, kinetic term symmetric and mass breaking, as Section 3.3 of [2] found for $\mathrm{Spin}(3,3)$. The broken directions are the two clock rotations: the kinetic term pairs one temporal direction with the spatial triple, and the breaking is the speed $|Y(n)|$ of the clock. They flip the charge, and their half turn acts as CP (Ad_{e_2} is the charge conjugation of Theorem 5.1 of [1], Ad_{e_1} the parity), reversing the sense of the rest rotation as in Proposition 3.8(iii) of [2]. Interpretively, the temporal $SU(2)$ joins particle and antiparticle through the sphere of clocks at fixed mass form.

Proposition 6.3 (The mass symmetry and the kinetic symmetry)**.** *Let $P_\pm = \frac{1}{2}(\mathbf{1} \pm \omega_S)$.*

(i) $\mathfrak{so}(3)_+ \cap \mathfrak{g}_{\text{massless}} = \mathbb{R}J$.

(ii) $e_2\Omega = -(e_1e_2)\,\omega_S$ *and* $e_3\Omega = (e_3e_1)\,\omega_S$. ω_S *commutes with* $\mathfrak{so}(3)_+$, *and* $P_\pm$ *are orthogonal projectors of rank four.*

(iii) $\mathfrak{so}(3)_+ = \{YP_+ + YP_-\}$ *and* $\mathfrak{su}(2)_{\mathrm{PG}} = \{YP_+ + (JYJ^{-1})P_-\}$, $Y \in \mathfrak{so}(3)_+$: *both are graphs of automorphisms of* $\mathfrak{so}(3)$ *over the two eigenspaces of* ω_S, *the identity and the half turn* Ad_J *about* J. *The Lie algebra they generate is* $\mathfrak{so}(3)_+P_+ \oplus \mathfrak{so}(3)_+P_- \cong \mathfrak{so}(4)$, *and they meet in the fixed algebra of* Ad_J, *which is* $\mathbb{R}J$.

(iv) $g = P_+ + JP_- = \exp(\frac{\pi}{2}JP_-)$ *is orthogonal and* $g\,\mathfrak{so}(3)_+\,g^{-1} = \mathfrak{su}(2)_{\mathrm{PG}}$; $2g = \mathbf{1} + J + \omega_S - J\omega_S$ *has components of grades* $0, 2, 3, 5$, *so* $g \notin \mathrm{Pin}(3,3)$.

(v) $\mathfrak{so}(3)_+$ *preserves* T *and the kinetic term breaks it to* $\mathbb{R}J$; $\mathfrak{su}(2)_{\mathrm{PG}}$ *preserves the kinetic forms and the mass breaks it to* $\mathbb{R}J$.

Proof. (ii) $e_2\Omega = e_2e_1e_4e_5e_6 = -e_1e_2\,\omega_S$, and similarly for e_3. ω_S is disjoint from each temporal bivector, and the grades 3 and 2 give a commuting sign. (iii) Substitute $\omega_S = P_+ - P_-$. J commutes with J and anticommutes with e_1e_2 and e_3e_1, so Ad_J is the half turn about J. A bracket of two graphs of automorphisms ϕ_1, ϕ_2 contains $[Y, Y']P_+ + [\phi_1 Y, \phi_2 Y']P_-$, and the generated algebra is computed (e6-2d). (iv) $(JP_-)^2 = -P_-$ and $gYg^{-1} = YP_+ + JYJ^{-1}P_-$, because $P_\pm$ commute with Y and J. (i) follows from (iii), and (v) is Proposition 6.2(i) together with Proposition 3.6(i)–(ii) of [2]. □

This is fixed (e6-2d); (v) restates Proposition 5.5 of [1] and Proposition 3.6 of [2]. The comparison has a sharp answer. The two $\mathfrak{su}(2)$'s are not the same, and neither contains the other. They are the untwisted and the Ad_J-twisted diagonals of $\mathfrak{so}(3)_+P_+ \oplus \mathfrak{so}(3)_+P_-$, and they share the Cartan direction J. One is the symmetry of the mass that the kinetic term breaks, the other the symmetry of the kinetic term that the mass breaks. Both leave the same $\mathfrak{u}(1)_J$ [32, 33]. The orthogonal map that exchanges them is not in the Clifford group.

6.3 A temporal b-field and the binding root

Proposition 6.4 (A b-field on the temporal triple)**.** *Let* $b = (R, 0)$ *on* $V^+ \oplus V^-$, *with* $R = [\mathbf{r}]_\times \neq 0$, $\omega = |\mathbf{r}|$, $\kappa > 0$ *and* $S = b + \kappa G$.

(i) $b = b_l + b_p$ *with* $b_l = \frac{1}{2}(R, R) \in \mathfrak{l}_G$, *a field-strength rotation, and* $b_p = \frac{1}{2}(R, -R) \in \mathfrak{p}_G$, *a counter-rotation; they commute. In the notation of equation (11) of [2],* $\mathbf{a} = \mathbf{c} = \mathbf{r}/2$, *so* $\mathbf{a} \cdot \mathbf{c} = \omega^2/4$.

(ii) $e^{\theta b}Ge^{-\theta b} = G_{C(\theta)}$ *with* $C(\theta) = e^{-\theta R}$. $G_C^2 = -\mathbf{1}$ *for every* C, *and* G_C *is* η*-self-adjoint iff* C *is orthogonal; so* $\kappa G_{C(\theta)}$ *stays in the self-adjoint summand and the rates,* $\pm i\kappa$, *are unchanged.* $[b, G] = 2b_pG \neq 0$, *so* b *is not removable along a worldline.*

(iii) $\det(\lambda - S) = (\lambda^2+\kappa^2)(\lambda^4 + (2\kappa^2+\omega^2)\lambda^2 + \kappa^4)$. *The lines are* κ, *on the plane* $\langle \hat{\mathbf{r}}, G\hat{\mathbf{r}}\rangle$, *and* $s \pm \omega/2$, *with* $s = (\kappa^2 + \omega^2/4)^{1/2}$ *and* $(s+\omega/2)(s-\omega/2) = \kappa^2$; *at* $(\kappa,\omega) = (2,3)$ *they are* $1, 2, 4$.

(iv) Not every solution is bounded. At $(\kappa,\omega) = (2,3)$ *and* $(6,5)$, M *has Jordan blocks exactly at the frequencies* $s \pm \omega/2$ *and is semisimple at* κ.

(v) On spinors $e_j e_k = \frac{1}{2}(e_j e_k - e_{j+3}e_{k+3}) + \frac{1}{2}(e_j e_k + e_{j+3}e_{k+3})$ *for temporal* j, k. *The first summand, a paired rotation, commutes with* N; *the second changes* N *by* ± 2 *only.*

Proof. (i) G commutes with $\operatorname{diag}(R,R)$ and anticommutes with $\operatorname{diag}(R,-R)$ in the $(\mathbf{B},\mathbf{E})$ blocks. (ii) $e^{\theta b} = \operatorname{diag}(e^{\theta R}, \mathbf{1})$, and $\operatorname{diag}(Q,\mathbf{1})\, G\, \operatorname{diag}(Q^{-1},\mathbf{1}) = G_{Q^{-1}}$. $G_C^\dagger = \eta G_C^\top \eta$ has blocks $-C^\top$ and $C^{-\top}$, which equal those of G_C iff $C^\top = C^{-1}$. The rates are the spectrum of the self-adjoint part, a tensor (Theorem 9.3(iii) of [2]). Removability fails by Proposition 5.9 of [2]. (iii) b_l commutes with b_p and with G. By Larmor's theorem (Proposition 5.7 of [2]) the substitution $X = e^{-b_l\tau}Y$ removes b_l and shifts the exponents by the eigenvalues $0, \pm i\omega/2$ of $-b_l$. On the support of b_p the remaining shift $b_p + \kappa G$ has the frequency $(\kappa^2+\omega^2/4)^{1/2}$ of Theorem 5.8(ii) of [2], and on its kernel $\langle \hat{\mathbf{r}}, G\hat{\mathbf{r}}\rangle$ the frequency κ. The characteristic polynomial is also computed symbolically in $\mathbf{r}$. (iv) By Theorem 5.10(i) of [2], boundedness needs $\mathbf{a}\cdot\mathbf{c} = 0$. The Jordan structure is that of $b_p + \kappa G$ (Theorem 5.8(ii) of [2]), transported by the bounded frame $e^{-b_l\tau}$, and is computed exactly. (v) With $e = a^+ + a^-$ and $e' = a^+ - a^-$, the sum is $2(a_j^+ a_k^+ + a_j^- a_k^-)$ and the difference $2(a_j^+ a_k^- + a_j^- a_k^+)$. □

This is fixed (e6-3a to e6-3e); the statement on the rates restates Theorem 9.3(iii) of [2]. A b-field on the temporal triple alone is therefore half binding-compatible and half binding-odd. Its paired half is the compact part of $\mathfrak{l}_G$, a Zeeman term removable by a rotating frame [38]. Its counter half lies in $\mathfrak{p}_G$. The latter gives the gauge-covariant remainder $D_\tau G = [b, G] = 2b_p G$, it moves the binding root to the root $G_{e^{-\theta R}}$ of a rotated conversion, and it tears the binding, with the screened frequency s and Jordan blocks. The rate $\kappa = c/\ell$ is never touched. In Proposition 6.4(iii) the two halves act independently, because $\mathbf{a} \parallel \mathbf{c}$: the lines are the screened frequency $\sqrt{\kappa^2 + |\mathbf{c}|^2}$ shifted by the Zeeman amounts $\pm|\mathbf{a}|$. Theorem 5.10 of [2] needed $\mathbf{a} \perp \mathbf{c}$ for screening; a field on one triple has the opposite geometry. On spinors the same split is the split of Proposition 1.6(ii): the paired half hops quanta within a constituent and the counter half creates or annihilates them in pairs, so the temporal rotations, J included, do not conserve N (Proposition 1.8(ii); Lemma 2.4).

6.4 c on the spatial triple: the polar form of the conversion

Proposition 6.5 (Conversions and the temporal rotations)**.** *Let* $C\colon V^+ \to V^-$ *be invertible,* $z_C = C\mathbf{B} + i\mathbf{E}$ *and* $q_C = z_C \cdot z_C$.

(i) $G_C^2 = -\mathbf{1}$, G_C *acts on* z_C *as multiplication by* i, *and* $e^{\theta G_C}$ *preserves* E_C, *which depends on* C *only through* $C^\top C$.

(ii) $G_C^\top N_c + N_c G_C = \left(\begin{smallmatrix} 0 & C^\top - c^2 C^{-1} \\ C - c^2 C^{-\top} & 0 \end{smallmatrix}\right)$, *so* G_C *preserves* $E_c = \frac{1}{2}(c^2|\mathbf{B}|^2 + |\mathbf{E}|^2)$ *iff* $C^\top C = c^2\mathbf{1}$, *that is iff* $C = cR$ *with* $R \in O(3)$.

(iii) For $C = cR$, *the elements of* $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$ *commuting with* G_C *are the twisted diagonal rotations* (Q, RQR^{-1}). $G_{cR} = \operatorname{Ad}_{(R^{-1},\mathbf{1})} G_c = \operatorname{Ad}_{(\mathbf{1},R)} G_c$ *for* $R \in SO(3)$, *and* $\operatorname{Ad}_{(R_+,R_-)} G_c = G_{cR_-R_+^{-1}}$ *for* $(R_+,R_-) \in O(3)\times O(3)$. *The Lie algebra preserving* η *and* E_c *is* $\mathfrak{so}(3)\oplus\mathfrak{so}(3)$, *and adding* $[Y, G_c] = 0$ *leaves the diagonal* $\mathfrak{so}(3)$. *Hence* $\{G_C : C^\top C = c^2\mathbf{1}\}$ *is one orbit of* $O(3)\times O(3)$, *of dimension three.*

(iv) A temporal rotation $(R_+, \mathbf{1})$ *carries* G_C *to* $G_{CR_+^{-1}}$ *and* N_C *to* $N_{CR_+^{-1}}$. *It conjugates* $C^\top C$ *to* $R_+ C^\top C R_+^{-1}$ *and fixes* $CC^\top$*; in the polar form* $C = (CC^\top)^{1/2}U$ *it moves only* U. *A spatial rotation* $(\mathbf{1}, R_-)$ *carries* C *to* $R_- C$, *fixes* $C^\top C$ *and moves only* U' *in* $C = U'(C^\top C)^{1/2}$.

(v) For $C = cR$: E_c *and* $\operatorname{Re} q_C = c^2|\mathbf{B}|^2 - |\mathbf{E}|^2$ *do not depend on* R, *while* $\operatorname{Im} q_C = 2c\,(R\mathbf{B})\cdot\mathbf{E}$ *does. At* $c = 1$, $\mathbf{B} = e_1$, $\mathbf{E} = e_2$ *and* R *the rotation through* θ *about* e_3, $q_C = 2i\sin\theta$*: the same field point is null for* $\theta = 0$ *and at rest for* $\theta = \frac{\pi}{2}$, *where* $E_0 = E_6 = 1$.

Proof. (i) G_C^2 has blocks $-C^{-1}C$ and $-CC^{-1}$. On z_C, $G_C X$ has $C\mathbf{B}' = -\mathbf{E}$ and $\mathbf{E}' = C\mathbf{B}$, so $z_C(G_C X) = iz_C(X)$, and $|z_C|^2 = 2E_C$. (ii) is a block computation. Its vanishing means $C^\top C = c^2\mathbf{1}$, and then C/c is orthogonal. (iii) The lower-left block of $[G_C, \operatorname{diag}(Q, Q')]$ is $CQ - Q'C$. The conjugation formulas are block products. An η-isometry preserving the positive form N_c preserves the eigenspaces $V^\pm$ of $\eta^{-1}N_c = \operatorname{diag}(c^2, -1)$, and is orthogonal on each. The stabiliser dimensions are exact null spaces (e6-4c). (iv) $(CR_+^{-1})(CR_+^{-1})^\top = CC^\top$. (v) Expand $z_C \cdot z_C$; in the example $R\mathbf{B} = (\cos\theta, \sin\theta, 0)$. □

This is fixed (e6-4a to e6-4e). Parts (i)–(iii) extend Proposition 2.7 of [2] and Theorem 3.2 of [1] from $C = c\,\mathbf{1}$ to the orbit. They are elementary linear algebra: Schur's lemma applied to a twisted diagonal action gives $C = cR$. Part (v) is the passive form of Proposition 2.14(2) of [2], where the anti-diagonal part of a compact transport turns a null field into one at rest at fixed E_6.

Remark 6.6 (What "c^2 on $(-,-,-)$, b on $(+,+,+)$" fixes)**.** (a) *Fixed.* The complete invariant of a conversion under the temporal rotations is $CC^\top$, a symmetric form on the spatial triple: C and C' lie in one orbit of the orthogonal group of the temporal triple iff $C'C'^\top = CC^\top$ (Proposition 6.5(iv); e6-4f). A b-field on the temporal triple, a flow in $SO(3)$, can change anything about the conversion except this and the sign of $\det C$ (e6-4f). The paper's c is the isotropic case $CC^\top = c^2\mathbf{1}$, which is also $C^\top C = c^2\mathbf{1}$, and that is exactly the case in which the binding root preserves the weighted energy (Proposition 6.5(ii)). In this sense, which is the reading of (c), c^2 sits on the spatial triple and the temporal b-field cannot reach it. The energy E_C sees the other symmetric factor $C^\top C$, a form on the temporal triple that the temporal rotations conjugate (Proposition 6.5(i), (iv)); the two factors coincide exactly in the isotropic case. What the temporal b-field moves is the orthogonal factor R. The energy E_c does not see R; the binding root G_{cR} (which spatial direction is paired with which temporal one), the pseudoscalar $\operatorname{Im} q_C$ and the rest energy $\frac{1}{2}|q_C|$ do.

(b) *Fixed: a mirror.* With the triples exchanged the statement still holds: a spatial b-field keeps $C^\top C$ on the temporal triple and moves the other orthogonal factor. The conversion alone therefore does not decide which triple carries c^2 and which the b-field. The Dirac data decide it, through the clock in V^+ and the kinetic term (Proposition 6.2): in the interior reading $\mathfrak{so}(3)_- \subset \mathfrak{l}_{\text{int}}$ is part of the Lorentz algebra, while $\mathfrak{so}(3)_+$ meets $\mathfrak{l}_{\text{int}} \oplus \mathbb{R}J$ only in J. The frame reading does not decide it, since its Lorentz algebra $\mathfrak{l}_G$ contains only the diagonal rotations.

(c) *Relation to the clause "c^2 is the symmetric part of $E = mc^2$".* Section 2.5 of [2] gives the clause its exact forms: c is the unique isotropic conversion, its weight in E_6 a unit convention (Propositions 2.7 and 2.8 there), and the shift splits as $\operatorname{End}(V) = \mathfrak{so}(3,3) \oplus \{Y^\dagger = Y\}$ with κG in the self-adjoint summand, $(\kappa G)^2 = -c^2/\ell^2$, and the connections in $\mathfrak{so}(3,3)$ (Section 5 there). The polar decomposition of the conversion is a further, compatible form: its symmetric factor carries c^2 and is invariant under the temporal b-field, its orthogonal factor is moved by it, and for every orthogonal C, κG_C stays in the self-adjoint summand with the same spectrum (Proposition 6.4(ii)). It adds nothing else: it does not fix c, does not make c a symmetry (the stabiliser $O(3)\times O(3)$ of η and E_c is not a Lorentz group), and

selects no rest energy. A constant R is a choice of identification of the triples (Theorem 3.2 of [1] selects $R = \mathbf{1}$ once a common $SO(3)$ is fixed); only a τ-dependent $R(\tau)$, a temporal b-field, has gauge-invariant content, its $D_\tau G \neq 0$. Reading the symmetric factor as "c^2" and the orthogonal factor as the place of the b-field is interpretive.

6.5 The six-momentum read by a clock

Let $P = \sum_i (\mathsf{E}_i/c) e_i + \sum_a p_a e_{a+3}$ with $\eta(P,P) = \mu^2 c^2$, so $\mu^2 c^4 = |\mathbf{E}|^2 - |\mathbf{p}|^2 c^2$, as in Proposition 2.10 of [2]. For a clock n the energy read is $E_n = n \cdot \mathbf{E}$ and $m_4(n)^2 c^4 = E_n^2 - |\mathbf{p}|^2 c^2 = \mu^2 c^4 - |n \times \mathbf{E}|^2$ (Proposition 3.13 of [31]). A temporal b-field of constant rate ν about a unit axis $\hat{\mathbf{e}}$ acts on P by $\mathbf{E}(t) = R_{\hat{\mathbf{e}}}(\nu t)\mathbf{E}$, with $\mathbf{p}$ fixed.

Proposition 6.7 (The four-dimensional mass under a temporal b-field)**.** *(i) The flow keeps μ and $|\mathbf{E}|$ constant.*

(ii) For a fixed clock n, $E_n(t) = A + B\cos(\nu t - \phi)$, where $A = (n \cdot \hat{\mathbf{e}})(\hat{\mathbf{e}} \cdot \mathbf{E})$ and $B = |n_\perp|\,|\mathbf{E}_\perp|$ with $n_\perp$ and $\mathbf{E}_\perp$ the parts orthogonal to $\hat{\mathbf{e}}$. Hence $m_4(n)^2 c^4 = \mu^2 c^4 - |\mathbf{E}|^2 + E_n(t)^2$ oscillates. Its maximum is $\mu^2 c^4 - |\mathbf{E}|^2 + (|A| + B)^2 \leq \mu^2 c^4$. Its minimum is $\mu^2 c^4 - |\mathbf{E}|^2 + (|A| - B)^2$ if $|A| \geq B$ and $\mu^2 c^4 - |\mathbf{E}|^2$ otherwise. It is constant iff $B = 0$, that is iff $n \parallel \hat{\mathbf{e}}$ or $\mathbf{E} \parallel \hat{\mathbf{e}}$.

(iii) The rotation about the clock, whose spin lift is proportional to $J(n)$ (Proposition 3.4(iii) of [31]), keeps E_n. On the reduced shell $\mathbf{E} \parallel n$ of Theorem 30.1 of [1] (Proposition 3.10 of [31]) the flow preserves the constraint iff $\hat{\mathbf{e}} \parallel n$.

(iv) With $c = 1$, $\mathbf{E} = (2,3,6)$ and $\mathbf{p} = (0,0,3)$, so that $\mu^2 = 40$: the rotation about e_3 read by e_1 gives $E_n = 2\cos\nu t - 3\sin\nu t$ and $m_4^2 \in [-9,4]$; read by the symmetric clock $\widehat{W} = (e_1+e_2+e_3)/\sqrt{3}$ (Proposition 3.5 of [1]) it gives $m_4^2 \in [\frac{35}{3} - 4\sqrt{26}, \frac{35}{3} + 4\sqrt{26}]$. The rotation about $\widehat{W}$ read by $\widehat{W}$ keeps $m_4^2 = \frac{94}{3}$ while the relative-time energies $\mathbf{E}_\perp$ turn.

Proof. Rodrigues' formula gives $n \cdot \mathbf{E}(t) = A + (n \cdot \mathbf{E} - A)\cos\nu t + n \cdot (\hat{\mathbf{e}} \times \mathbf{E})\sin\nu t$, and the squared amplitude $(n \cdot \mathbf{E} - A)^2 + (n \cdot (\hat{\mathbf{e}} \times \mathbf{E}))^2$ equals $(|n|^2 - (n \cdot \hat{\mathbf{e}})^2)(|\mathbf{E}|^2 - (\hat{\mathbf{e}} \cdot \mathbf{E})^2)$ (Lagrange's identity in the plane orthogonal to $\hat{\mathbf{e}}$). $|A| + B \leq |\mathbf{E}|$ by the Cauchy–Schwarz inequality. The rest is substitution; the reduced-shell statement says that a vector stays on the line $\mathbb{R}n$ under all rotations about $\hat{\mathbf{e}}$ iff the line is the axis. □

This is fixed (e6-5a, symbolic for three rational axes; e6-5b, exact). The reading needs care. "The four-dimensional mass oscillates" holds only if the clock is held fixed while the temporal b-field turns the temporal momentum. Equivalently, the clock turns against a fixed momentum along the circle of the clock sphere of Proposition 3.13 of [31] about the axis $\hat{\mathbf{e}}$ through n, a great circle only when $n \perp \hat{\mathbf{e}}$ (e6-5c). Read in the sense of Remark 3.14 of [31], the energy read by the clock is the projection of the undivided temporal energy, and the b-field moves energy periodically between the clock and the relative times. This is the dynamical form of the obstruction of Proposition 2.10 of [2]: splitting the temporal energy lowers the mass. The six-dimensional mass is constant throughout. The rotation about the clock is the charge $J(n)$ and changes nothing; the two clock rotations are what make m_4 oscillate, and they are also what the kinetic term breaks (Proposition 6.2). On the reduced shell, the gauge-fixed reading of [1], a temporal b-field transverse to the clock is not admissible at fixed clock.

6.6 Between constituents

Write $C_+^{12} = -\frac{1}{2}\sum_{i\leq 3} E_{(1,i)}E_{(2,i)}$ and $C_-^{12} = \frac{1}{2}\sum_{a\geq 4} E_{(1,a)}E_{(2,a)}$ for the temporal and the spatial parts of the copy rotation, $C_b = C_+^{12} + C_-^{12}$, and $C_\times = C_+^{12} - C_-^{12}$. We drop the superscript.

Proposition 6.8 (Temporal and spatial copy rotations)**.** *(i) $C_\pm$ are antisymmetric and commute, and each is a sum of three commuting terms of square $-\frac{1}{4}\mathbf{1}$. Hence $e^{2\pi C_\pm} = -\mathbf{1}$ and $e^{2\pi C_b} = \mathbf{1}$. The eigenvalues of $C_\pm$ are $\pm\frac{i}{2}$ (24 times each) and $\pm\frac{3i}{2}$ (8 times each).*

(ii) On vectors neither C_+ nor C_- commutes with G_2, C_b does, and $C_\times$ is compact and anticommutes with G_2.

(iii) $C_+ = \frac{1}{2}C_b + \frac{1}{2}C_\times$. C_b conserves N and $C_\times = -\sum_i (a^+_{1,i}a^+_{2,i} + a^-_{1,i}a^-_{2,i})$ changes it by ± 2, one quantum in each constituent. From the Fock vacuum, $e^{\theta C_+}|0\rangle$ has weight $\binom{3}{k}\cos^{2(3-k)}\frac{\theta}{2}\sin^{2k}\frac{\theta}{2}$ in $N = 2k$, none in odd N, and $N^{(1)} = N^{(2)}$ throughout. So $\langle N\rangle = 3(1-\cos\theta)$, and $\theta = \pi$ fills all six modes.

(iv) As a constant field between two binding frames, $S = \kappa G_2 + \beta V_+$, where V_+ is the vector form of C_+, has $\det(\lambda - S) = \left(\lambda^4 + (2\kappa^2+\beta^2)\lambda^2 + \kappa^4\right)^3$. The rates are κ and the lines are $s \pm \beta/2$. At $(\kappa,\beta) = (2,3)$, M has Jordan blocks at both lines.

Proof. (i) $E_{(1,i)}E_{(2,i)}$ squares to $-\mathbf{1}$ for every i, and the terms for distinct i involve four distinct anticommuting generators. Then $e^{2\pi C_\pm} = \prod(\cos\pi \mp \sin\pi\, E_{(1,i)}E_{(2,i)}) = (-1)^3$. The multiplicities follow from the minimal polynomial $(\lambda^2+1)(\lambda^2+9)$ of $2C_\pm$ and $\mathrm{Tr}(2C_\pm)^2 = -192$. (ii) $V_\pm$ maps $e_a^{(1)}$ to $e_a^{(2)}$ on the temporal (respectively spatial) directions, and G_2 exchanges the two kinds; $V_+ - V_- = \varepsilon \otimes \eta$. (iii) As in Proposition 6.4(v): $E_{(1,i)}E_{(2,i)} \pm E_{(1,i+3)}E_{(2,i+3)}$ is a pair term or a hopping term. Since $[C_b, C_\times] = 0$ and $C_b|0\rangle = 0$, $e^{\theta C_+}|0\rangle = e^{\theta C_\times/2}|0\rangle$, a product of three rotations in the planes $\langle|0\rangle, a^+_{1,i}a^+_{2,i}|0\rangle\rangle$; this is checked exactly at $\cos\frac{\theta}{2} = \frac{3}{5}$. (iv) As in Proposition 6.4(iii), with $\varepsilon \otimes \mathbf{1}_3$ in place of $[\hat{\mathbf{r}}]_\times$. □

Part (i) restates Proposition 9.6(i) of [2]; the rest is fixed (e6-6a, e6-6c, e6-6e). Part (iii) is the compact counterpart of the copy boost of Proposition 2.6. The pairs emerge unitarily, with probability $\binom{3}{k}\cos^{2(3-k)}\frac{\theta}{2}\sin^{2k}\frac{\theta}{2}$ in place of the hyperbolic weights there. As Section 2.4 says, compactness does not protect the number of quanta; commutation with the binding root does, and C_+ fails it. Part (iv) puts C_+ outside the binding-compatible class of Theorem 9.3 of [2]: as a constant field it tears the binding of two manifolds as a temporal b-field tears that of one.

Proposition 6.9 (Mass transfer between constituents)**.** *Let $H_g = m_1E_{(1,1)} + m_2E_{(2,1)}$, $M = (m_1^2+m_2^2)^{1/2}$, $u_1 = e_1 \otimes \mathbf{1}$ and $u_2 = \mathbf{1} \otimes e_1$.*

(i) $\mathrm{Ad}(e^{\theta C_+})E_{(1,1)} = \cos\theta\, E_{(1,1)} + \sin\theta\, E_{(2,1)}$ and $\mathrm{Ad}(e^{\theta C_+})E_{(2,1)} = \cos\theta\, E_{(2,1)} - \sin\theta\, E_{(1,1)}$. Hence $\mathrm{Ad}(e^{\theta C_+})H_g = m_1(\theta)E_{(1,1)} + m_2(\theta)E_{(2,1)}$, where $(m_1(\theta), m_2(\theta))$ is (m_1,m_2) rotated through θ. The spectrum stays $\{\pm M\}$, and $(M,0)$ goes to $(M\cos\theta, M\sin\theta)$. C_- commutes with H_g.

(ii) The ungraded $H_u = m_1u_1 + m_2u_2$ has spectrum $\{\pm m_1 \pm m_2\}$, each value 16 times, and this multiset determines $\{|m_1|,|m_2|\}$. Rotating (m_1,m_2) inside the ungraded family changes the spectrum unless $\sin 2\theta = 0$, from $\{\pm M\}$ at $(M,0)$ to $\{\pm M(\cos\theta+\sin\theta), \pm M(\cos\theta-\sin\theta)\}$. $(\cos\theta\, u_1 + \sin\theta\, u_2)^2 = \mathbf{1} + \sin 2\theta\, u_1u_2$, so for $\sin 2\theta \neq 0$ no invertible matrix conjugates u_1 into $\cos\theta\, u_1 + \sin\theta\, u_2$, whereas $(\cos\theta\, E_{(1,1)} + \sin\theta\, E_{(2,1)})^2 = \mathbf{1}$. The product u_1u_2 of the two ungraded clocks squares to $\mathbf{1}$ and commutes with both. The flip $P(x\otimes y) = y\otimes x$ is orthogonal with $Pu_1P^{-1} = u_2$: the exchange $(m_1,m_2) \mapsto (m_2,m_1)$ keeps the spectrum, it is the only change of $(|m_1|,|m_2|)$ that does, and no continuous family at fixed spectrum changes $(|m_1|,|m_2|)$.

(iii) $\mathrm{Ad}(e^{\theta C_+})$ keeps the spectrum of H_u, but it maps u_1 to $\cos\theta\, u_1 + \sin\theta\, \omega \otimes e_1$. This lies in $\mathrm{Cl}(3,3)\otimes\mathbf{1} + \mathbf{1}\otimes\mathrm{Cl}(3,3)$ only for $\sin\theta = 0$.

Proof. (i) For $i = 2,3$, $E_{(k,1)}$ anticommutes with both factors of $E_{(1,i)}E_{(2,i)}$, so these terms commute with $E_{(k,1)}$. The term $X = E_{(1,1)}E_{(2,1)}$ anticommutes with $E_{(1,1)}$ and $E_{(2,1)}$, so $e^{-\theta X/2}E_{(1,1)}e^{\theta X/2} = e^{-\theta X}E_{(1,1)} = \cos\theta\, E_{(1,1)} + \sin\theta\, E_{(2,1)}$. C_- commutes with $E_{(k,1)}$ by the same count. (ii) u_1 and u_2 are commuting involutions whose products have trace zero, so each joint sign occurs on 16 dimensions. The largest and smallest absolute values in the multiset are

$|m_1|+|m_2|$ and $||m_1|-|m_2||$. Conjugation preserves the spectrum, and $\{\pm\cos\theta\pm\sin\theta\}\neq\{\pm1\}$ for $\sin 2\theta\neq 0$. P exchanges the tensor factors, and since the spectrum fixes $\{|m_1|,|m_2|\}$, a continuous family at fixed spectrum keeps $(|m_1|,|m_2|)$. (iii) $\omega\otimes e_1$ is a tensor of two non-identity blades, independent of every $A\otimes\mathbf{1}$ and $\mathbf{1}\otimes B$. □

This is fixed (e6-6b, e6-6d, e6-6f). The checks are exact at $\cos\frac{\theta}{2}=\frac{3}{5}$, where the rotated ungraded spectrum is $\{\pm\frac{17}{25},\pm\frac{31}{25}\}M$. Part (i) is Proposition 1.2(ii) realised by the temporal copy field. The single bivector $B=E_{(2,1)}E_{(1,1)}$ used there is one of the three terms of $2C_+$, and the other two leave the clocks alone.

Remark 6.10 (Does a second particle acquire mass from the first?)**.** The algebra is fixed. The temporal copy rotation turns the clock $E_{(1,1)}$ of the first constituent into the clock $E_{(2,1)}$ of the second, and from $(M,0)$ it produces $(M\cos\theta, M\sin\theta)$ at fixed graded mass: a massless second copy acquires mass from the first. Reading this as "a second particle acquires mass from the first through the b-field" needs two further structures. One is a clock per copy, $E_{(k,1)}$ as the γ^0 of copy k. The other is the graded reading, and there Proposition 1.2 applies: the copies have no joint mass shell, and the rest energy is the single value M. In that reading the process is a rotation of the mass direction of one particle between two internal clock directions, the masses adding in quadrature as in Proposition 8.11(iv) of [2]. It is not the creation of a second massive particle. In the ungraded reading, where energies add and there are two particles (Proposition 1.2(iii)–(iv)), the continuous transfer is obstructed: no conjugation turns the clock u_1 into $\cos\theta\,u_1+\sin\theta\,u_2$ with $\sin 2\theta\neq 0$, and the additive spectrum fixes the unordered pair of masses (Proposition 6.9(ii)). The only spectrum-preserving change is the exchange of the two masses by the flip of the tensor factors, a relabelling of the constituents rather than a transfer. The transfer is moreover not binding-compatible, and half of it creates quanta in pairs across the constituents (Proposition 6.8(iii)). So the mass transfer between constituents is fixed algebra; the reading of it as the emergence of a second massive particle is interpretive in the one-particle reading and, as a continuous process, obstructed in the two-particle reading.

6.7 Up the ladder: the mass forms of a rung

Theorem 6.11 (Mass forms of a rung)**.** *At rung n let T_n be the product, in increasing order, of the $3n$ temporal generators $E_{(k,i)}$ ($i\le 3$), S_n that of the $3n$ spatial ones, and Γ_n the volume element of* $\mathrm{Cl}(3n,3n)$ *(here S_n is the spatial volume, not the spinor module $(\mathbb{R}^8)^{\otimes n}$ of equation (2) of [1]).*

(i) For every n, $Y^\top T_n+T_nY=0=Y^\top S_n+S_nY$ for every bivector Y, $T_nS_n=\pm\Gamma_n$, and the forms on $(\mathbb{R}^8)^{\otimes n}$ invariant under $\mathfrak{so}(3n,3n)$ are exactly $\mathrm{span}\{T_n,S_n\}$.

(ii) $T_n^\top=(-1)^{3n(3n-1)/2}T_n$ and $S_n^\top=(-1)^{3n+3n(3n-1)/2}S_n$. For $n\equiv 1,2,3,0 \pmod 4$ the pair (T_n,S_n) is (antisymmetric, symmetric), (antisymmetric, antisymmetric), (symmetric, antisymmetric), (symmetric, symmetric). Hence the Grassmann mass kernel invariant under $\mathfrak{so}(3n,3n)$ is unique iff n is odd. It is T_n for $n\equiv 1$ and S_n for $n\equiv 3 \pmod 4$. For $n\equiv 2$ the kernels form the plane $\mathrm{span}\{T_n,S_n\}$*, and for $n\equiv 0$ there is none.*

(iii) At $n=2$, $T_2=-\omega_S\otimes T$ and $S_2=-T\otimes\omega_S$. The four products $M_1\otimes M_2$, $M_k\in\{T,\omega_S\}$, span the forms invariant under $\mathfrak{so}(3,3)^{(1)}\oplus\mathfrak{so}(3,3)^{(2)}$; $T\otimes T$ and $\omega_S\otimes\omega_S$ are symmetric, and $\mathfrak{so}(6,6)$ keeps exactly the antisymmetric ones, T_2 and S_2. The per-copy $T\otimes\mathbf{1}$ is not invariant even under the boosts of copy 2.

(iv) The forms invariant under the temporal rotations $\mathfrak{so}(3n)_+$ are their commutant, of dimension 16, 128 and 1024 at $n=1,2,3$, with (symmetric, antisymmetric) dimensions $(6,10)$, $(64,64)$ and $(528,496)$. They contain T_n and S_n.

(v) *The forms invariant under the diagonal* $\mathfrak{so}(3,3)$*, the same bivector on every constituent, have dimension* 2*,* 14 *and* 168 *at* $n = 1, 2, 3$*, with (symmetric, antisymmetric) dimensions* $(1,1)$*,* $(10,4)$ *and* $(84,84)$*.*

Proof. (i) At every rung the temporal generators are symmetric and the spatial ones antisymmetric, $E_A^\top = \eta_{AA} E_A$, since $\omega^\top = \omega$. By the blade rule in $\mathrm{Cl}(3n, 3n)$, $T_n E_A T_n^{-1} = (-1)^{3n-1} \eta_{AA} E_A = (-1)^{3n-1} E_A^\top$ for every A, so for $Y = E_A E_B$, $T_n Y T_n^{-1} = E_A^\top E_B^\top = (E_B E_A)^\top = -Y^\top$, which is the invariance. The same holds for S_n with the sign $(-1)^{3n}$. Conversely, if M is invariant, then $Y^\top = -T_n Y T_n^{-1}$ gives $Y(T_n^{-1} M) = (T_n^{-1} M) Y$ for every bivector. The commutant of the bivectors is spanned by the blades it contains. A blade e_I commutes with $E_A E_B$ iff $|I \cap \{A, B\}|$ is even, and this holds for all $A \neq B$ iff $I = \emptyset$ or I is everything. So $M \in T_n \,\mathrm{span}\{\mathbf{1}, \Gamma_n\} = \mathrm{span}\{T_n, S_n\}$. (ii) The transpose of a product of m symmetric (respectively antisymmetric) generators is the reversed product times 1 (respectively $(-1)^m$), and reversal costs $(-1)^{m(m-1)/2}$. (iii) $E_{(2,1)} E_{(2,2)} E_{(2,3)} = \omega^3 \otimes T = \omega \otimes T$ and $T\omega = -\omega_S$. Within each copy the invariant forms are $\mathrm{span}\{T, \omega_S\}$, and blade conjugation is diagonal. (iv) A compact Y is antisymmetric, so invariance is commutation; the dimensions are exact orbit counts. (v) Let $c(\mu)$ be the weight multiplicities of a module for the split Cartan subalgebra spanned by the pole operators $e_i e_{i+3}$, which are diagonal. The multiplicity of the trivial module is $\sum_{w \in W(D_3)} \det(w)\, c(\rho - w\rho)$ [39]. It is evaluated on Sym^2 and Λ^2 of $(\mathbb{R}^8)^{\otimes n}$ with the exact weights; at $n = 2$ a direct exact solution on the 216 entries of opposite weight agrees. □

The theorem is fixed (e6-7a to e6-7e). At $n = 1$, (i) restates Corollary 18.4 of [1] and the pairing of Section 28 there, and the antisymmetric count 10 of (iv) restates the kernel space of $\mathfrak{so}(T)$ in Theorem 18.3 there. The pattern of (ii) is Chevalley's: on the spinors of $\mathbb{R}^{m,m}$ the two invariant forms have symmetries $(-1)^{m(m-1)/2}$ and $(-1)^{m(m+1)/2}$ [3, 6], here with $m = 3n$ (inherited). What the arena adds is that the two forms are the temporal and the spatial volume of the rung, with the Klein factors of (iii). The uniqueness of the mass in $T = e_1 e_2 e_3$ is therefore a statement of the first rung. At the paper's three-constituent rung the unique $\mathfrak{so}(9,9)$-invariant Grassmann kernel is the spatial volume S_3, and T_3 is a symmetric (bosonic) form. Under the diagonal action the mass is not unique from $n = 2$ on, and under the temporal rotations alone not even at $n = 1$. At the second rung the antisymmetric invariant kernels pair the temporal volume of one constituent with the spatial volume of the other (iii): no invariant mass kernel belongs to one constituent alone.

6.8 The two assignments compared

Remark 6.12 (The temporal b-field and the b-field of [2])**.** (a) *How they relate* (fixed). In Sections 5 and 9 of [2] the b-field is the η-antisymmetric part of the shift $S = \kappa G_n + b$, and κG_n is its η-self-adjoint part, $(\kappa G)^2 = -c^2/\ell^2$; κG maps V^+ to V^- and back, and that is where c enters. Section 9 of [2] studies the binding-compatible fields, $[b, G_n] = 0$. A b-field on the temporal triple is part of the connection, compact and gauge-admissible (Corollary 2.4 of [1]), but it is not binding-compatible: half of it lies in $\mathfrak{p}_G$ (Proposition 6.4(i)). It belongs to the compact class of Section 5 of [2] with $\mathbf{a} = \mathbf{c}$, it transports κG to $\kappa G_{e^{-\theta R}}$ without changing the rates, and it tears the binding (Proposition 6.4). Between constituents the same holds for the temporal copy rotation (Proposition 6.8(iv)), which lies outside the Schrödinger class of Theorem 9.3 of [2].

(b) *A non-abelian Yang–Mills* $\mathfrak{su}(2)$ *with Cartan direction* J (fixed; the Cartan choice is the clock's). $\mathfrak{so}(3)_+ \cong \mathfrak{su}(2)$ is compact. It acts on V as isospin one on V^+ and trivially on V^-, and on spinors as isospin one-half (Proposition 6.2(iii)). The algebra fixes no Cartan direction, since all elements of $\mathfrak{so}(3)_+$ are conjugate. The kinetic term fixes it to $J = J(e_1)$, the rotation about the clock, which it keeps. The two off-diagonal directions are the clock rotations; they flip the charge, and their half turn acts as CP.

(c) *A mass term for the off-diagonal components* (identity fixed; reading interpretive; mechanism open). The breaking of the kinetic forms by $Y \in \mathfrak{so}(3)_+$ is $Y^\top K^a + K^a Y = 2Y(n)\gamma^a$, with total weight $96|Y(n)|^2$ (Proposition 6.2(i)). This is a quadratic form on $\mathfrak{so}(3)_+$ with kernel $\mathbb{R}J(n)$, invariant under the stabiliser of the clock: an invariant statement of the breaking. It has the form of the adjoint-Higgs mass term $g^2|[b, \langle\Phi\rangle]|^2$ of the Georgi–Glashow $SO(3)$ model [34], with the clock in the role of the Higgs vector [35, 36, 37]: the two charge-flipping components would be massive and $J(n)$ massless. That needs a clock field $n(x)$ with a covariant kinetic term, and the arena supplies none: a dynamics that selects the clock is open (Section 3 of [31]). Gauge invariance forbids a bare mass term (inherited), and the Dirac mass gives none, since $\mathfrak{so}(3)_+$ preserves T (Proposition 6.1). In Remark 9.8(e) of [2] the rate matrix played the role of the adjoint vacuum value; here the clock does.

(d) *What the assignment fixes, constructs, obstructs and leaves open.* Fixed: every finite-dimensional statement of Sections 6.1 to 6.7. Constructed: the clock, which fixes the Cartan direction and the kinetic term, and the temporal b-field itself. Obstructed: its removal along a worldline, bounded binding under it, continuous mass transfer at fixed spectrum in the ungraded reading, and a unique Grassmann mass kernel at even rungs. Open: a dynamics for the clock, hence a mass for the off-diagonal components.

(e) *What it implies for the emergence of a second particle* (interpretive, drawn from the fixed statements). The temporal b-field gives the second rung a process that moves mass from one constituent to the other, but only as a rotation of one mass vector in the graded reading, in which the rung carries one particle. In the reading that has two particles it is obstructed as a continuous process; only the exchange of the two masses, a relabelling, keeps the spectrum. The uniqueness of the mass in T, which the assignment invokes, does not survive the ladder in a form that assigns a mass to each particle: the invariant kernels of the second rung pair the temporal volume of one constituent with the spatial volume of the other.

6.9 Answer

The assignment "c on $(-,-,-)$, b on $(+,+,+)$" has a precise and correct form. The temporal rotations cannot change the symmetric factor $CC^\top$ of the conversion, which lives on the spatial triple and is c^2 in the isotropic case; they keep the mass form T and the rest flow, and they move the orthogonal factor of the conversion, the clock and the binding root. The price is the binding and the kinetic term: a temporal b-field keeps the rates, splits the lines, cannot be gauged away and tears the binding, and the kinetic term breaks it to $\mathfrak{u}(1)_J$ in proportion to the speed of the clock. The mass is unique in $T = e_1e_2e_3$ on the first rung only; up the ladder the unique invariant Grassmann kernel is the temporal volume T_n for $n \equiv 1$ and the spatial volume S_n for $n \equiv 3 \pmod 4$, and at even rungs there is a plane of them or none. Between two constituents the temporal b-field rotates graded mass from one clock to the other at fixed total, a statement about one particle, not two.

References

[1] D. Bencheikh, *The electromagnetic sphere: structure, matter, dynamics and causal structure of the split-signature Clifford algebra* $\mathrm{Cl}(3,3)$ *and its tower*, 17 October 2026 (`ES_electromagnetic_sphere_2026-10-17.pdf`, sources in `tex/`).

[2] *The undivided energy and its differentiation:* $E = mc^2$ *in the split-signature arena*, investigation note to [1], 19 October 2026 (`investigations/rest_energy/main.pdf`).

[3] C. Chevalley, *The Algebraic Theory of Spinors*, Columbia University Press, New York, 1954.

statement	class	checks
conventions; the triples, T, ω_S, Fock modes	restated (ES17 eqs. (1), (2), Thm 18.1)	e6-0
Proposition 6.1(i): T and ω_S invariant forms under $\mathrm{Spin}(3,3)$	restated (ES17 Cor. 18.4, Sec. 28; RE Prop. 3.5)	e6-1a
Proposition 6.1(ii), (iii): commutes with $\mathfrak{so}(3)_+ \oplus \mathfrak{so}(3)_-$ only; commutant $M_4(\mathbb{C})$, 32 blades	fixed ($M_4(\mathbb{C})$ inherited)	e6-1b
Proposition 6.1(iv): $\mathrm{Ad}_T = \eta$; boosts conjugate T to $e^{2sY}T$	restated (RE Prop. 2.11); fixed	e6-1c
Proposition 6.2(i): kinetic breaking $2Y(n)\gamma^a$, weight $96\lvert Y(n)\rvert^2$; $\mathfrak{so}(3)_+ \cap (\mathfrak{l}_{\mathrm{int}} \oplus \mathbb{R}J) = \mathbb{R}J$	fixed (covariance restated, RE Prop. 3.5(ii))	e6-2a
Proposition 6.2(ii): clock rotation of $H(0)$; transport to the clock n	fixed (transport restated, TR Prop. 3.4, Thm 3.5)	e6-2b
Proposition 6.2(iii), (iv): rest flow kept; isospin $\frac{1}{2}$; CP half turn; octant duality	fixed; (iv) restated (TR Prop. 4.12)	e6-2c
Proposition 6.3: $\mathfrak{so}(3)_+$ and $\mathfrak{su}(2)_{\mathrm{PG}}$ two diagonals of $\mathfrak{so}(4)$, meeting in $\mathbb{R}J$	fixed ((v) restated, ES17 Prop. 5.5)	e6-2d
Proposition 6.4(i), (ii): half $\mathfrak{l}_G$, half $\mathfrak{p}_G$; $G \to G_{e^{-\theta R}}$; rates kept; not removable	fixed (rates restated, RE Thm 9.3(iii))	e6-3a, e6-3b
Proposition 6.4(iii), (iv): lines κ, $s \pm \omega/2$; Jordan blocks	fixed	e6-3c, e6-3d
Proposition 6.4(v): paired half conserves N, counter half changes it by ± 2	fixed	e6-3e
Proposition 6.5: $C = cR$; twisted diagonal; orbit $O(3)$; $CC^{\top}$ invariant; q_C	fixed	e6-4a to e6-4e
Remark 6.6: "c^2 on $(-,-,-)$, b on $(+,+,+)$"	(a), (b) fixed; reading interpretive	e6-4d, e6-4f
Proposition 6.7: m_4 oscillates at fixed μ; constant iff on the axis	fixed; the reading interpretive	e6-5a to e6-5c
Proposition 6.8: $C_\pm$, $e^{2\pi C_\pm} = -\mathbf{1}$; pairs of quanta; tearing between frames	fixed ((i) restated, RE Prop. 9.6(i))	e6-6a, e6-6c, e6-6e
Proposition 6.9: graded mass transfer at fixed M; ungraded obstruction	fixed	e6-6b, e6-6d, e6-6f
Remark 6.10: "a second particle acquires mass"	interpretive (one-particle reading); obstructed as a continuous process (two-particle reading)	—
Theorem 6.11(i), (ii): invariant forms $\mathrm{span}\{T_n, S_n\}$; period four	fixed (Chevalley's pattern inherited)	e6-7a, e6-7b
Theorem 6.11(iii)–(v): Klein factors; temporal and diagonal counts	fixed	e6-7a, e6-7c to e6-7e
Remark 6.12: relation to RE; Yang–Mills $\mathfrak{su}(2)$; clock as adjoint Higgs	(a), (b) fixed; (c) identity fixed, reading interpretive, mechanism open; (e) interpretive	e6-2a

Table 14: Claims of Section 6, their classes and their checks, which are `PASS` lines of `verify/e6_bfield_triples.py`. "Restated": fixed in [1], [2] or [31], re-verified here.

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